

**Stuck in the Cycle:
Expected Returns in International
Government Bond Markets**

Master's Thesis

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Anthropic's Claude	Language editing of text I had written myself (grammar, phrasing and clarity)	All chapters	Chat logs retained
Anthropic's Claude	Refactoring and documentation of the analysis code and debugging. Reorganising and making R-scripts more efficient.	R analysis code underlying the empirical chapters (Data & Methodology, Replication, Results, Robustness, Strategy)	Chat logs retained
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Abstract

In our thesis, we examine whether the bond return predictability that Cieslak and Povala (2015) document for the United States also applies to international government bonds. Under the expectations hypothesis, excess returns on long-term bonds should not be forecastable, yet a large literature shows that they are. The cycle factor of Cieslak and Povala (2015), which measures the deviation of yields from a slow-moving trend-inflation anchor, provides a coherent economic mechanism behind this predictability. However, it has so far only been studied for the United States. We therefore construct local cycle factors for eleven G10 markets (five sharing the euro), aggregate them into a GDP-weighted global cycle factor, and additionally build a novel FX-adjusted global cycle factor for an unhedged US-dollar investor. We evaluate all three factors in three empirical phases, both in-sample and out-of-sample. We find that the local cycle factor predicts excess returns in all eleven markets with the sign pattern of the original study and an average in-sample R^2 of 25%. The global cycle factor subsumes the local factors in most markets and is the only factor that survives out of sample. It beats the recursive historical mean in ten of eleven markets, whereas the local factor does so in only two. For an unhedged US-dollar investor, currency conversion erodes most of the predictability. The FX-adjusted factor recovers much of it in sample but retains only a small, conditional edge in real time. Converting the surviving global cycle factor into a strategy, a timing rule on the global factor raises the Sharpe ratio of a hedged global bond portfolio from 0.34 to 0.40.

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Chapter 1

Introduction

Investors earn more on long-term government bonds than on rolling over short-term bills, and this compensation varies predictably over time. According to classical theory, this should not be the case. Under the expectations hypothesis (EH), the yield on an n -year bond is the average of expected future short rates plus at most a constant term premium,¹

$$y_t^{(n)} \approx \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}_t[y_{t+i}^{(1)}],$$

so excess returns from holding longer maturities should not be forecastable, because whatever is known today about future short rates is already priced into the curve. The EH is the null hypothesis of bond pricing. However, it fails empirically. Four decades of evidence on US treasuries document that excess returns are forecastable with information available at the start of the holding period. This evidence begins with the forward–spot spread of Fama and Bliss (1987) and the yield spread regressions of Campbell and Shiller (1991), whose slope coefficients carry the wrong sign relative to the EH benchmark. The modern statement of the puzzle comes from Cochrane and Piazzesi (2005), who show that a single tent-shaped linear combination of forward rates, the CP factor, predicts one-year excess returns across maturities with an R^2 of 0.35 to 0.44. From their research we take away that bond risk premia are time-varying and countercyclical.

The CP factor sets a strong statistical benchmark. However, it tells us little about the economics, since it is a reduced-form combination of prices that explains predictability with other prices. Consequently, the question remains which macroeconomic forces drive the premium. Ludvigson and Ng (2009) show that factors extracted from a large macro panel forecast bond returns above the information in the yield curve, and Cieslak and Povala (2015) provide a clear economic mechanism. They decompose nominal yields into a slow-moving trend-inflation component and transitory cycles. The first is a discounted moving average of past core inflation which serves as an anchor and the latter are the deviations from it. A single cycle factor built from these detrended yields predicts US Treasury excess returns more strongly than the CP factor and subsumes it. The economic intuition is that the trend pins down where yields will settle, so the distance of today’s curve from its macro anchor measures the premium investors demand

¹This is the log-linearised expectations hypothesis. Iterating the one-period relation $p_t^{(n)} = \mathbb{E}_t[p_{t+1}^{(n-1)}] - y_t^{(1)}$ under a constant term premium and using $y_t^{(n)} = -p_t^{(n)}/n$ gives the average-expected-short-rate expression.

for holding duration. The structural counterpart of this view is the “falling stars” evidence of Bauer and Rudebusch (2020). Once the slow drift in the long-run inflation and real-rate anchors is accounted for, the predictable component of returns remains.

However, this macro-anchored evidence is almost exclusively evidence about the United States. If the trend-cycle mechanism exists, the same mechanism should also operate in other developed bond markets, and a failure to do so would cast doubt on the economic interpretation. The international literature is encouraging, though it has pursued different predictors. Dahlquist and Hasseltoft (2013) take the forward-rate factor abroad. They construct local CP factors for four markets, aggregate them into a GDP-weighted global CP factor, and show that the global factor predicts bond returns in every country and largely drives out the local factors. These findings suggest that government-bond risk premia are internationally integrated. They also address a problem that is central to the international setting. An unhedged US-dollar investor earns the foreign bond return plus a currency return. Because this currency component erodes predictability, Dahlquist and Hasseltoft (2013) build an FX-adjusted global factor to recover it. On the macro side, Zhang et al. (2021) document that the falling-stars trends improve return forecasts in four markets outside the United States, and related evidence extends to China (Zhang et al., 2022).

When we place these strands side by side, we can identify a clear gap. The cycle factor of Cieslak and Povala (2015), a factor with a strong economic rationale, has not yet been taken international. There is no country-level evidence across the G10, no GDP-weighted *global* cycle factor analogous to the global CP factor of Dahlquist and Hasseltoft (2013), and no FX-adjusted version of such a factor for the dollar investor. Zhang et al. (2021) come closest. However, they add raw inflation trends to yield-curve regressions in four markets rather than constructing the cycle factor itself, and they build no global or currency-adjusted aggregate. A second gap is methodological. Generated regressors such as the cycle factor are fitted on the very sample over which they are evaluated, so in-sample R^2 can substantially overstate what a real-time investor could have exploited. An out-of-sample test in the spirit of Campbell and Thompson (2008) is therefore part of the question. Whether macro-anchored bond return predictability is a global phenomenon, and whether it survives in real time, is thus still an open question. In our thesis, we aim to answer it.

In our thesis, we ask one main question:

Does the Cieslak and Povala (2015) cycle factor apply to international government bonds?

If the trend-cycle mechanism is a fundamental property of bond pricing rather than a US idiosyncrasy, it should predict excess returns in other developed markets with the same sign structure and a comparable magnitude. To sharpen the international dimension, we pose three sub-questions:

1. *What is the role of the local cycle factor relative to a GDP-weighted global cycle factor?*
Integrated bond markets suggest a shared risk factor across countries (Dahlquist and Hasseltoft, 2013), and a GDP-weighted average of the national cycle factors could be the summary of such a factor.

2. *Does the global cycle factor subsume the predictive power of the local cycle factors?* If markets are integrated, investors are compensated for global rather than local risk. In this case, adding the local cycle to a regression that already contains the global factor should not raise the predictive power.
3. *Can an FX-adjusted global cycle factor recover the predictability that currency conversion destroys and what are its dynamics?* For an unhedged US-dollar investor the realised return carries a currency leg, so the plain global factor might lose much of its power on dollar returns. We would also expect new insights in the dynamics of this factor compared to the unadjusted global cycle factor.

We try to answer these questions in three empirical phases (Section 3.11). In Phase I, we test the cycle factor market by market across the G10. In Phase II, we run the horse race between the global and the local factors. In Phase III, we adopt the perspective of an unhedged US-dollar investor, for whom currency risk intervenes between the foreign bond and the realised dollar return, and we ask whether a purpose-built FX-adjusted global cycle factor can restore the predictability that currency conversion destroys. Because the cycle factors are generated regressors, we evaluate every phase both in-sample and under a fully-recursive out-of-sample protocol, in which the entire construction, that is the trend, the cycles, the predictive coefficients, and the global weights, is rebuilt each month from historically available data only.

With our thesis, we contribute a three-factor hierarchy and the evidence to evaluate it. The first factor is the *local cycle factor* $CF_{i,t}$, which we construct for each of eleven G10 markets as Cieslak and Povala (2015) construct it for the United States, on a monthly panel of zero-coupon yields, core inflation, exchange rates, and GDP spanning 1989–2024. The second, the *global cycle factor* GCF_t , aggregates the national factors with time-varying GDP weights. With this factor, we transplant the integration test of Dahlquist and Hasseltoft (2013) from the forward-rate family to the macro-anchored cycle. The third is a new *FX-adjusted global cycle factor* $FXGCF_t$, which fits a USD cycle factor per country. Each country’s *dollar* excess return is projected on its own cycles and aggregated with GDP weights, isolating the component of the cycles that is informative for a US investor’s currency-converted returns.

Three features distinguish our evidence from similar factors in previous research. The first is breadth. Relative to the four markets of Dahlquist and Hasseltoft (2013) and Zhang et al. (2021), we cover the full G10 cross-section of eleven markets, including the euro-area periphery. The second is the predictive object. While Zhang et al. (2021) add inflation and real-rate trends to yield-curve regressions, we carry the cycle-factor construction itself abroad, so that the US benchmark and the international evidence are directly comparable factor by factor. The third is real-time evaluation. We re-estimate all three factors fully recursively and score them with the out-of-sample R^2 . We complement this with a crisis-subsample analysis, a horse race against the global forward-rate factor, a core-versus-headline-inflation comparison and a market-timing strategy. We also replicate the published US and international results (Chapter B) to validate our data processing pipeline.

We arrive at four main findings. First, we find that the cycle mechanism also works outside the United States. The local cycle factor predicts one-year excess returns with the exact sign pattern that Cieslak and Povala (2015) document for the United States in all eleven markets, significantly

so in ten, with an average in-sample R^2 of 25% and uniform loadings across the maturity curve. Second, we find that the predictability is overwhelmingly global. The GDP-weighted global cycle factor is significant in every market and matches the explanatory power of the eleven separately estimated local factors. Once it is included, genuinely local content survives in only three markets (Italy, the Netherlands, Belgium). Integration is therefore the rule, although it weakens under severe sovereign stress. Third, we find that currency risk is a considerable obstacle for the dollar investor. Dollar conversion erodes roughly two-thirds of the global factor's predictive power. The FX-adjusted factor recovers much of it in sample, substantially raising the average dollar-investor R^2 from 9% to 14% and restoring significance in ten of eleven markets, provided it is built bottom-up from the country-level USD regressions. The two factors are distinct only in the high-rate regime. They correlate 0.54 before 2008 but 0.95 afterwards, as policy rates converge to the zero lower bound and the cross-country interest differentials that make currency returns predictable compress. Out of sample, consistent with this, the FX adjustment delivers only a small and conditional edge. Fourth, and in our view most importantly, only the global factor survives in real time. Out of sample, the local factors lose to the recursive historical mean almost everywhere, while the global cycle factor beats it in ten of eleven markets. This real-time edge holds across crisis subsamples, beats the global forward-rate factor, and depends materially on measuring the trend with core rather than headline inflation. Moreover, it is economically exploitable. A timing strategy based on it raises the Sharpe ratio of a hedged global bond portfolio from 0.34 to 0.40 with a positive certainty-equivalent gain.

The remainder of our thesis is organised as follows. In Chapter 2, we review the three literatures our thesis draws together, namely US bond return predictability, macro-anchored yield decompositions, and international bond risk premia, and locate the gap. In Chapter 3, we describe the G10 panel and develop the technical framework, which covers the yield decomposition, the factor hierarchy, the return definitions for the local and the dollar investor, and the predictive regressions. In Chapter 4, we present the three-phase main results. In Chapter 5, we convert the surviving predictability into a real-time portfolio strategy, and in Chapter 6, we stress-test our findings. In Chapter 7, we interpret the findings and state their limitations, before we conclude in Chapter 8. In Chapter B, we validate our empirical engine against the published results of Cieslak and Povala (2015) and Dahlquist and Hasseltoft (2013).

Chapter 2

Related Literature

In this chapter, we position our thesis in the literature. It draws together three strands of research. The first is the rejection of the expectations hypothesis in US Treasury markets and the search for the right return-forecasting state variable. The second is the macro-finance work that anchors yields to slow-moving inflation trends. The third is the international evidence on common, global bond risk premia. A fourth concern is methodological, as out-of-sample evaluation cuts across all three and shapes our research design. We review each in turn, moving from the US evidence to the international gap our thesis aims to fill. The technical framework we build on these ideas (the yield decomposition, the factor hierarchy, the return definitions, and the predictive regressions) is developed in Chapter 3.

2.1 The Expectations Hypothesis and Bond Return Predictability

The expectations hypothesis holds that long yields are averages of expected future short rates plus at most a constant term premium, so that excess returns on long bonds should not be forecastable. The empirical rejection of this hypothesis is one of the oldest and most robust findings in fixed income. Fama and Bliss (1987) show that the forward–spot spread forecasts the one-year excess return on the matched-maturity bond. Their paper introduces both the founding result and the zero-coupon data construction that became the standard in the field. Campbell and Shiller (1991) contribute to the puzzle. In their regressions of yield changes on the scaled yield spread, the EH predicts slope coefficients of one. However, the estimates are significantly negative and increasingly so with maturity. When the spread is high, long yields tend to fall, which amplifies rather than offsets the return on the long bond. Together, these results establish that the term premium is not constant. It is time-varying and, as the subsequent literature documented, also countercyclical, rising in recessions and falling in booms.

Once predictability is accepted, the next question is which variable forecasts returns. We can distinguish three positions. The first position holds that the information is contained in the yield curve itself and can be distilled into a single combination of forward rates (Section 2.2). The second position holds that macroeconomic variables matter beyond the curve. Ludvigson and Ng (2009) extract factors from a large panel of macro series and show that real-activity and inflation

factors forecast excess returns over and above forward rates and yields. This is the empirical origin of “unspanned” macro risk, which is predictive information that the current yield curve does not reveal. Joslin et al. (2014) give this idea a no-arbitrage form, with output and inflation unspanned by the yield curve yet shaping the prices of level, slope, and curvature risk. The third position, articulated by Duffee (2011), holds that a substantial part of risk-premium variation is hidden from the current curve altogether. Such a hidden factor can move expected short rates and term premia in offsetting directions, leaving the yield curve nearly unchanged. The change in term premia can still be used to forecast returns. The macro-anchored cycle factor at the centre of our thesis (Section 2.3) can be seen as a sharpened version of the second position, since it relies on one specific, economically interpretable macro variable, namely trend inflation.

Another challenge is an econometric one. Bauer and Hamilton (2018) re-examine the published evidence for predictors beyond the level, slope, and curvature of the yield curve and argue that much of it is a small-sample artefact. Persistent regressors, overlapping returns, and the small-sample bias that Stambaugh (1999) identified for predictive regressions with a persistent, predetermined regressor make standard tests over-reject the spanning null, and under a bootstrap designed for this null the case for unspanned predictors weakens considerably. This critique applies with full force to any study of the cycle factor, because the cycle is a persistent, estimated regressor in an overlapping twelve-month regression. The methodological choices we make in Chapter 3 are, therefore, in large part a response to this.

2.2 Forward-Rate Predictors: Cochrane and Piazzesi (2005)

Cochrane and Piazzesi (2005) transformed the predictability literature by showing that a single linear combination of forward rates, with the well-known tent shape in its loadings, forecasts one-year excess returns on two- to five-year US Treasuries with an in-sample R^2 of 0.35 to 0.44. This roughly triples the explanatory power of the single-spread regressions that preceded it. The same factor prices the entire maturity spectrum, as the loadings of individual bonds’ returns on the factor rise smoothly with maturity. This smooth maturity pattern points to one return-forecasting factor rather than maturity-specific premia. For the research that followed, two features of the CP factor stand out. It is largely orthogonal to the level, slope, and curvature factors that summarise the cross-section of yields, which started the spanning debate of Section 2.1. It is a reduced-form object, that is, a combination of prices that predicts returns on those same prices, with no economic mechanism attached. It tells us that premia vary, without saying why. Rebonato and Nyholm (2025) trace the factor’s predictive power to the cointegration among the near-unit-root forward rates rather than to the tent shape of the loadings, and connect this mechanism to the cycle factor we turn to next. This open question is where the macro-anchored literature, and ultimately our thesis, comes in. The CP factor nonetheless remains the benchmark. We therefore replicate it in Chapter B and compare its global version with the global cycle factor in Chapter 6.

2.3 The Macro-Anchored Cycle Factor: Cieslak and Povala (2015)

Cieslak and Povala (2015) provide the economic mechanism behind this predictability. Their starting point is a trend-cycle view of nominal yields. While the level of yields is anchored by long-run inflation expectations, which move only slowly across decades, the return-relevant variation is the transitory deviation of yields from this anchor. Operationally, trend inflation is a discounted moving average of past year-on-year core CPI inflation. This construction goes back to Kozicki and Tinsley (2001), who showed that long-rate dynamics are dominated by a shifting perceived “endpoint” tied to inflation expectations, and that discounting past inflation tracks survey measures of these expectations closely (Kozicki and Tinsley, 2012). Projecting each maturity’s yield on this trend yields maturity-specific *cycles*. A single *cycle factor*, the fitted combination of the short-maturity cycle and the average cycle, then forecasts one-year excess returns with in-sample fits that exceed the forward-rate benchmark, peak at long maturities, and subsume the CP factor in direct comparisons. Beyond its fit, the construction is attractive. Economically, the factor has a clear interpretation. The premium is the distance of the curve from its macro anchor, and it mean-reverts at business-cycle frequency. Econometrically, the detrending reduces the near-unit-root persistence of yields that complicates predictive regressions, because the cycle is stationary while the yield is not.

The structural (model-based) counterpart to this reduced-form approach is Bauer and Rudebusch (2020). In their shifting-endpoint term-structure model, the risk-neutral long-run mean of the short rate moves with trend inflation and the equilibrium real rate ($i^* = \pi^* + r^*$). Accounting for these “falling stars” simultaneously raises measured return predictability, produces more plausible term-premium estimates, and attributes the great post-1980s yield decline largely to the falling anchors rather than to the premium. We can view the reduced-form cycle factor and the structural falling-stars model as two views of the same mechanism. It is this mechanism-based character that leads to the testable international prediction we make use of in this thesis. If the mechanism is fundamental, it should not be unique to the US.

2.4 Global Integration: Dahlquist and Hasseltoft (2013)

That international bond returns are predictable, and predictable by common instruments, has been known since Ilmanen (1995), who documents forecastability in six markets with a small set of global variables and finds that expected returns are highly correlated across countries. The forward-rate factor of Cochrane and Piazzesi (2005) has itself been taken abroad. Kessler and Scherer (2009) and Sekkel (2011) find that it forecasts excess returns across developed markets, though with more modest and more country-specific power than in the United States. The modern factor-based statement of this result is Dahlquist and Hasseltoft (2013), which forms the international backbone of our thesis. They construct local CP factors for the United States, Germany, Switzerland, and the United Kingdom, aggregate them into a GDP-weighted *global* CP factor, and establish two results that define the integration agenda. First, the global factor predicts excess returns in every market. Second, in joint regressions it largely subsumes the local factors, meaning that once the common component is included, country-specific forward-rate

information adds little. They interpret this as evidence of market integration. Internationally diversified investors price a common risk, so compensation accrues to global rather than to local exposure. Furthermore, they tie the global factor to US premia and the international business cycle. Zhu (2015) extends this result to real time and shows that the GDP-weighted global CP factor predicts out of sample across developed markets where local predictors fail. We later mirror this pattern with the global cycle factor, which predicts out of sample where the local cycle factors do not (Section 4.4).

Dahlquist and Hasseltoft (2013) also address currency risk, which is the issue that separates international from domestic bond economics. A US-dollar investor who holds foreign bonds unhedged earns the local bond return plus a currency return that is an order of magnitude more volatile. The currency strand of the literature establishes that this exposure is at best partially compensated. Campbell et al. (2010) show that for a global bond investor the risk-minimising currency position is close to a full hedge, and Lustig et al. (2019) show that the currency carry premium shrinks as foreign bond maturity lengthens, so that long-maturity unhedged positions earn little compensation for their currency risk. Against this background, Dahlquist and Hasseltoft (2013) construct an *FX-adjusted* global factor, which is a version of the global factor fitted to dollar-denominated returns, and show that it recovers much of the predictability that the currency conversion destroys, while remaining only moderately correlated with the unadjusted factor. The FX-adjusted cycle factor we build is the direct analogue of this construction.

2.5 International Macro-Anchored Predictability and the Gap

The two research areas described above, macro-anchored predictability and global aggregation, have only begun to be combined. Zhang et al. (2021) take the falling-stars logic to international data. Using the same discounted moving-average inflation trend plus an estimated real-rate trend, they show that adding both trends to excess-return regressions delivers notable out-of-sample gains in Canada, Germany, Japan, and the United Kingdom. Related evidence extends the trend-augmented approach to Chinese government bonds (Zhang et al., 2022) and documents, for emerging markets, that global factors dominate local ones in explaining bond risk premia (Iania et al., 2021). These papers establish that the macro anchor matters outside the United States. However, they stop short of combining the trends into a single macro-anchored factor. They add raw trends to yield-based regressions market by market rather than constructing the cycle factor itself. They cover four markets instead of the G10 and they build neither a global aggregate nor a currency-adjusted factor. Closest in spirit, Cejnek and Randl (2020) aggregate Cochrane and Piazzesi (2005) bond factors across four regions and, in an extension, apply the Cieslak and Povala (2015) decomposition to bond and dividend claims, but they do not construct a global cycle factor for government bonds.

From this review, we can identify a specific gap in the literature. To our knowledge, no study takes the Cieslak and Povala (2015) cycle factor itself across the G10, no study builds a GDP-weighted *global cycle factor* in the manner of Dahlquist and Hasseltoft (2013), and no study constructs an FX-adjusted global cycle factor for the unhedged dollar investor. The existing international macro-anchored evidence has also not been subjected to the full out-of-sample discipline that the methodological literature demands. Because of this gap, we want to add to

the field by constructing the local, global, and FX-adjusted global cycle factors for eleven G10 markets and by evaluating them in the three-phase design of Section 3.11. We hold every result to the real-time standard, i.e. estimating it out-of-sample (Campbell and Thompson, 2008).

Chapter 3

Data and Methodology

In this chapter, we describe the data and develop the methodology of our empirical analysis. We first describe the data we use to construct the cycle factors and to run the predictive regressions of Chapter 4. We assemble all series in a single workbook (`data.xlsx`) and process them into a monthly country–maturity panel, implemented in `data_preparation.R`. We begin with the cross-sectional universe and the four raw inputs (yields, inflation, GDP, and exchange rates) with their sources and units, then present the sample coverage, our cleaning and alignment choices, and summary statistics. We then develop the technical framework. Namely the expectations-hypothesis null, the yield decomposition, the three-factor hierarchy, the excess-return definitions, and the predictive specifications. We set out the estimation and inference used to evaluate them.

3.1 Universe and Sources

Our cross-section is the G10 government-bond universe, comprising eleven countries. That is Belgium (BE), Canada (CA), Switzerland (CH), Germany (DE), France (FR), the United Kingdom (GB), Italy (IT), Japan (JP), the Netherlands (NL), Sweden (SE), and the United States (US). Five of these countries share the euro (BE, DE, FR, IT, NL). The remaining currencies are the Canadian dollar (CAD), Swiss franc (CHF), pound sterling (GBP), Japanese yen (JPY), Swedish krona (SEK), and US dollar (USD). The US dollar serves as the home currency for the international investor.

We obtain four raw inputs for our analysis, drawn from two data providers:

Zero-coupon yields (Bloomberg).

We obtain end-of-month yields from Bloomberg’s fitted zero-coupon government curves, one curve per country, at maturities of 1, 2, 4, 5, 9 and 10 years. Each country maps to a Bloomberg curve identifier (e.g. I25 for the US, I22 for the UK, I18 for Japan), which we record in the workbook’s `curve_translation` sheet. Yields are continuously compounded and quoted in percent per annum.

Core CPI (FRED).

We use monthly seasonally adjusted core consumer price indices, one per country. The workbook stores the *index level*. We compute year-on-year inflation $\pi_{i,t} = (\text{CPI}_{i,t}/\text{CPI}_{i,t-12} - 1) \times 100$ in-sample, and trend inflation $\pi_{i,t}^e$ then follows from the EWMA in (3.3).

Table 3.1**Effective monthly coverage of the yield panel, by country.**

Country	Code	Currency	Sample	Months
United States	US	USD	1989-03 – 2024-04	422
Japan	JP	JPY	1989-04 – 2024-04	421
Belgium	BE	EUR	1994-12 – 2024-04	353
Canada	CA	CAD	1994-12 – 2024-04	353
Switzerland	CH	CHF	1994-12 – 2024-04	353
Germany	DE	EUR	1994-12 – 2024-04	353
France	FR	EUR	1994-12 – 2024-04	353
United Kingdom	GB	GBP	1994-12 – 2024-04	353
Italy	IT	EUR	1994-12 – 2024-04	353
Netherlands	NL	EUR	1994-12 – 2024-04	353
Sweden	SE	SEK	1994-12 – 2024-04	353

Nominal GDP (LSEG Refinitiv).

We use annual nominal GDP in local currency (e.g. `aUSWDALB5A`) to form the time-varying global weights $w_{i,t}$ of (3.7). Before weighting, we convert GDP to a common US-dollar basis using year-end exchange rates.

Spot exchange rates (Bloomberg).

We obtain end-of-month spot rates quoted as US dollars per unit of foreign currency (`EURUSD`, `CADUSD`, `JPYUSD`, `GBPUSD`, `SEKUSD`, `CHFUSD`), with the USD rate fixed at unity. These series give us the log currency return in the dollar-investor excess return (3.9) and the FX conversion of GDP.

3.2 Sample and Coverage

Our panel is monthly and sampled at month-end. Although the raw yield workbook extends back to 1980 and the inflation indices to 1955, the usable span is governed at the early end by the availability of each country’s Bloomberg zero-coupon curve and at the recent end by the annual GDP weights used for the global factor. The United States and Japan enter the sample first, in March and April 1989 respectively. The remaining nine countries begin in December 1994, and the estimation sample runs through April 2024, although the raw yield curves themselves extend to April 2026 (Figure A.6). This yields 422 monthly observations for the US, 421 for Japan, and 353 for each of the other countries. The excess-return regressions use somewhat fewer after the factor construction and the twelve-month return horizon. In Table 3.1 we summarise the effective coverage.

Our panel is therefore unbalanced. We handle this throughout by computing cross-country objects only over the countries that are present at each date. The global weights $w_{i,t}$ in (3.7) are, thus, renormalised each month to sum to one over the available panel, and we compute the summary correlations below on a pairwise-complete basis. For the five euro-area members we apply the euro rate throughout, including the pre-1999 years before the single currency, so their pre-1999 dollar-investor currency returns reflect a single vendor-backfilled euro series rather than the legacy national currencies. In Figure 3.1 we visualise the coverage, and in Figure A.6 we

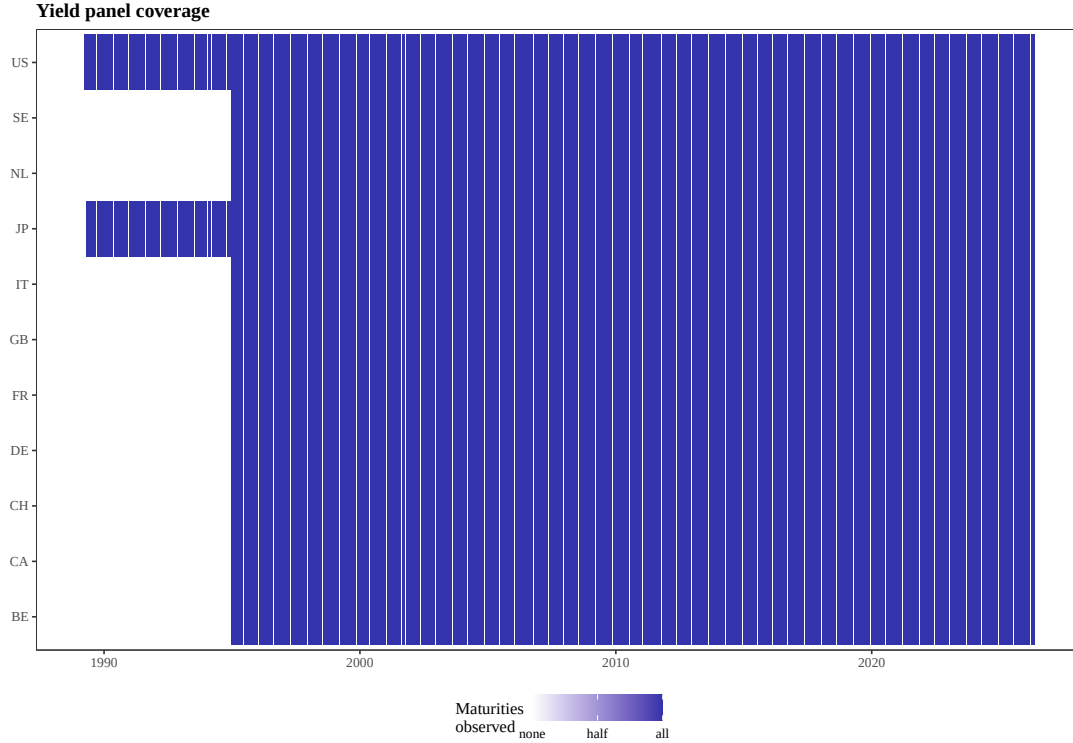


Figure 3.1: Sample coverage by country and maturity. The United States and Japan extend back to 1989. The remaining countries begin in late 1994.

show the resulting 10-year yield series.

3.3 Cleaning and Variable Construction

We align the raw inputs to a common monthly grid keyed on year-month and reshape the panel to long form (country $\times$ maturity $\times$ month). Four steps in our construction need spelling out:

1. **Trend inflation.** We compute year-on-year core inflation from the CPI index and smooth it with the backward-looking EWMA (3.3) with window $M = 120$ months and decay $v = 0.987$.
2. **Cycles.** For each country-maturity pair, we project yields on trend inflation by OLS (3.2). The residual is the cycle $c_{i,t}^{(n)}$. The one-year cycle and the average cycle over $\{2, 4, 5, 9, 10\}$ years (3.4) are the inputs to the cycle factor.
3. **Excess returns.** We form one-year holding-period excess returns (3.1) for $n \in \{2, 5, 10\}$ (the maturities for which the adjacent shorter yield $y^{(n-1)}$ is observed), standardise them by duration $D^{(n)} = n$, and average them (3.8). For the dollar investor, we add the realised currency return and net the US short rate (3.9).
4. **GDP weights.** We convert annual nominal GDP to US dollars at the year-end exchange rate and divide it by the cross-sectional total to obtain $w_{i,t}$. We hold the weights constant within a calendar year and renormalise them over the available panel.

Table 3.2**Summary statistics of zero-coupon yields by country and maturity**

Country	$y_t^{(1)}$		$y_t^{(2)}$		$y_t^{(4)}$		$y_t^{(5)}$		$y_t^{(9)}$		$y_t^{(10)}$	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Belgium	1.88	1.86	2.07	1.92	2.46	2.01	2.64	2.04	3.25	2.08	3.37	2.07
Germany	1.78	1.86	1.89	1.91	2.19	2.02	2.33	2.04	2.83	2.12	2.91	2.08
France	1.90	1.94	2.04	1.97	2.39	2.01	2.55	2.01	3.12	2.01	3.23	1.99
Italy	2.61	2.58	2.96	2.65	3.43	2.60	3.64	2.58	4.28	2.46	4.40	2.43
Netherlands	1.79	1.86	1.95	1.92	2.30	2.01	2.46	2.02	2.99	2.06	3.08	2.04
Switzerland	0.85	1.36	0.86	1.36	1.10	1.45	1.23	1.48	1.63	1.56	1.71	1.57
United Kingdom	3.07	2.46	3.16	2.43	3.38	2.32	3.47	2.26	3.77	2.10	3.83	2.05
Sweden	2.26	2.32	2.43	2.40	2.73	2.45	2.85	2.46	3.23	2.49	3.30	2.48
Canada	2.77	1.90	2.91	1.89	3.21	1.93	3.32	1.93	3.69	1.92	3.75	1.91
Japan	0.90	1.89	0.97	1.81	1.19	1.84	1.33	1.89	1.76	1.88	1.85	1.84
United States	3.12	2.38	3.37	2.40	3.73	2.28	3.90	2.24	4.35	2.10	4.44	2.08

This table reports the mean and standard deviation of monthly zero-coupon yields $y_t^{(n)}$, in percent per annum, by country and maturity $n \in \{1, 2, 4, 5, 9, 10\}$ years, over the full per-country sample: the United States from 1989-03 and Japan from 1989-04; all other countries from 1994-12; all through 2024-04 (see Table 3.1). Average yields rise with maturity while volatility is broadly flat across the curve. After Dahlquist and Hasseltoft (2013), Table 1.

Missing values arise from the unbalanced panel and from the twelve-month leads and lags in the return construction. We propagate them as NA and drop them pairwise in each regression, rather than imputing them.

3.4 Summary Statistics

In Table 3.2 we report the mean and standard deviation of yields for every country–maturity pair. We can note two features that are consistent across countries. First, average yield curves slope upward in maturity. Second, volatility is broadly flat across the maturity spectrum. The short end is somewhat more volatile in the United States, the United Kingdom, Italy, and Japan, whereas in much of the euro area the long end is marginally more volatile. In Figure A.2 we show the average curve shapes. Summary statistics for the non-yield inputs (core inflation, the currency return, and the GDP weight) are collected in Table A.3.

In Table A.2 we report the cross-country correlation matrix of the ten-year yield, computed pairwise-complete over the full per-country sample. We can see that yields co-move strongly internationally. The correlations are uniformly high and are tightest within the euro-area bloc (BE, DE, FR, IT, NL). This co-movement is what motivates a *global* cycle factor. If the levels of long yields share a common component across countries, it seems plausible that the predictable component of their returns does so as well. We examine this integration hypothesis in Chapter 4.

3.5 The Expectations Hypothesis as a Null

Having described the data, we now develop the formal apparatus that it feeds. Throughout, i indexes countries in the G10 cross-section $\mathcal{G}$, time t is measured in months, and bond maturity n in years. The empirical maturity menu is $N = \{1, 2, 4, 5, 9, 10\}$ years, and all yields are continuously compounded and expressed in percent per annum. We let $y_t^{(n)}$ denote the time- t continuously

compounded yield on an n -year zero-coupon bond and $p_t^{(n)} = -n y_t^{(n)}$ the corresponding log price. The one-year holding-period excess return on an n -year bond,

$$rx_{t+12}^{(n)} = p_{t+12}^{(n-1)} - p_t^{(n)} - y_t^{(1)} = n y_t^{(n)} - (n-1) y_{t+12}^{(n-1)} - y_t^{(1)}, \quad (3.1)$$

should therefore satisfy $\mathbb{E}_t[rx_{t+12}^{(n)}] = \text{const}$, i.e. the expected excess return is the same at every date. However, the empirical bond-pricing literature rejects this restriction. The remainder of this chapter builds the predictors we use to document and interpret this predictability internationally.

3.6 Yield Decomposition and Interest-Rate Cycles

The central argument of Cieslak and Povala (2015) is that nominal yields are driven by two forces. The first is a persistent *trend-inflation* component $\pi_{i,t}^e$ that anchors the level of interest rates over long horizons, and the second consists of transitory *cycles* $c_{i,t}^{(n)}$ that capture business-cycle conditions, real-rate movements, and variation in risk appetite. For each country i and maturity n , we decompose yields by the linear projection

$$y_{i,t}^{(n)} = \alpha_{i,n} + \beta_{i,n} \pi_{i,t}^e + c_{i,t}^{(n)}, \quad c_{i,t}^{(n)} \equiv \varepsilon_{i,t}^{(n)}, \quad (3.2)$$

so that the cycle is, by definition, the residual of yields on trend inflation. We estimate this projection separately for every country–maturity pair by ordinary least squares.¹

We construct trend inflation as a backward-looking, exponentially weighted moving average of realised year-on-year core inflation $\pi_{i,t}$:

$$\pi_{i,t}^e = \frac{1-v}{1-v^M} \sum_{j=0}^{M-1} v^j \pi_{i,t-j}, \quad M = 120, \quad v = 0.987. \quad (3.3)$$

The window spans $M = 120$ months (ten years), and the geometric decay $v = 0.987$ has a half-life of roughly four and a half years. Because the average never looks forward, $\pi_{i,t}^e$ is observable at t and introduces no look-ahead bias. The normalising constant $(1-v)/(1-v^M)$ makes the weights sum to one.

We run the decomposition in (3.2) country-by-country and maturity-by-maturity, so the loadings $(\alpha_{i,n}, \beta_{i,n})$, and thus the cycles, are local objects. The cross-country structure only enters later, through the global factor. The decay parameter $v = 0.987$ follows Cieslak and Povala (2015). In Chapter 6, we examine the sensitivity of our results to this choice and to the window length M .

3.7 The Local Cycle Factor

Cieslak and Povala (2015) summarise the cross-section of cycles with two objects. The first is the one-year cycle $c_{i,t}^{(1)}$, which tracks the cyclical position of the short rate, and the second is the

¹Empirically the cycle is the fitted OLS residual $\hat{\varepsilon}_{i,t}^{(n)}$. We keep the population notation $(\alpha_{i,n}, \beta_{i,n})$ in (3.2) and reserve hats for the fitted factor in (3.6).

average cycle

$$\bar{c}_{i,t} = \frac{1}{|N \setminus \{1\}|} \sum_{n \in N, n \neq 1} c_{i,t}^{(n)}, \quad (3.4)$$

which captures the common, longer-maturity cyclical variation and attenuates maturity-specific measurement error. We define the *local cycle factor* as the fitted value of the one-year-ahead excess return projected on these two predictors:

$$\bar{r}x_{i,t+12} = \gamma_0 + \gamma_{i,1} c_{i,t}^{(1)} + \gamma_{i,2} \bar{c}_{i,t} + \varepsilon_{i,t+12}, \quad (3.5)$$

$$CF_{i,t} = \hat{\gamma}_{i,1} c_{i,t}^{(1)} + \hat{\gamma}_{i,2} \bar{c}_{i,t}. \quad (3.6)$$

Here, $\bar{r}x_{i,t+12}$ is the duration-standardised, maturity-averaged excess return that we define in Section 3.9. $CF_{i,t}$ is, by construction, the component of the cycle cross-section that is relevant for predicting returns. Because the factor is a fitted value rather than an observed regressor, we treat it throughout as a generated regressor. That means it is fitted and evaluated on the same data, which is the reason for the in-sample inference of Section 3.12 and the fully-recursive out-of-sample tests of Section 3.12.

3.8 The Global Cycle Factor and GDP Weights

If government-bond markets are internationally integrated, investors should be compensated for exposure to a common source of risk rather than for country-specific fluctuations (Dahlquist and Hasseltoft, 2013). To capture this idea, we aggregate the local cycle factors into a single global factor. We use time-varying GDP weights so that larger economies contribute proportionally more:

$$GCF_t = \sum_{i \in \mathcal{G}} w_{i,t} CF_{i,t}, \quad w_{i,t} = \frac{\text{GDP}_{i,t}}{\sum_{j \in \mathcal{G}} \text{GDP}_{j,t}}. \quad (3.7)$$

We recompute the weights $w_{i,t}$ each month over the set of countries that are present in the estimation panel at t , so that $\sum_i w_{i,t} = 1$ holds by construction even when coverage is not fully available across all countries (Section 3.2). We convert nominal GDP to US dollars before weighting. The global factor GCF_t is the international counterpart of the local $CF_{i,t}$ and provides us with a test of market integration. Under full integration, the global factor should subsume the predictive content of the local factor (Section 3.11).

3.9 Excess Returns: Local and US-Dollar Investor

Local-currency returns. The one-year excess return on an n -year bond for a domestic investor is given by (3.1). Because return volatility scales roughly with duration, we follow Cieslak and Povala (2015) and work with *duration-standardised* returns, $\widetilde{r}x_{i,t+12}^{(n)} = rx_{i,t+12}^{(n)}/D^{(n)}$, and then average across the maturity set to obtain a single, scale-comparable return series:

$$\bar{r}x_{i,t+12} = \frac{1}{K} \sum_{n \in \mathcal{N}} \frac{rx_{i,t+12}^{(n)}}{D^{(n)}}, \quad K = |\mathcal{N}|, \quad D^{(n)} = n. \quad (3.8)$$

The duration weight $D^{(n)} = n$ is the Macaulay duration of a continuously compounded zero-coupon bond. For the construction, we use the averaging set $\mathcal{N} = \{2, 5, 10\}$ years. These are the maturities for which a clean one-year-ahead excess return is identified given the data menu, since each requires the adjacent shorter yield $y^{(n-1)}$, i.e. $y^{(1)}, y^{(4)}, y^{(9)}$, which are all available.

US-dollar investor returns. We now consider a US investor who borrows for one year in US dollars, converts to foreign currency, holds the foreign bond, and converts the proceeds back. For this investor, the dollar excess return adds the realised currency return and nets out the US short rate:

$$rx_{i,t+12}^{(n),\text{USD}} = rx_{i,t+12}^{(n)} + y_{i,t}^{(1)} + (s_{i,t+12} - s_{i,t}) - y_{\text{US},t}^{(1)}, \quad (3.9)$$

where $s_{i,t} = \log(\text{USD per unit of currency } i)$, so that $s_{i,t+12} - s_{i,t}$ is the one-year log currency return. We apply the same duration-standardisation and maturity-averaging as in (3.8) to (3.9) and obtain $\overline{rx}_{i,t+12}^{\text{USD}}$. This dollar return is the object of interest for the international investor and the input to the FX-adjusted factor below.

3.10 The FX-Adjusted Global Cycle Factor

A global factor built from local-currency returns ignores the currency risk that an international investor bears. Because of this, we construct an *FX-adjusted global cycle factor* $FXGCF_t$ that targets dollar returns directly, in analogy to the FX-adjusted global factor (FXGCP) of Dahlquist and Hasseltoft (2013). We build the factor bottom-up, exactly as the global factor in (3.7), but from country-level factors that are fitted to dollar returns. For each country, we first estimate the *USD cycle factor* $FXCF_{i,t}$ as the fitted value of the country’s dollar excess return on its own cycle predictors:

$$\overline{rx}_{i,t+12}^{\text{USD}} = \delta_{i,0} + \delta_{i,1} c_{i,t}^{(1)} + \delta_{i,2} \bar{c}_{i,t} + \varepsilon_{i,t+12}, \quad (3.10)$$

$$FXCF_{i,t} = \hat{\delta}_{i,1} c_{i,t}^{(1)} + \hat{\delta}_{i,2} \bar{c}_{i,t}. \quad (3.11)$$

We then aggregate the national USD cycle factors with the same GDP weights as in (3.7):

$$FXGCF_t = \sum_{i \in \mathcal{G}} w_{i,t} FXCF_{i,t}. \quad (3.12)$$

By construction, (3.10)–(3.12) mirror the global factor exactly. $FXGCF_t$ is to the dollar returns what GCF_t is to the local-currency returns, so any difference between the two factors can be attributed to the return currency alone. An alternative is the “top-down” construction of Dahlquist and Hasseltoft (2013), in which the GDP-weighted average dollar return is projected on the GDP-weighted average cycle predictors. We examined this variant and found that it is nearly collinear with the unadjusted global factor in our sample (correlation 0.99), so it adds almost no incremental information about currency risk. We therefore adopt the bottom-up construction as our main specification and compare the two constructions in Section 6.5.

3.11 Hypotheses and Predictive Specifications

We map the three-factor hierarchy onto our research questions through a sequence of one-year-ahead predictive regressions. Because the data are monthly but the forecast horizon is twelve months, all of these regressions use *overlapping* returns. Consecutive observations share eleven months of the realised return, so the residual is serially correlated by construction as an MA(11) process. It is central to the HAC inference of Section 3.12. We estimate each specification by OLS and report the slope point estimate with robust inference.

Phase I — Does the cycle factor predict returns locally? For each country, we estimate

$$\overline{rx}_{i,t+12} = \alpha_i + \beta_i CF_{i,t} + \varepsilon_{i,t+12}, \quad (3.13)$$

and test $H_0: \beta_i = 0$. A rejection of this null establishes that the macro-anchored cycle mechanism of Cieslak and Povala (2015) also operates outside the United States.

Phase II — Does the global factor subsume the local factor? We run the horse race

$$\overline{rx}_{i,t+12} = \alpha_i + \beta_i CF_{i,t}^\perp + \gamma_i GCF_t + \varepsilon_{i,t+12}, \quad (3.14)$$

$$\overline{rx}_{i,t+12} = \alpha_i + \gamma_i GCF_t + \varepsilon_{i,t+12}. \quad (3.15)$$

Under full market integration, the global factor should drive out the local factor. To make the comparison clean, we orthogonalise the local factor with respect to the global factor before the joint regression. That is, $CF_{i,t}^\perp$ is the residual of the construction regression

$$CF_{i,t} = a_i + b_i GCF_t + CF_{i,t}^\perp, \quad (3.16)$$

so that β_i in (3.14) measures the incremental local content. Adding $CF_{i,t}^\perp$ to (3.15) should not raise the predictive power under integration, i.e. $\beta_i = 0$.

Phase III — Does currency risk break the model, and does the FX-adjusted factor restore it? For the dollar investor, we estimate

$$\overline{rx}_{i,t+12}^{\text{USD}} = \alpha_i + \beta_i CF_{i,t} + \varepsilon_{i,t+12}, \quad (3.17)$$

$$\overline{rx}_{i,t+12}^{\text{USD}} = \alpha_i + \gamma_i GCF_t + \varepsilon_{i,t+12}, \quad (3.18)$$

$$\overline{rx}_{i,t+12}^{\text{USD}} = \alpha_i + \delta_i FXGCF_t + \varepsilon_{i,t+12}. \quad (3.19)$$

With specifications (3.17)–(3.18), we ask whether the local-currency factors retain their predictive power once currency risk is introduced. With (3.19), we ask whether the purpose-built FX-adjusted factor $FXGCF_t$ recovers the predictability that an unadjusted global factor loses on currency-converted returns. Together, (3.13)–(3.19) constitute the empirical programme that we test in Chapter 4 and stress-test in Chapter 6.

3.12 Estimation and Inference

We estimate every predictive regression by ordinary least squares and tailor the inference to the overlapping-return structure. The t -statistics use heteroskedasticity- and autocorrelation-consistent (HAC) Newey–West standard errors (Newey and West, 1987) with 18 lags. For multi-regressor specifications we add a joint HAC Wald test that all slopes are zero. Because the cycle factors are generated regressors, we re-evaluate them fully recursively, rebuilding the entire factor chain (decomposition, local regression, and global aggregation) from data available at each date t so that both the factor and the forecasting regression respect t (a “doubly out-of-sample” forecast), and scoring it by the Campbell and Thompson (2008) out-of-sample R^2 against the recursive prevailing mean,

$$R_{\text{os}}^2 = 1 - \frac{\sum_t (\overline{rx}_{t+12} - \widehat{rx}_{t+12}^{\text{factor}})^2}{\sum_t (\overline{rx}_{t+12} - \widehat{rx}_{t+12}^{\text{mean}})^2}, \quad (3.20)$$

pooled across countries so that each competes against its own recursive mean and applied identically to the Cochrane–Piazzesi comparison factors for the head-to-head evidence of Chapter 6.

3.13 Deviations from the Original Studies

For transparency, we collect the substantive departures from Cieslak and Povala (2015) and Dahlquist and Hasseltoft (2013) in this section. We revisit each of them where it bears on the results.

- **Maturity menu.** Yields are available at $\{1, 2, 4, 5, 9, 10\}$ years rather than the finer grid of the original papers. We form individual excess returns for $n \in \{2, 5, 10\}$, the maturities with an observed adjacent shorter yield.
- **Sample.** The G10 panel spans 1989/1994–2026 (Section 3.2), which is shorter than with the original US samples, so we expect the replicated levels to differ from the published numbers.
- **Inference.** We replace the exact reverse-regression delta method with HAC (18-lag Newey–West) t -statistics, for the identification reason given in Section 3.12.
- **FX-adjusted factor.** We build $FXGCF_t$ bottom-up, as the GDP-weighted average of per-country USD cycle factors, rather than top-down or as a projection on GCF_t (Section 3.10). We compare the constructions in Section 6.5.

These choices keep our construction internally consistent and reproducible while remaining as close to the source methodology as the data permit.

Chapter 4

Empirical Findings

In this chapter, we present the empirical core of our thesis. In Chapter B, we established that our estimation engine reproduces the United States evidence of Cieslak and Povala (2015) and the international evidence of Dahlquist and Hasseltoft (2013). The local cycle factor recovers the published sign pattern and a return-forecasting factor that correlates 0.69 with the average cycle, compared to the 0.61 that Cieslak and Povala (2015) report. Building on this validation, we now turn to the three research questions in the order set out in Section 3.11. In Phase I, we ask whether the macro-anchored cycle mechanism predicts bond returns market by market across the G10. In Phase II, we ask whether a single GDP-weighted global factor subsumes these local factors, to test international market integration. In Phase III, we ask whether currency risk overturns the predictability for a US-dollar investor and, if so, whether the FX-adjusted global factor restores it. We then study the dynamics of the adjusted and unadjusted global factors, asking when the FX adjustment adds independent information rather than merely rescaling the global factor.

In all regressions, we forecast the one-year-ahead, duration-standardised, maturity-averaged excess return $\overline{rx}_{i,t+12}$ of (3.8) from time- t predictors. We estimate them by ordinary least squares and evaluate them with the 18-lag Newey–West HAC inference of Section 3.12. The result tables and figures are generated by `main_results.R`.

4.1 The Local Cycle Factor across the G10

The first question we examine is whether the cycle factor of Cieslak and Povala (2015), which was designed for and tested on US Treasuries, also predicts government-bond returns in each member of the G10. The relevant specification is the local predictive regression (3.13), $\overline{rx}_{i,t+12} = \alpha_i + \beta_i CF_{i,t} + \varepsilon_{i,t+12}$. Because the local cycle factor $CF_{i,t}$ is the fitted value of the excess return on the two cycle predictors (3.5)–(3.6), the regression on CF has slope identically one and an R^2 equal to the fit of that underlying forecasting regression. In Table 4.1, we therefore report the informative object directly. The country-by-country projection of $\overline{rx}_{i,t+12}$ on the one-year cycle $c_{i,t}^{(1)}$ and the average cycle $\bar{c}_{i,t}$, of which CF is the fitted combination how Cieslak and Povala (2015) do it.

We find that the cycle mechanism extends to the international cross-section, and it does so with a uniform structure. In Table 4.1 we can see that the one-year cycle enters negatively

Table 4.1

Phase I: local cycle-factor predictability across the G10

Country	$c_{i,t}^{(1)}$		$\bar{c}_{i,t}$		R^2	Wald p	N
	$\hat{\gamma}_1$	t	$\hat{\gamma}_2$	t			
Belgium	-0.42	(-3.42)	0.56	(4.61)	0.301	0.000	353
Germany	-0.34	(-3.27)	0.47	(4.82)	0.256	0.000	353
France	-0.39	(-4.02)	0.51	(4.26)	0.126	0.000	353
Italy	-0.26	(-1.46)	0.59	(3.19)	0.287	0.000	353
Netherlands	-0.36	(-3.18)	0.56	(5.46)	0.344	0.000	353
Switzerland	-0.11	(-1.17)	0.26	(1.77)	0.073	0.201	353
United Kingdom	-0.41	(-4.36)	0.54	(4.54)	0.262	0.000	353
Sweden	-0.26	(-1.91)	0.43	(3.33)	0.238	0.001	353
Canada	-0.22	(-2.37)	0.41	(4.36)	0.320	0.000	353
Japan	-0.33	(-1.70)	0.50	(2.50)	0.260	0.005	412
United States	-0.40	(-3.21)	0.79	(4.30)	0.326	0.000	412
G10 panel	-0.33	(-10.70)	0.51	(15.46)	0.256	0.000	4001

The dependent variable is the duration-standardized, maturity-averaged one-year excess return $\overline{rx}_{i,t+12}$. Each row reports the predictive regression $\overline{rx}_{i,t+12} = \gamma_0 + \gamma_1 c_{i,t}^{(1)} + \gamma_2 \bar{c}_{i,t} + \varepsilon_{i,t+12}$ underlying the local cycle factor (3.6); $CF_{i,t}$ is the fitted value of this regression. Newey–West HAC t -statistics (12-month overlap) in parentheses. Wald p is the p -value of the joint test that both cycle slopes are zero. “G10 panel” pools all countries with country fixed effects. R^2 is in-sample. Sample coverage as in Table 3.1.

($\hat{\gamma}_1 < 0$) and the average cycle positively ($\hat{\gamma}_2 > 0$) in every one of the eleven markets. This is the sign pattern that Cieslak and Povala (2015) document for the United States, in which the level of the short-rate cycle and the slope of the cycle curve carry opposite-signed information about future returns. The joint Wald test rejects the no-predictability null at the one-percent level in ten of the eleven markets. The only exception is Switzerland, where the cycle slopes are correctly signed but jointly insignificant ($p = 0.20$). When we pool the panel with country fixed effects, both loadings are highly significant ($t = -10.7$ on the one-year cycle and $t = 15.5$ on the average cycle).

In Figure A.1, we plot the in-sample R^2 of (3.13) by country. It ranges from 7% in Switzerland to 34% in the Netherlands, averages 25% across the G10, and reaches 33% in the United States, with eight of the eleven markets above an R^2 of 24%.

The predictability also holds at each individual maturity and not only for the maturity-averaged return. In Table A.1, we repeat the exercise of Cieslak and Povala (2015), Table 4, for the G10. We regress the duration-standardised individual excess return $rx_{i,t+12}^{(n)}/n$ on the local cycle factor for $n \in \{2, 5, 10\}$ years, market by market. The factor loads positively at all three maturities in every market and significantly so in ten of the eleven, Switzerland is again the exception. The loadings are flat to rising in maturity, near 0.6–0.9 at two years, 1.1–1.3 at five, and 1.0–1.2 at ten, with pooled values of 0.78, 1.16 and 1.06 respectively. This is the signature of a single return-forecasting factor. We see the same effect in Cieslak and Povala (2015) for the United States. The per-maturity R^2 tracks the maturity-averaged figures and is largest at the five- and ten-year points, reaching 35% at ten years for the United States and 36% at five years for the Netherlands.

This settles the first research question. The macro-anchored cycle of Cieslak and Povala

Global cycle factor (GDP-weighted, eq 7–8)

Heavy line: global cycle factor GCF. Light lines: country-level local factors.

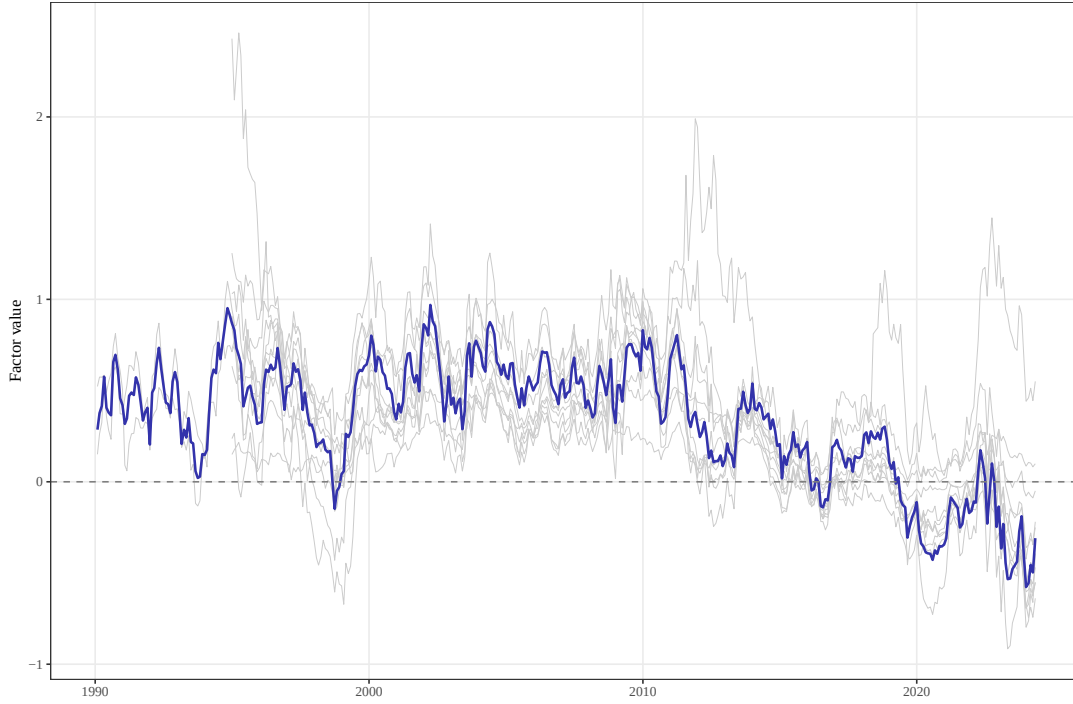


Figure 4.1: The GDP-weighted global cycle factor GCF_t (3.7) (dark blue line) against the eleven country-level local cycle factors $CF_{i,t}$ (light lines).

(2015) is not unique to the United States. It predicts one-year bond excess returns, with the theoretically predicted sign structure and economically large R^2 , throughout the G10, and it does so at every point of the maturity curve.

4.2 Does the Global Factor Subsume the Local Factor?

If government-bond markets are internationally integrated, the predictable component of returns should reflect a common source of risk rather than eleven separate national cycles (Dahlquist and Hasseltoft, 2013). The GDP-weighted global cycle factor GCF_t of (3.7) is the summary of this common component. In Figure 4.1, we can see that it tracks the centre of mass of the country-level factors. It correlates most strongly with the large core economies (the United States, 0.96; Germany, 0.94), but this is not purely a weight effect. Sweden and Canada also correlate strongly (0.94 and 0.89) despite small weights, while Italy is the outlier at 0.21, so the global factor captures common business-cycle co-movement rather than the biggest economies mechanically (Table A.4). In Phase II, we test integration directly through the horse race (3.14)–(3.15).

Under full integration, the global factor should drive out the local factor, so that $CF_{i,t}$ adds no predictive power once GCF_t is included. To isolate the local content, we orthogonalise the local factor against the global factor. Following the construction of Dahlquist and Hasseltoft (2013), $CF_{i,t}^\perp$ is the residual of $CF_{i,t}$ on GCF_t . We then estimate the joint regression $\bar{r}x_{i,t+12} = \alpha_i + \beta_i CF_{i,t}^\perp + \gamma_i GCF_t + \varepsilon$ per country. Table 4.2 reports the results.

We find that the evidence favours a strong but incomplete form of integration. In Table 4.2

Table 4.2
Phase II: global versus local cycle factor

Country	R^2			HAC t		Wald p	p (BH)
	CF	GCF	Joint	$CF^\perp$	GCF		
Belgium	0.301	0.208	0.301	2.39	3.70	0.000	0.000
Germany	0.256	0.284	0.284	0.13	4.29	0.000	0.000
France	0.126	0.259	0.260	-0.22	4.16	0.000	0.000
Italy	0.287	0.074	0.313	3.72	2.05	0.000	0.000
Netherlands	0.344	0.264	0.350	3.25	3.98	0.000	0.000
Switzerland	0.073	0.197	0.198	-0.20	3.04	0.012	0.012
United Kingdom	0.262	0.260	0.284	1.77	3.95	0.000	0.000
Sweden	0.238	0.292	0.295	-0.52	4.53	0.000	0.000
Canada	0.320	0.352	0.359	0.71	4.14	0.000	0.000
Japan	0.260	0.272	0.298	1.54	3.88	0.000	0.000
United States	0.326	0.329	0.335	0.87	4.35	0.000	0.000
G10 panel	0.280	0.244	0.293	5.99	13.81	0.000	—

The dependent variable is the local-currency $\overline{r}x_{i,t+12}$. The CF and GCF columns under R^2 report the single-factor fits of the local regression (3.13) and the global regression (3.15); the Joint column reports the horse race $\overline{r}x_{i,t+12} = \alpha_i + \beta_i CF_{i,t}^\perp + \gamma_i GCF_t + \varepsilon_{i,t+12}$ (3.14), where $CF_{i,t}^\perp$ is the part of the local factor orthogonal to GCF_t , following Dahlquist and Hasseltoft (2013). The HAC t columns report Newey–West t -statistics on the two slopes of the joint regression. Wald p is the p -value of the joint test of no predictability, and p (BH) applies the Benjamini–Hochberg false-discovery-rate adjustment across the eleven markets. A significant GCF together with an insignificant $CF^\perp$ signals that the global factor subsumes the local factor. The final row adds country fixed effects on the pooled G10 panel (Benjamini–Hochberg not applicable).

we can see that the global factor is statistically significant in every market, with HAC t -statistics between 2.1 and 4.5, and that on its own it explains as much of the return variation as the local factor does. The cross-country average R^2 is 25% for the local factor and a nearly identical 25% for the global factor. A single, GDP-weighted international factor thus reproduces the predictability of eleven separately estimated national factors. Once the global factor is included, the orthogonalised local factor is significant in only three of the eleven markets, namely Italy ($t = 3.7$), the Netherlands ($t = 3.3$) and Belgium ($t = 2.4$), and is statistically indistinguishable from zero in the other eight. We can observe this contrast in Figure 4.2. The global bars reach the 1.96 threshold everywhere, whereas the local bars only do so for the three markets named.

In Figure A.5, we can trace the same conclusion through the R^2 ladder. For the eight markets in which the local factor is subsumed, adding $CF^\perp$ to the global factor lifts the R^2 only trivially. For the three markets with surviving local content, the increment is material. Italy rises from 7.4% on the global factor alone to 31.3% jointly, and the Netherlands from 26.4% to 35.0%. The joint Wald test also rejects in all eleven markets, and the rejections survive the Benjamini–Hochberg correction for multiple testing across the panel. This suggests that the combined cycle information is priced everywhere. On the pooled G10 panel with country fixed effects, both factors enter significantly ($t = 6.0$ on $CF^\perp$ and $t = 13.8$ on GCF), with a joint R^2 of 0.29 of which the global factor accounts for the larger part (Table 4.2).

The three exceptions are consistent with our framework. Italy, Belgium and the Netherlands retain idiosyncratic cyclical predictability that the GDP-weighted global factor, which is dominated by the largest economies, does not capture. The broad message, however, is

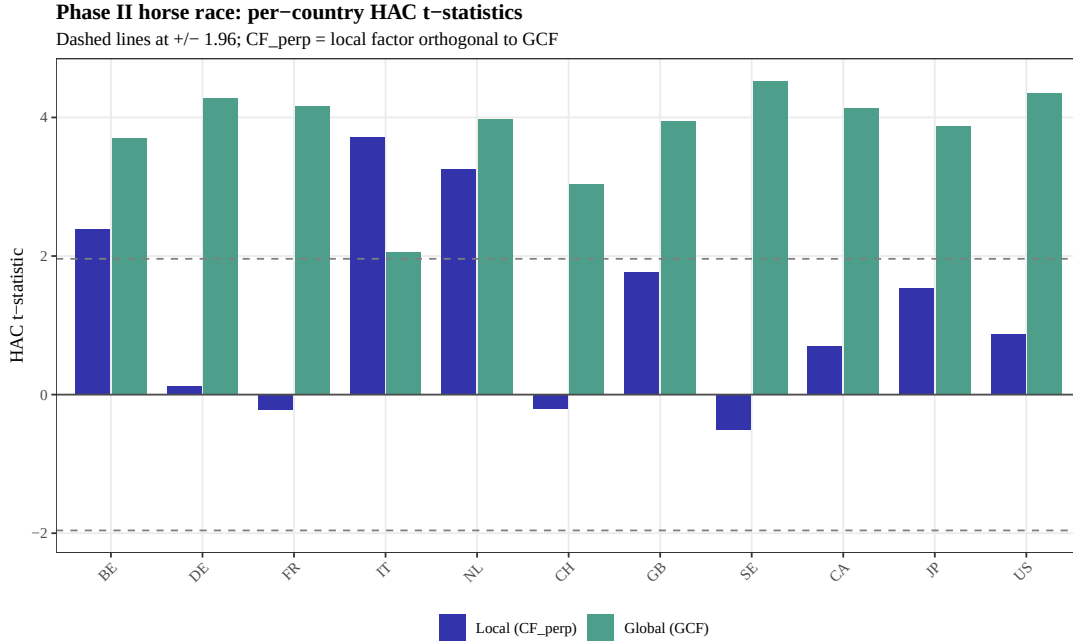


Figure 4.2: Phase II horse race: per-country HAC t -statistics on the orthogonalised local factor $CF^\perp$ and the global factor GCF in the joint regression (3.14). Dashed lines mark ± 1.96 .

that of Dahlquist and Hasseltoft (2013). The predictable component of international bond returns is overwhelmingly global. This answers the second research question. The global cycle factor subsumes the local factor in the majority of the G10 and leaves statistically robust local predictability in only a minority of markets.

4.3 The Global Investor and the FX-Adjusted Factor

In the first two phases, we adopted the perspective of a domestic investor: returns are measured in local currency and currency risk is set aside. For a US investor who holds foreign bonds unhedged, the relevant object is the dollar excess return (3.9), which augments the local return with the realised currency return and nets out the US short rate. Throughout, “dollar” refers to the US dollar. In Phase III, we ask two questions for this investor. First, how far does currency risk erode the global-factor predictability we documented in Phase II? Second, does the FX-adjusted global cycle factor $FXGCF_t$, which we build bottom-up as the GDP-weighted average of the per-country USD cycle factors (Section 3.10), recover what an unadjusted factor loses? Table 4.3 reports the dollar-investor regressions (3.18) and (3.19).

We find that currency risk is a dominant influence on the dollar investor’s returns and that it sharply weakens the global factor. The cross-country average predictive R^2 of the global factor falls from 25% on local-currency returns to 9% on dollar returns, and the factor’s HAC t -statistic drops below the conventional threshold for the entire euro-area bloc, Switzerland and Japan. Predictability survives currency conversion in the United States, whose dollar return mechanically equals its local return because it carries no currency leg ($R^2 = 33\%$), in Canada ($t = 2.8$), and, more marginally, in Sweden and the United Kingdom ($t \approx 2.0$). The currency return is at most weakly related to the bond cycle in every market: its correlation with the local

Table 4.3

Phase III: the US-dollar investor, global versus FX-adjusted factor

Country	R^2 (CF)	GCF_t			$FXGCF_t$			N
		$\hat{\gamma}$	t	R^2	$\hat{\delta}$	t	R^2	
Belgium	0.032	1.76	(1.57)	0.048	1.98	(2.58)	0.130	353
Germany	0.065	1.77	(1.59)	0.050	2.00	(2.62)	0.137	353
France	0.083	1.77	(1.61)	0.050	1.95	(2.57)	0.129	353
Italy	0.109	1.67	(1.51)	0.040	1.57	(2.01)	0.075	353
Netherlands	0.011	1.78	(1.59)	0.050	2.00	(2.62)	0.135	353
Switzerland	0.146	1.55	(1.52)	0.045	2.00	(2.97)	0.161	353
United Kingdom	0.114	1.74	(1.96)	0.071	0.99	(1.37)	0.049	353
Sweden	0.098	2.53	(1.96)	0.074	1.91	(2.13)	0.090	353
Canada	0.114	2.16	(2.83)	0.127	1.53	(2.73)	0.136	353
Japan	0.025	2.20	(1.82)	0.057	1.85	(2.80)	0.115	412
United States	0.326	1.41	(4.36)	0.329	0.94	(5.39)	0.415	412

The dependent variable is the US-dollar excess return of (3.9), duration-standardized and maturity-averaged to $\overline{r}_{i,t+12}^{\text{USD}}$. The first column repeats the R^2 of the dollar return regressed on the local cycle factor $CF_{i,t}$ (3.17) for reference. The GCF block reports the regression on the global factor (3.18) and the $FXGCF$ block the regression on the bottom-up, GDP-weighted FX-adjusted global factor (3.19). Newey–West HAC t -statistics in parentheses. The United States row carries no currency leg, so its dollar return coincides with its local-currency return. In this sample $\text{Corr}(GCF_t, FXGCF_t) = 0.81$, so the FX-adjusted factor carries information that is distinct from the unadjusted global factor.

cycle factor never exceeds 0.34 in absolute value (Table A.5). Currency conversion therefore adds largely independent noise across the board rather than being absorbed by the bond factor, and the signal survives only where that noise is smallest relative to the underlying predictability. We can see this collapse in Figure 4.3, where the dark blue bars for the foreign markets are only a fraction of their local-currency heights. This is the international-bond analogue of the result that long-horizon currency-converted returns are hard to predict. The volatility of the one-year currency return erodes most of the cyclical signal in the bond.

The FX-adjusted factor was designed for this problem, and we find that it helps substantially. Across the panel, it raises the dollar-investor R^2 in ten of the eleven markets and lifts the t -statistic in nine. The cross-country average R^2 recovers from 9% under the global factor to 14% under the FX-adjusted factor, a substantial gain in the dollar-investor fit. Markets that were insignificant under GCF_t also become significant under $FXGCF_t$. The euro-area core moves to $t \approx 2.6$ (Italy to $t = 2.0$), Switzerland to $t = 3.0$, and Japan to $t = 2.8$, so that the factor is significant in ten of eleven markets (the exception is the United Kingdom). In Figure 4.4, we can see why new information can be captured. The FX-adjusted factor correlates 0.81 with the unadjusted global factor, so the two series share the global cycle but the FX-adjusted factor carries distinct information about the currency leg of the dollar return. We compare the constructions in Section 6.5 and take up the economic interpretation in Chapter 7.

Our answer to the third research question is therefore twofold. Currency risk does break the global-factor predictability for an unhedged dollar investor in most markets and cuts the average R^2 from 25% to 9%. The FX-adjusted global cycle factor restores a large part of this predictability in-sample, raising the average fit to 14% and restoring significance in ten of eleven markets. Before we ask whether this in-sample recovery survives in real time, we look more closely at how the two global factors relate to one another over the sample, because this relationship is

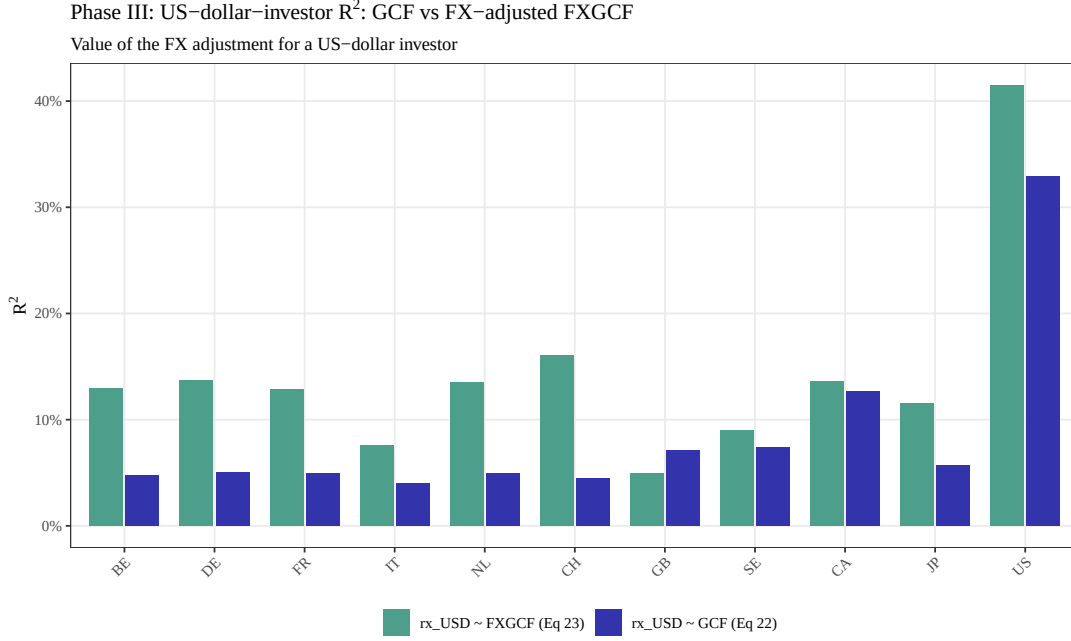


Figure 4.3: Phase III: US-dollar-investor in-sample R^2 by country, the global factor (3.18) versus the FX-adjusted global factor (3.19).

what decides whether the FX adjustment can matter at all.

The global factor and its FX-adjusted counterpart summarise the same cross-section of cycles, measured in two different currencies. The unadjusted factor GCF_t prices the *hedged*, local-currency return, so it is a summary of *interest-rate-cycle risk*, the compensation for holding duration when yields sit far from their macro anchor. The FX-adjusted factor $FXGCF_t$ prices the *unhedged* dollar return, so it carries that same duration signal *plus* the currency leg that a dollar investor cannot avoid. Because the two are built identically except for the currency of the return (3.10)–(3.12), their difference isolates the currency component,

$$FXGCF_t - GCF_t = \sum_{i \in \mathcal{G}} w_{i,t} (FXCF_{i,t} - CF_{i,t}), \quad (4.1)$$

which we call the *wedge*. Reading GCF_t as the interest-rate factor and the wedge as the FX factor lets us ask not just whether the FX adjustment helps on average, but *when* the two factors carry different information, and why.

Table 4.4 collects their properties. The wedge’s standard deviation (0.35) is as large as that of the interest-rate factor itself (0.33) and it is highly persistent (first-order autocorrelation 0.99, half-life of about five years), so the FX component is a slow-moving second factor in its own right. Yet the full-sample correlation between the two factors, 0.81, hides considerable time variation. Figure 4.5 plots the 36-month rolling correlation of GCF_t and $FXGCF_t$. It swings from near zero in the early 1990s and around the global financial crisis to essentially one for most of the post-2008 decade. Split at 2008, the two factors correlate 0.54 in the first half of the sample and 0.95 in the second. In other words, the FX adjustment matters in the earlier sample and barely at all in the later one.

The currency leg carries independent information only to the extent that currency returns

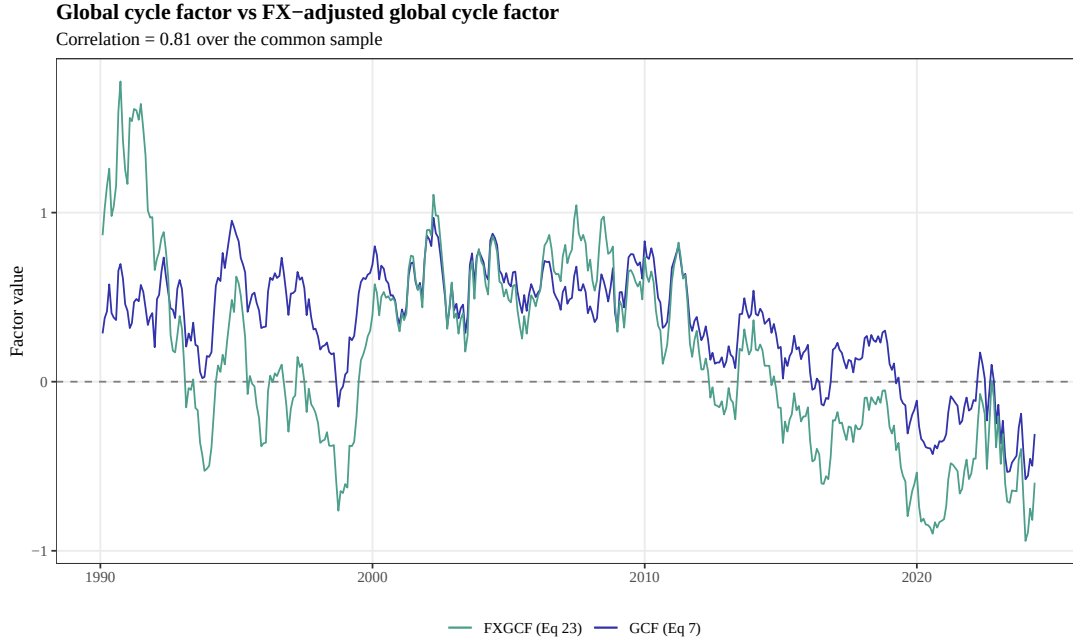


Figure 4.4: The global cycle factor GCF_t (3.7) and the FX-adjusted global cycle factor $FXGCF_t$ (3.12) over the common sample. The two series share the global cycle but are distinct (correlation 0.81).

are themselves predictable, and the classic source of that predictability is the cross-country interest-rate differential through the failure of uncovered interest parity (Fama, 1984). When rates are high and spread apart, differentials are wide, the carry channel is active and the currency component moves on its own. When rates converge to a synchronised zero lower bound, differentials compress and the currency component adds little that the bond cycle does not already capture. Figure 4.6 shows this pattern. The rolling correlation of the two factors falls when the GDP-weighted G10 short rate is high and rises as it approaches zero, with the two moving in opposite directions (correlation -0.42). The regimes in Table 4.4 tell the same story in levels. The low-correlation first half of the sample is the high-rate regime (mean short rate 3.9%, cross-country yield dispersion 1.37), and the near-collinear second half is the low-rate regime (mean short rate 0.9%, dispersion 0.71). The brief decoupling at the right edge of Figure 4.6, as rates rose again after 2022, fits the same logic.

This dynamic reconciles two facts. It explains why our full-sample FX adjustment recovers less than Dahlquist and Hasseltoft (2013) report. Their forward-rate factor retains the rate-level information that drives carry and their sample is weighted towards the high-differential decades. Our detrended cycle factor removes the level and our sample is dominated by the post-2008 low-rate regime, in which the currency component seems nearly redundant. It also anticipates the out-of-sample evidence below. A factor whose extra information is concentrated in one rate regime, and largely absent in the regime that fills the second half of the sample, should help only intermittently in real time.

Table 4.4
Properties and dynamics of the two global cycle factors

<i>Panel A. Univariate properties (full sample, 412 months)</i>					
Factor	Mean	SD	AR(1)	Half-life (months)	SD pre / post 2008
GCF_t	0.34	0.33	0.96	16.3	0.20 / 0.34
$FXGCF_t$	0.16	0.56	0.97	27.0	0.52 / 0.47
Wedge $FXGCF_t - GCF_t$	-0.18	0.35	0.99	59.0	0.44 / 0.18
<i>Panel B. Correlation and the rate regime</i>					
Regime	N	Corr(GCF , $FXGCF$)	Short rate	Dispersion	
Full 1990–2024	412	0.81	2.5%	1.06	
Pre-2008	216	0.54	3.9%	1.37	
Post-2008	196	0.95	0.9%	0.71	

The global cycle factor GCF_t prices the hedged local-currency premium; the FX-adjusted factor $FXGCF_t$ prices the unhedged US-dollar premium; their difference (the “wedge”) isolates the currency component, $FXGCF_t - GCF_t = \sum_i w_{i,t}(FXCF_{i,t} - CF_{i,t})$. Both factors are GDP-weighted aggregates of the per-country USD/local cycle factors under the bottom-up construction (3.12). AR(1) is the monthly first-order autocorrelation and the half-life is $\ln(0.5)/\ln(\text{AR}(1))$. Panel B splits the sample at the global financial crisis; “Short rate” is the mean GDP-weighted G10 one-year yield and “Dispersion” the mean cross-country standard deviation of one-year yields in each regime. The 36-month rolling correlation of the two factors and the rolling short-rate level correlate -0.42 .

4.4 Out-of-Sample Predictability

The evidence so far is in-sample. Because the cycle factors are generated regressors, fitted on the same data they are then scored on, an in-sample R^2 can overstate what a real-time investor could have utilised. In this section, we therefore re-evaluate the three phases out of sample, using the fully-recursive protocol of Section 3.12. The entire chain is rebuilt at each date from data available only up to that date, and every recursive factor forecast is scored by the out-of-sample R^2 (3.20), popularized by Campbell and Thompson (2008). A positive value means that the factor’s real-time forecasts are more accurate than simply extrapolating the historical average, zero means a tie, and a negative value means that the factor does worse than the mean. Out of sample, even small positive values matter economically and negative values are common. Table 4.5 is best read down each column for the sign and the cross-factor ranking, not across the two columns for a change in magnitude. We compare the cycle factor against the forward-rate factor, and stress-test the estimation scheme, in Chapter 6.

Figure 4.7 shows the pooled picture across the four specifications, and we arrive at three main messages, one per phase. First, we find that the local cycle factor does not survive the transition to real time. Its pooled R^2_{os} is -0.07 , and it beats the prevailing mean in only two of the eleven markets (the United States and Canada). Second, the global factor is the exception, and a notable one. Its pooled R^2_{os} is $+0.05$ and it is positive in ten of the eleven markets, as we can see in Figure 4.8. Aggregating the eleven national cycles into one GDP-weighted factor averages away the country-specific estimation noise that causes the local factor to fail, so the global factor predicts out of sample where the local factor cannot. We consider this strong support in our thesis for integration. The global cycle factor is the more efficient in-sample factor, it is the only cycle factor with real-time predictive content. This real-time predictability



Figure 4.5: Dynamics of the two global factors: the 36-month rolling correlation of the interest-rate factor GCF_t (3.7) and the FX-adjusted factor $FXGCF_t$ (3.12). The dashed line marks the full-sample correlation (0.81); the dotted line marks 2008.

is also economically exploitable. In Chapter 5, we turn the global factor into a bond-timing strategy for a currency-hedged investor and show that it raises the Sharpe ratio of a global bond portfolio and delivers a positive certainty-equivalent gain out of sample. Third, for the dollar investor, the out-of-sample picture qualifies the strong in-sample finding of Phase III. The recursive R_{OOS}^2 is negative for the unadjusted global factor (-0.07), which beats the recursive mean in only two markets. The FX-adjusted factor erases this loss. It turns positive at the pooled level ($+0.021$) and beats the mean in six of the eleven markets. The pooled margin is, however, an order of magnitude smaller than the in-sample gain, and, as Chapter 6 shows, it is sensitive to the estimation scheme. The FX adjustment therefore does what it was designed to do, but the real-time edge it delivers for the currency-converted investor is more modest than the in-sample fit suggests.

The recursive factors themselves show us the mechanism behind this divergence. In Figure A.7, we overlay the global factor with its fully-recursive counterpart GCF_t^{OOS} . The recursive series converges to the full-sample factor and tracks it closely, which is the expected behaviour of a stable real-time estimator. The recursive *local* factors converge far less cleanly. Their period-by-period correlation with the full-sample local factor is materially below one for several markets (right panel of Figure A.7), which is why their out-of-sample forecasts deteriorate while the aggregated global factor remains reliable.

For the dollar investor, Figure 4.9 traces the same contrast across the panel. The unadjusted global factor beats the recursive mean in only two markets, whereas the FX-adjusted factor turns positive in six, recovering part of the currency-converted predictability in real time.

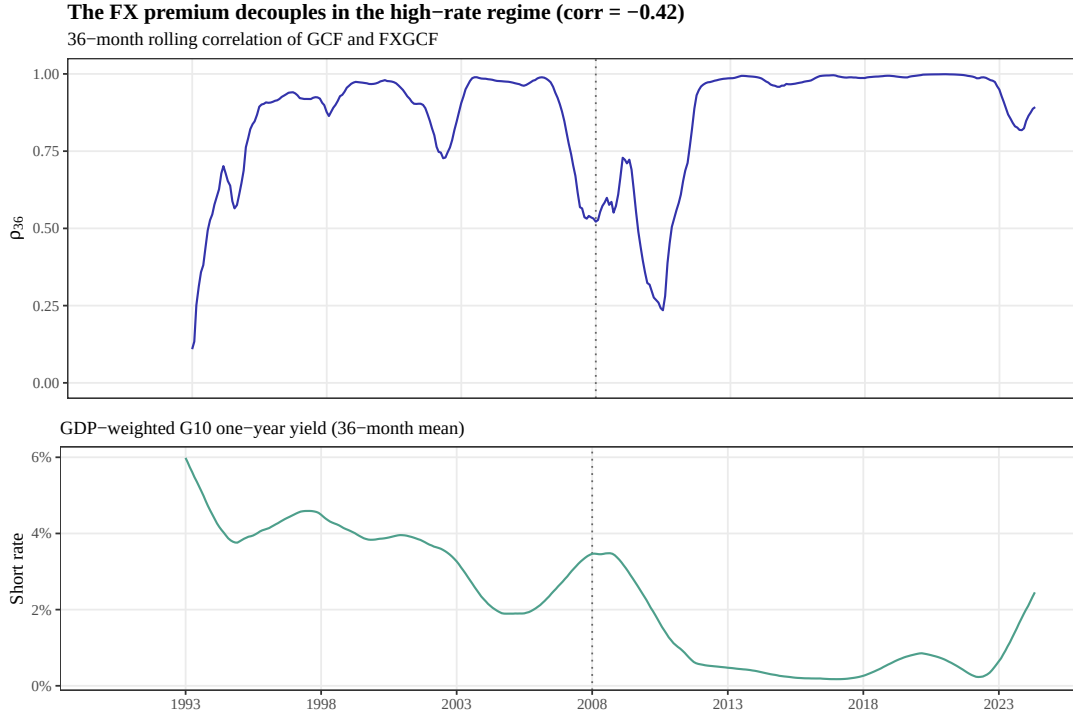


Figure 4.6: The FX premium decouples in the high-rate regime. Top: the 36-month rolling correlation of GCF_t and $FXGCF_t$. Bottom: the GDP-weighted G10 one-year yield (36-month mean). The two series correlate -0.42 : the factors are most distinct when rates are high and differentials wide, and collapse onto each other at the synchronised zero lower bound.

4.5 Summary of Findings

The three phases combine into a consistent account of international bond-return predictability through the lens of the macro-anchored cycle. First, the cycle factor of Cieslak and Povala (2015) generalizes to the G10. It predicts one-year excess returns with the theoretically predicted sign structure in all eleven markets and with an average in-sample R^2 of 25%. Second, we find that this predictability is overwhelmingly a global phenomenon. A single GDP-weighted global cycle factor is significant in every market and matches the explanatory power of the local factors. Once it is included, local predictability survives in only three of the eleven markets. The evidence thus supports strong, though not complete, integration of government-bond risk premia. This is the central message of Dahlquist and Hasseltoft (2013), now established for the cycle factor rather than the forward-rate factor. Third, the global investor's problem is currency risk. Dollar conversion erases most of the global factor's predictive power for the unhedged investor. The bottom-up FX-adjusted factor recovers much of it in-sample, substantially raising the average dollar-investor R^2 from 9% to 14% and restoring significance in ten of eleven markets.

Fourth, these conclusions survive out of sample and, in one respect, even strengthen. The global cycle factor beats the recursive prevailing mean robustly (pooled $R^2_{\text{OOS}} = +0.05$, positive in ten of eleven markets), while the local cycle factor and the *unadjusted* dollar-investor factor do not (both negative, positive in only two markets each). The FX-adjusted factor sits in between. It turns the unadjusted dollar factor's loss into a pooled gain ($+0.021$, positive in six of eleven markets), but this real-time edge is much smaller than its in-sample fit and, as

Table 4.5**Out-of-sample summary: recursive Campbell–Thompson R^2 by phase**

Phase	Specification	In-sample R^2 (mean)	R^2_{oos} (pooled)	Markets OOS+
I — local	$\bar{r}\bar{x} \sim CF$	0.254	−0.073	2/11
II — global	$\bar{r}\bar{x} \sim GCF$	0.254	0.048	10/11
III — USD, global	$\bar{r}\bar{x}^{\text{USD}} \sim GCF$	0.086	−0.070	2/11
III — USD, FX-adj.	$\bar{r}\bar{x}^{\text{USD}} \sim FXGCF$	0.143	0.021	6/11

The in-sample R^2 is the cross-country mean of the single-factor fits (3.13), (3.15), (3.18) and (3.19), i.e. variance explained around the full-sample mean. The out-of-sample R^2_{oos} is the pooled statistic $1 - \text{SSE}_{\text{factor}}/\text{SSE}_{\text{mean}}$ described in (3.20); a positive value means the factor beats the real-time historical average. Pooling sums each country’s squared forecast errors against its own recursive mean before forming the ratio, and “Markets OOS+” counts the markets with a positive R^2_{oos} . Both the factor construction and the forecasting regression respect time- t information, so the forecasts are doubly out of sample. The two R^2 columns use different benchmarks and are not comparable in level. Per-country detail is developed in Chapter 6.

Chapter 6 shows, sensitive to the estimation scheme. Real-time predictability is therefore led by the global cycle factor, with the FX adjustment recovering a real but conditional edge for the currency-converted investor. The contrast between the local factor’s strong in-sample fit and its negative out-of-sample R^2 is a considerable sign that global predictability deserves our attention.

We develop the summary out-of-sample evidence above in full, per country, across subsamples, and against the forward-rate factor, in Chapter 6, and we take up the economic interpretation of all four findings in Chapter 7.

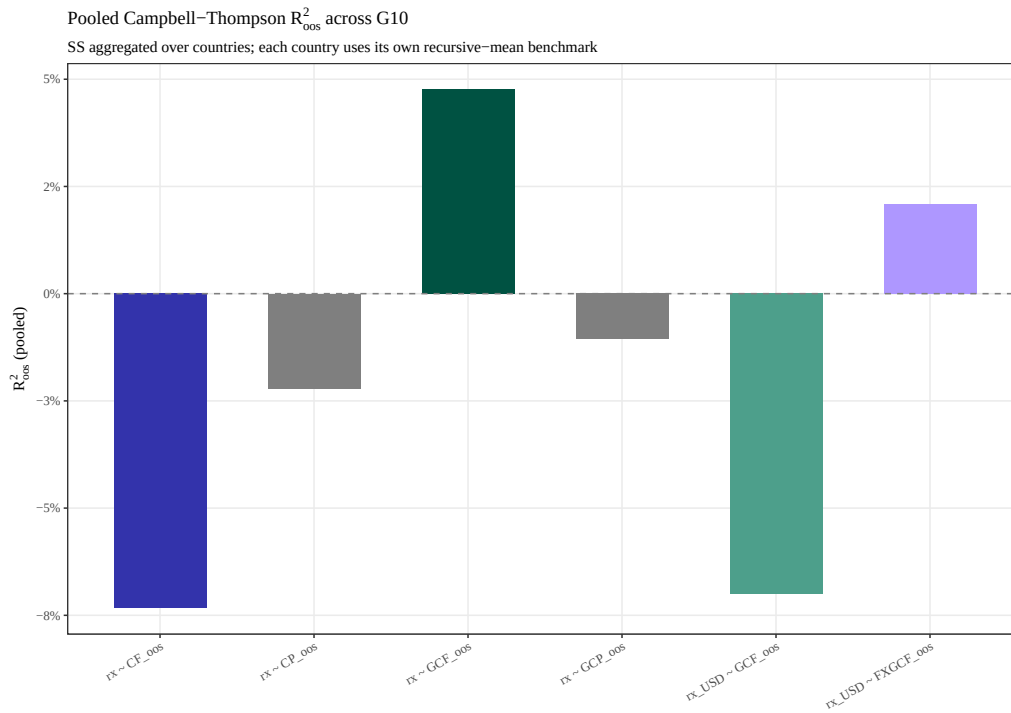


Figure 4.7: Pooled out-of-sample R^2_{oos} (3.20) of the four headline specifications, fully-recursive factor forecast against the recursive prevailing mean. Squared errors are aggregated across the G10, and each country is scored against its own recursive mean. Only the global cycle factor and, marginally, the FX-adjusted factor are above zero.

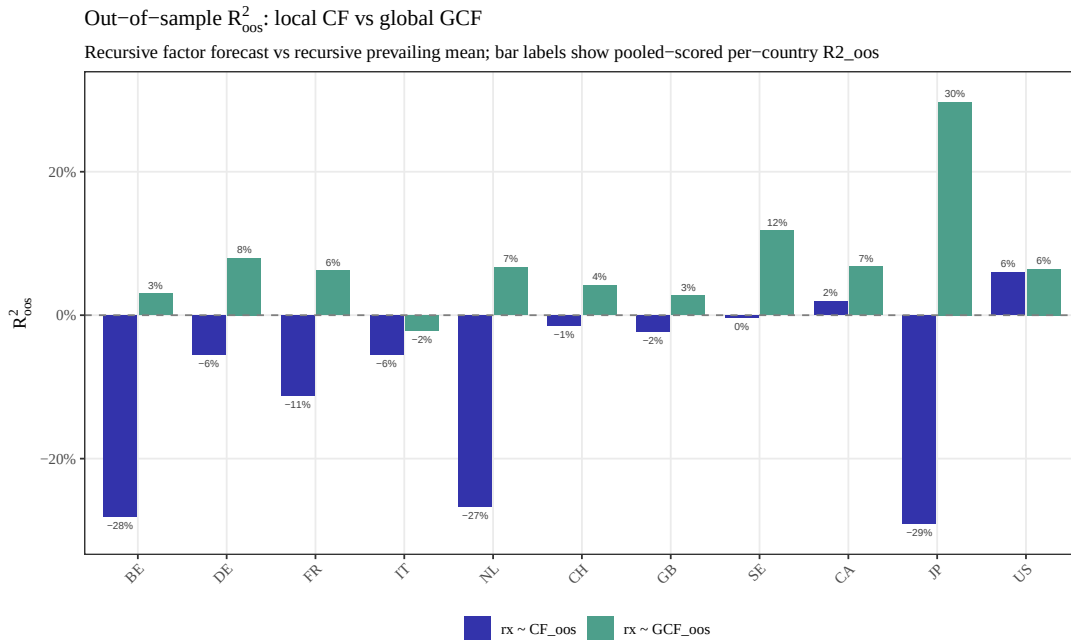


Figure 4.8: Out-of-sample R^2_{oos} by country, the local cycle factor versus the global cycle factor. Positive bars beat the recursive prevailing mean; the global factor is positive in ten of eleven markets, the local factor in only two.

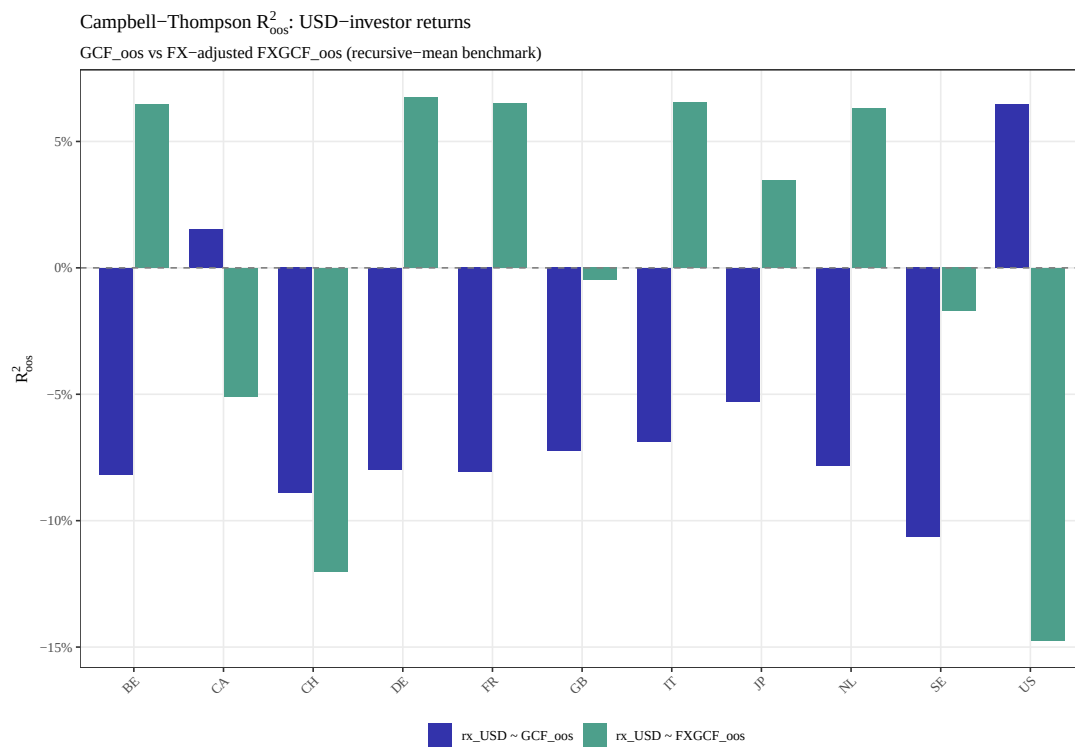


Figure 4.9: Out-of-sample R^2_{oos} by country for the US-dollar investor, unadjusted global factor (3.18) versus the FX-adjusted global factor (3.19).

Chapter 5

Portfolio Construction

In Section 4.4 we established the one statistical result that matters for an investor is the global cycle factor. A positive out-of-sample R^2 is the condition under which a real-time investor can convert predictability into utility (Campbell and Thompson, 2008). In this chapter, we ask whether this conversion is economically meaningful. To answer this question, we build a fully out-of-sample market-timing strategy around the global cycle factor and compare it against a passive benchmark. We keep this exercise self-contained. We use only the recursive factor and forecasts of Section 3.12, entering positions with information known only at time t .

5.1 Strategy Design

Three features of our evidence guide the design of the strategy. First, the predictability that survives out of sample is in *local-currency* returns (Section 4.3). We therefore run the strategy for a *currency-hedged* global investor (GCF) and analyse the unhedged US-dollar investor (FXGCF) separately in Section 5.5. Local-currency excess returns, in which each bond position is funded in the local money market, are the standard representation of an FX-risk-hedged international bond portfolio and are equivalent to managing the currency exposure with FX forwards (Randl et al., 2025). Second, the global cycle factor is a single time series, so it carries no information about which country to favour and can only time aggregate bond exposure. The investable asset is accordingly a single GDP-weighted, ten-year global government-bond portfolio, whose one-year excess return is the weighted average of the national returns. Third, the signal must be usable in real time, so both the factor and the forecasting regression respect time t .

Each month, the investor forms a real-time forecast of the portfolio's twelve-month excess return from the recursive regression $rx_{t+12} = a + b \text{GCF}_t^{\text{oos}} + \varepsilon$, and sets a mean-variance exposure

$$w_t = \frac{1}{\gamma} \frac{\widehat{\mathbb{E}}_t[rx_{t+12}]}{\widehat{\sigma}_t^2}, \quad (5.1)$$

where $\widehat{\sigma}_t^2$ is the recursive return variance and γ the risk aversion. As benchmarks, we use the same investor with the recursive prevailing-mean forecast, and a passive buy-and-hold position. The Sharpe ratio is invariant to a constant rescaling of w_t , so we report it directly. The certainty-equivalent (CER) return is not scale-invariant, so we report it at equal average exposure across strategies. This makes the variance penalty comparable.

Table 5.1**Economic value of the global cycle factor: timing a 10Y global bond portfolio**

Strategy	Mean (%)	Vol (%)	Sharpe	CER, $\gamma = 5$ (%)	CER, $\gamma = 10$ (%)
<i>GCF</i> -timed	3.10	7.69	0.40	1.62	0.15
Recursive-mean timing	2.47	6.77	0.36	1.32	0.17
Buy-and-hold	2.27	6.74	0.34	1.14	0.00

This table reports the real-time performance of timing a 10Y GDP-weighted global bond portfolio with the global cycle factor over December 2005–April 2024 ($n = 221$ months; the portfolio’s out-of-sample R^2 against the recursive mean is 0.12). Exposure follows the mean–variance rule (5.1), $w_t \propto \hat{\mathbb{E}}_t[rx_{t+12}]/\hat{\sigma}_t^2$, with the forecast taken from the recursive GCF^{oos} regression, so the strategy is doubly out-of-sample. The Sharpe ratio is scale-invariant; mean, volatility, and the certainty-equivalent (CER) return are reported at equal average exposure across strategies so that the variance penalty is comparable. CER is the annual certainty equivalent of a mean–variance investor with risk aversion γ . Returns are local-currency (currency-hedged) excess returns based on overlapping twelve-month holding periods.

5.2 Implementation

In this section, we cover the implementation. Each month t , using only information available at t : (i) we re-estimate the full generated-regressor chain on outcomes realised by t (a twelve-month lag), with a sixty-month minimum training history (Section 3.12). This regards trend inflation (3.3), the yield-cycle decomposition (3.2), the local-factor regressions (3.5), and the GDP aggregation (3.7). (ii) We form the forecast $\hat{\mathbb{E}}_t[rx_{t+12}] = \hat{a}_t + \hat{b}_t GCF_t^{\text{oos}}$ from the recursive regression, (iii) we estimate the variance $\hat{\sigma}_t^2$ on an expanding window, (iv) we set the target weight to the mean–variance exposure (5.1), truncated at zero (long-only) and rescaled to unit average exposure for reporting, and (v) we rebalance monthly, with one-way turnover $|w_t - w_{t-1}|$.

Anything the strategy earns is therefore attributable to the real-time information content of the global cycle factor.

5.3 Performance

In Table 5.1, we report the out-of-sample performance over 2005–2024, the period for which the fully-recursive factor is available. The global bond portfolio itself has an out-of-sample R^2 of 0.127 against the recursive mean. We find that this statistical edge translates into economic value. Timing the portfolio with the global cycle factor raises the annual Sharpe ratio from 0.34 for the passive position to 0.40, and from 0.36 for an investor who times on the recursive mean. For a mean–variance investor with risk aversion $\gamma = 5$, the certainty-equivalent gain is 0.48% per year over buy-and-hold and 0.31% over the mean-timing benchmark. At $\gamma = 10$, however, the more risk-averse investor slightly prefers the mean-timing benchmark (0.17% against 0.15%), as the sharper variance penalty offsets the timing gain.

In Figure 5.1, we can see the cumulative wealth of the timed and passive portfolios on a monthly, one-month-holding basis.¹ The timing strategy ends the period ahead, and most of its outperformance accrues from avoiding the worst of the bond-market drawdowns rather than from amplifying the good years. Figure 5.2 shows the mechanism. The global cycle factor cuts bond

¹The wealth and drawdown figures rebalance monthly and hold for one month, using a one-month excess return on the 10Y global bond portfolio.

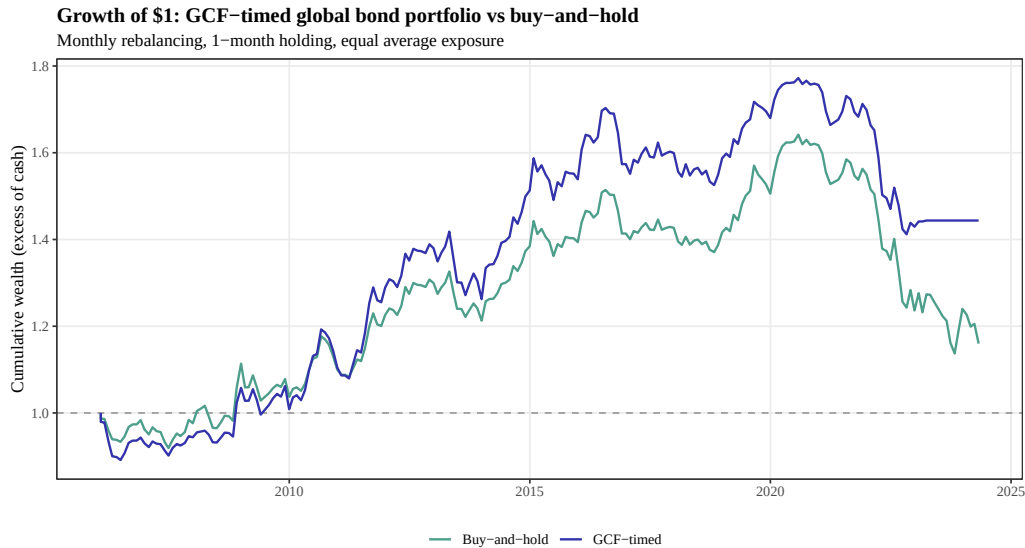


Figure 5.1: Cumulative wealth (excess of cash) of the GCF-timed global bond portfolio versus buy-and-hold, monthly rebalancing with a one-month holding period at equal average exposure.

exposure when it signals a low risk premium and restores it when the signal turns favourable, whereas the mean-timing investor holds a nearly constant position.

The drawdown profile confirms that the value of the signal is defensive. In Figure 5.3, we plot the drawdown from peak of the same monthly wealth curves. The maximum drawdown of the timed portfolio is 20.3%, against 24.9% for the mean-timing investor and 30.7% for buy-and-hold. The deepest drawdowns fall in the 2022–2023 global hiking cycle, the worst bond-market episode in the sample. The timed investor enters it with reduced exposure because the global cycle factor had already turned (Figure 5.2). This is the behaviour the mechanism predicts. The factor forecasts low premia when yields sit far below their inflation anchor, as they did in 2021. We quantify where the strategy trades, and what this trading costs, in Section 5.4.

5.4 Subperiods, Turnover, and Transaction Costs

A full-sample Sharpe ratio does not show us in which periods a strategy actually generates its performance. In Table 5.2, we therefore split the evaluation window in half and additionally isolate the 2021–2022 hiking cycle. The split separates two very different regimes. In the strong bond decade of 2005–2014, timing did not pay off on a risk-adjusted basis. The timed portfolio earned a full percentage point a year more than buy-and-hold (6.5% against 5.3%) but took proportionally more risk, so its Sharpe ratio was lower (0.93 against 1.09). But the ranking reverses where an investor most needs it to. Over 2015–2024, a decade in which the passive portfolio lost money, the timed strategy preserves a higher mean and the highest Sharpe ratio, and in the 2021–2022 drawdown it has the least negative Sharpe ratio. The full-sample advantage of Table 5.1 is therefore not a uniform edge but a defensive one. It is concentrated in the regime of falling and volatile bond prices. We can observe this characterisation in Figure 5.4. In the rolling five-year Sharpe ratios, the passive portfolio holds the upper hand in the early, high-premium windows, the two series then track each other through the middle of the sample, and the timed

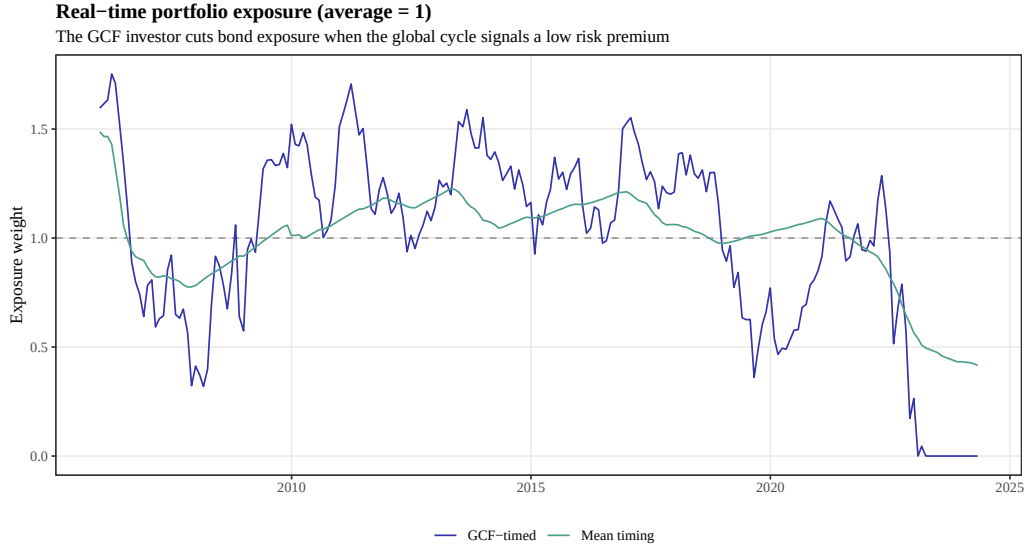


Figure 5.2: Real-time portfolio exposure (average normalised to one). The GCF-timed investor varies bond exposure with the global cycle signal.

portfolio pulls ahead only as the hiking cycle unfolds. Across all windows, the timed portfolio is not a clear winner. The full-sample edge is therefore real, but it comes from a handful of episodes rather than steady accrual.

The trading intensity of the strategy is modest. Because the global cycle factor is highly persistent, the target weight moves slowly. The average one-way turnover of the timed portfolio is 9.1% of notional per month (mean-timing: 1.3%), which corresponds to roughly one full portfolio turn per year. In Table 5.3, we convert this into net-of-cost performance by charging a proportional one-way cost on every unit of notional traded. At 10 basis points, which is already conservative for ten-year G10 government bond futures, where quoted half-spreads on the underlying market are of the order of one basis point (Fleming, 2003), the Sharpe ratio of the timed strategy falls from 0.41 to 0.39 and the certainty-equivalent gain over buy-and-hold shrinks by about eleven basis points a year. Even at punitive 50 basis points, the timed strategy still matches or beats buy-and-hold on both metrics, although at this cost level the near-static mean-timing rule becomes the better implementation. Transaction costs therefore qualify the magnitude of the economic value but do not overturn its existence.

5.5 The US-Dollar Investor and the FX-Adjusted Factor

A US-based investor who holds the global bond portfolio unhedged earns the dollar excess return (3.9), which adds the currency return to the local bond return. The signal for this investor is the FX-adjusted global cycle factor $FXGCF_t$, which we built in Section 3.10 to target dollar returns directly. In Table 5.4, we run the timing rule of Section 5.1 on the dollar portfolio, using the recursive $FXGCF^{oos}$ signal. The FX-adjusted signal shares the five-year recursive burn-in of the global factor, so we evaluate the dollar strategy over the same out-of-sample window (2005–2024) as the hedged strategy of Section 5.3.

The comparison tells us two things. Over this sample the unhedged dollar bond premium is



Figure 5.3: Drawdown from peak of the GCF-timed and buy-and-hold global bond portfolios, computed on the monthly one-month-holding wealth curves of Figure 5.1 at equal average exposure.

modest but positive. The passive portfolio earns about 1.4% a year in excess return for a Sharpe ratio near 0.18, as the strength of the dollar through the 2010s offsets much, though not all, of the foreign-bond return for a US investor. The two dollar signals also diverge at the aggregate level. Timing on the unadjusted recursive global factor works: the GCF-timed dollar strategy earns a Sharpe ratio of 0.24 against the passive 0.18, with an aggregate-portfolio out-of-sample R^2 of 0.078. The FX-adjusted factor, in contrast, does not time the *aggregate* dollar portfolio well. Its forecast of the GDP-weighted portfolio return has an out-of-sample R^2 of -0.01 , and the FXGCF-timed strategy earns a Sharpe ratio of 0.18, no better than buy-and-hold.

This does not contradict Phase III. It reflects where the FX-adjusted factor's strength lies. Its in-sample recovery (Section 4.3) and its pooled out-of-sample edge (Chapter 6) come from the per-country USD regressions, that is, from the cross-section of dollar returns. We therefore reach the same conclusion as elsewhere in our thesis. Timing adds value for the dollar investor as well, but currency risk, and not a failure of the cycle signal, caps the level it can reach. Both dollar strategies remain well below the 0.40 of the currency-hedged investor in Section 5.3. The economic value of the global cycle factor accrues overwhelmingly to the investor who hedges the currency exposure of the international bond position.

5.6 Caveats

We should read this result as a proof of concept rather than a tradable system, and note two limitations. First, the magnitude is modest. The portfolio shows a Sharpe-ratio improvement of about 0.06 and a certainty-equivalent gain below one percent a year. Its edge is time-sensitive, earned in the poor bond decade and the 2021–2022 drawdown rather than in the calm years. Second, the forecasts use overlapping twelve-month returns, so the supporting precision is overstated and the subperiod statistics are descriptive. Transaction costs (Table 5.3) erode but

Table 5.2**Subperiod performance of the GCF-timed global bond portfolio**

Strategy	Mean (%)	Vol (%)	Sharpe
<i>Panel A: 2005–2014</i>			
<i>GCF</i> -timed	6.45	6.97	0.93
Recursive-mean timing	5.44	5.35	1.02
Buy-and-hold	5.33	4.90	1.09
<i>Panel B: 2015–2024</i>			
<i>GCF</i> -timed	−0.16	6.94	−0.02
Recursive-mean timing	−0.43	6.78	−0.06
Buy-and-hold	−0.70	6.97	−0.10
<i>Panel C: 2021–2022 (hiking cycle)</i>			
<i>GCF</i> -timed	−9.06	5.66	−1.60
Recursive-mean timing	−8.93	5.43	−1.64
Buy-and-hold	−9.59	5.44	−1.76

This table splits the out-of-sample window of Table 5.1 into subperiods. Panels A and B are the two halves of the evaluation window; Panel C isolates the 2021–2022 global hiking cycle, the worst bond-market episode in the sample. Strategies, exposure rule, and equal-average-exposure reporting are identical to Table 5.1; returns are local-currency (currency-hedged) excess returns based on overlapping twelve-month holding periods.

do not eliminate the advantage. With these caveats, the predictability of Chapter 4 carries genuine, if moderate, economic value for a currency-hedged global bond investor. We take up the broader interpretation in Chapter 7.

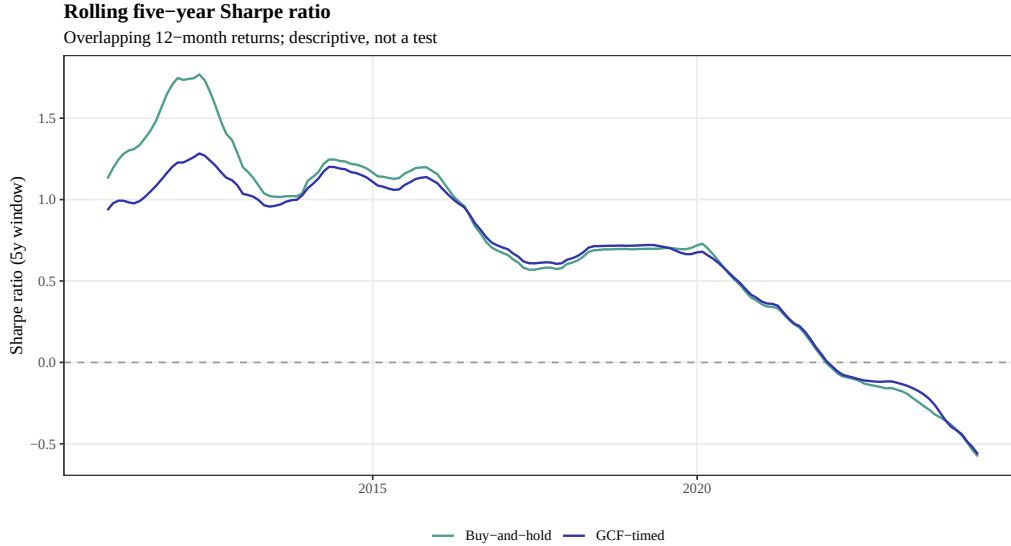


Figure 5.4: Rolling five-year Sharpe ratio of the GCF-timed and buy-and-hold portfolios (overlapping twelve-month returns; descriptive).

Table 5.3
Turnover and net-of-cost performance

Strategy	Turnover (%/month)	One-way cost (bp)			
		0	10	25	50
<i>Panel A: Sharpe ratio</i>					
GCF-timed	9.1	0.41	0.39	0.37	0.34
Recursive-mean timing	1.3	0.37	0.37	0.36	0.36
Buy-and-hold	–	0.34	0.34	0.34	0.34
<i>Panel B: CER, $\gamma = 5$ (%)</i>					
GCF-timed	9.1	1.66	1.55	1.39	1.11
Recursive-mean timing	1.3	1.35	1.33	1.31	1.27
Buy-and-hold	–	1.16	1.16	1.16	1.16

This table reports the strategies of Table 5.1 net of proportional transaction costs. Turnover is the average one-way monthly turnover $|w_t - w_{t-1}|$ of the equal-average-exposure weight path, in percent of the portfolio notional. The annualised cost drag $c \cdot 12 \cdot |w_t - w_{t-1}|$ is subtracted from each twelve-month return for a one-way cost of c basis points on the notional traded. Buy-and-hold is assumed costless after the initial purchase. The first month of the evaluation window is dropped (turnover undefined), so the zero-cost column differs marginally from Table 5.1.

Table 5.4
The US-dollar investor: timing dollar returns with the FX-adjusted factor

Strategy	Mean (%)	Vol (%)	Sharpe	OOS R^2
<i>FXGCF</i> -timed	1.46	7.93	0.18	−0.007
GCF-timed	1.93	7.97	0.24	0.078
Recursive-mean timing	1.55	7.86	0.20	–
Buy-and-hold	1.43	7.79	0.18	–

This table reports the performance of timing the unhedged US-dollar excess return (3.9) of the 10Y global bond portfolio over December 2005–April 2024 ($n = 221$ months). The *FXGCF*-timed and *GCF*-timed strategies use the recursive FX-adjusted and unadjusted global factors, $FXGCF^{OOS}$ and GCF^{OOS} , with the same mean–variance rule (5.1) and equal-average-exposure reporting as the hedged strategy of Table 5.1. OOS R^2 is the Campbell and Thompson (2008) statistic of the corresponding factor forecast.

Chapter 6

Robustness

The headline results of Chapter 4 rest on two simplifications that we will test with more scrutiny. First, we estimate and evaluate the predictive regressions in-sample, on the same data that we use to construct the generated cycle regressors, which can overstate the apparent fit. Second, we use the full estimation window and therefore average over three decades that include two severe episodes in the history of government-bond markets, the 2007–2009 global financial crisis and the 2010–2012 euro-area sovereign-debt crisis. The out-of-sample evidence is developed in Section 4.4. Here we test its robustness to the estimation scheme, the sample period, the predictor family, the inflation measure, and the construction choices. In Section 6.1, we vary the estimation scheme, contrasting expanding and rolling windows and the length of the training minimum. In Section 6.2, we re-estimate every headline specification over economically defined subsamples, with the two crises isolated as windows of their own, in-sample and out-of-sample. In Section 6.3, we ask whether our conclusions are specific to the macro-anchored cycle factor or hold equally for the forward-rate factor of Cochrane and Piazzesi (2005) and Dahlquist and Hasseltoft (2013). In Section 6.4, we rebuild the entire chain on regular all-items CPI in place of the core CPI, mirroring the inflation-series robustness check of Cieslak and Povala (2015). In Section 6.5, we close with the sensitivity of the results to our construction choices and state which findings are fragile and which survive. Throughout the chapter, we arrive at the same organising result as in Chapter 4. The GDP-weighted global cycle factor GCF_t is far more robust than any of its competitors.

6.1 Expanding versus Rolling Estimation Windows

Our out-of-sample test embeds two estimation choices. It uses expanding windows, meaning that the recursive regression at each date t is fitted on all history up to t , and it requires a five-year minimum training span before the first real-time forecast is produced. Both choices are conventional, but a reader is entitled to ask whether the global factor’s advantage is an artefact of our window choice. To answer this question, we rebuild the recursive chain, that is the yield–trend decomposition, the predictive regression and the prevailing-mean benchmark, under five estimation schemes. Expanding windows with a three-, five-, and seven-year minimum, and rolling windows of ten and fifteen years. Each scheme scores its factor against the prevailing mean constructed under the same scheme, so the comparison is internally consistent. Table 6.1

and Figure 6.1 report the pooled out-of-sample R^2 . The expanding/five-year column reproduces the baseline of Table 6.2 exactly, which serves as a check on the rebuilt machinery.

The length of the training minimum matters little for the global cycle factor and much more for the FX-adjusted one. The global factor delivers essentially the same out-of-sample R^2_{oos} under a three-, five-, or seven-year minimum, namely +0.05. It is positive in ten of eleven markets in every case. For the headline result, the sixty-month choice is therefore representative rather than a tuned parameter. A seven-year minimum also lifts the local cycle factor to approximately zero (−0.002) and the unadjusted dollar-investor factor from −0.07 to −0.01, because more training history stabilises each recursive coefficient before it is used. The FX-adjusted factor, however, is the most sensitive of the four to this choice. Its pooled gain of +0.021 holds under the baseline five-year minimum but turns negative under both a three-year (−0.12) and a seven-year (−0.10) minimum. Its eleven per-country USD regressions have less stable recursive coefficients than the single global aggregate, and shifting the training minimum shifts which early forecasts enter the evaluation window. We read this as a caution. The FX-adjusted factor’s real-time edge exists at the baseline but is not a robust feature of every reasonable scheme.

The window shape, by contrast, matters. The out-of-sample advantage of every specification comes from the expanding window choice. The global cycle factor’s R^2_{oos} falls from +0.05 to −0.10 under a ten-year rolling window and to −0.04 under a fifteen-year one, and it is positive in zero of eleven markets in both cases. The FX-adjusted factor is negative under the ten-year rolling window (−0.07) and approximately flat under the fifteen-year one (+0.01). However, we see this result as economically coherent rather than troubling. The cycle factors are anchored to slowly-moving trend inflation and their predictive loadings are stable through time (Section 6.2), so accumulating the full history sharpens those factors. A rolling window, in contrast, throws away informative old data and re-estimates on a shorter, noisier span, while the rolling prevailing mean it is judged against becomes more adaptive and harder to beat. The factors’ real-time advantage is therefore a property of expanding-window estimation, which benefits from a long, accumulated macro history, and it is not a generic feature of any real-time scheme. Across the five schemes the rankings are largely stable. Under all three expanding-window minimums the global cycle factor is the strongest specification and is positive, while the local and unadjusted dollar-investor factors are negative throughout. The FX-adjusted factor is the least stable of the four. It is positive only under the baseline five-year expanding minimum and, marginally, the fifteen-year rolling window, so we read its real-time edge as conditional on the estimation scheme. Under the rolling windows no specification retains a reliable edge.

6.2 Subsample Stability

The full-sample estimates could in principle be driven by a single regime. To rule this out, we re-estimate every headline specification over four non-overlapping subsamples, dated by the forecast-origin month. A pre-crisis window through June 2007, the global financial crisis (July 2007–December 2009), the euro-area sovereign-debt crisis (2010–2012), and a post-crisis window from 2013 onward. The two crisis windows are short, with thirty and thirty-six forecast origins respectively, so the t -statistics they produce are by construction noisier than the full-sample figures, and the in-sample R^2 in short, volatile windows is mechanically higher. The point of the

Table 6.1

Out-of-sample R^2_{os} under alternative estimation schemes

Specification	Expanding window			Rolling window	
	5-year min	3-year min	7-year min	10-year	15-year
$\overline{rx}_{i,t+12} \sim CF_{i,t}$	−0.073	−0.120	−0.002	−0.219	−0.119
$\overline{rx}_{i,t+12} \sim GCF_t$	0.048	0.051	0.048	−0.103	−0.038
$\overline{rx}_{i,t+12}^{\text{USD}} \sim GCF_t$	−0.070	−0.097	−0.009	−0.043	−0.061
$\overline{rx}_{i,t+12}^{\text{USD}} \sim FXGCF_t$	0.021	−0.124	−0.102	−0.073	0.013

Pooled R^2_{os} (3.20) of the fully-recursive factor forecast against the recursive prevailing mean, with the entire factor chain—the cycle decomposition, the predictive regression, and the benchmark—rebuilt under each estimation scheme. Expanding windows use all history up to t with a three-, five-, or seven-year training minimum; rolling windows use only the most recent ten or fifteen years. Each scheme scores its factor against the prevailing mean built under the same scheme, so columns are internally consistent; the expanding/five-year column reproduces the baseline of Table 6.2. The global cycle factor’s edge is invariant to the training-minimum length but is specific to the expanding window.

exercise is therefore the pattern across windows, not the precise level in any one of them.

In Table A.6, we report the in-sample results. The local-currency cycle predictability is present in every subsample and never collapses. Strikingly, it is strongest in the euro-area crisis, where the local factor reaches a mean R^2 of 0.50 (pooled $t = 6.7$, significant in nine of eleven markets) and the global factor 0.39 ($t = 2.4$). Bond risk premia are, thus, largest and most cyclical when sovereign stress is at its peak, which is when the cycle factor has the most to explain. The weakest window is the financial crisis itself, with a mean R^2 of 0.17 for the local and 0.05 for the global factor and pooled t -statistics near 1.7. In this window, the flight-to-quality fall in yields swamps the slow-moving cyclical signal. The post-crisis window is the most revealing for our integration thesis. There, the *global* factor strengthens to a mean R^2 of 0.24 ($t = 11.5$, significant in ten of eleven markets) while the local factor weakens to 0.13, which is consistent with the common factor becoming the dominant summary of bond risk premia as markets integrate further.

The dollar-investor story is more regime-dependent, and the subsample split reveals a vulnerability that the full sample hides. In the pre-crisis window, the dollar-investor predictability was in fact strong. The FX-adjusted factor delivered a mean R^2 of 0.29, significant in nine of eleven markets, and the unadjusted global factor 0.12 in eleven of eleven. However, this predictability erodes through both crises and disappears after 2013, where the mean R^2 falls to 0.04 for both dollar factors and the pooled t -statistics turn negative (−2.1 for FXGCF, −1.6 for GCF). The strong full-sample dollar-investor fit that we document in Section 4.3 is therefore largely a pre-2008 phenomenon. Once the post-crisis currency regime of near-zero rate differentials and synchronised monetary policy sets in (Section 4.3), the cyclical signal in dollar returns disappears. This is most likely a limitation of the FX-adjusted factor rather than a firm conclusion, and the subsample evidence states it more sharply than the pooled regression can.

The out-of-sample subsample evidence (Table 6.2, Figure 6.2) is the more demanding test, and the global cycle factor passes it almost everywhere. Its recursive R^2_{os} is positive in the pre-crisis window (+0.09), the euro-area crisis (+0.01), and the post-crisis window (+0.09), and it only turns negative in the financial crisis (−0.19), a regime in which we would not expect a

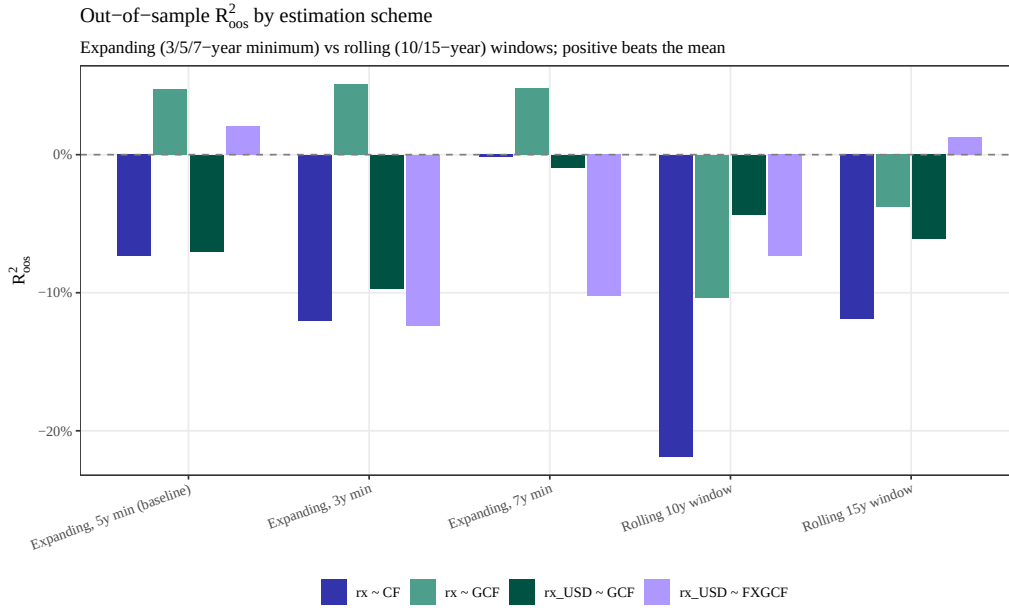


Figure 6.1: Out-of-sample R^2_{oos} by estimation scheme and specification. The global cycle factor (red) is above zero under all three expanding-window minimums but not under either rolling window. The baseline is the expanding-window, five-year-minimum scheme.

slow-moving macro factor to forecast the panic-driven repricing of 2008. The local cycle factor, by contrast, fails out of sample in every window, most severely in the financial crisis (-0.40). The two dollar-investor factors split the sample between them. The unadjusted global factor is positive only in the pre-crisis window ($+0.13$) and negative through both crises and after. The FX-adjusted factor is close to the opposite. It is the only specification that stays positive through the financial crisis ($+0.08$), when the flight into the dollar moved currency and bond returns together and the per-country USD regressions had something genuine to forecast, and it remains approximately flat in the euro-area crisis and the post-crisis window ($+0.002$ and $+0.006$). It is mildly negative pre-crisis (-0.02), where its forecasts rest on the shortest training histories. Its small full-sample advantage is therefore earned mainly in the 2008 repricing rather than built up evenly across regimes. For the headline factor, the message is unchanged. The global cycle factor’s real-time advantage does not come from a single benign regime. It is present before, between, and after the two crises, and it breaks down only in the financial crisis of 2008.

6.3 Cycle Factor versus the Cochrane–Piazzesi Factor

A concern is that our results reflect a generic “global factor” effect rather than anything specific to the macro-anchored cycle. The sharpest test we can perform is to compare the cycle factor directly with the return-forecasting factor of the literature, the forward-rate factor of Cochrane and Piazzesi (2005), aggregated to a global forward factor in the manner of Dahlquist and Hasseltoft (2013). We construct the local forward factor $CP_{i,t}$ as the fitted value of the one-year-ahead excess return on the available forward menu, and its GDP-weighted global counterpart GCP_t exactly as GCF_t is built from $CF_{i,t}$, so that the two factor families differ only in their predictor set: macro-anchored cycles versus forward rates.

Table 6.2**Out-of-sample R_{os}^2 across subsamples**

Subsample	$\bar{r}\bar{x} \sim CF$	$\bar{r}\bar{x} \sim GCF$	$\bar{r}\bar{x}^{\text{USD}} \sim GCF$	$\bar{r}\bar{x}^{\text{USD}} \sim FXGCF$
Full sample	−0.073	0.048	−0.070	0.021
Pre-crisis (−2007:06)	−0.228	0.087	0.125	−0.024
Financial crisis (2007–09)	−0.395	−0.187	−0.177	0.077
Euro crisis (2010–12)	0.050	0.014	−0.079	0.002
Post-crisis (2013–)	−0.085	0.091	−0.039	0.006

Each cell reports the pooled out-of-sample statistic R_{os}^2 (3.20) of the fully-recursive factor forecast against the recursive prevailing mean, scored over the country-months whose forecast origin lies in the indicated window, with squared errors aggregated across the G10 and each country scored against its own recursive mean. A positive value means the factor beats the real-time historical average. Both the factor and the predictive regression respect time t , so the exercise is doubly out-of-sample. All four factors share a five-year training minimum, so every window carries real-time forecasts. Specifications and subsample definitions as in Table A.6.

The two local factors carry overlapping but distinct information. They correlate 0.69 on average across the G10 (right panel of Figure A.3), and the two global factors correlate 0.74. In sample, the forward factor actually performs better. Its cross-country mean R^2 is 0.38 for the local and 0.35 for the global version, against 0.25 for both cycle factors (Figure A.3). At face value, this is not in the cycle factor’s favour. However, because the forward factor uses six predictors compared to the cycle factor’s two and therefore has far more scope to over-fit a fixed sample.

Out of sample, however, the verdict reverses (Figure 6.3). The global cycle factor is the only one of the four that beats the recursive prevailing mean, with a pooled $R_{\text{os}}^2 = +0.05$ and positive forecasts in ten of eleven markets. The global forward factor, despite its higher in-sample fit, fails out of sample (−0.01, positive in only two markets), as do the local forward factor (−0.02, four markets) and the local cycle factor (−0.07, two markets). The forward factor’s in-sample advantage is the over-fit that the out-of-sample test is designed to expose, and it does not survive the transition to real time. The macro anchoring of the cycle factor, which ties the predictable component of returns to a slowly-moving trend-inflation object rather than to the fast-moving forward curve, is therefore no accident, and it underpins the global factor’s real-time stability. The integration finding of Chapter 4 is stronger for the cycle factor than for the forward factor. Dahlquist and Hasseltoft (2013) established global integration in-sample for four markets with the forward-rate factor, whereas we establish it out-of-sample across the G10 with the macro-anchored cycle.

6.4 Core versus Headline Inflation

The cycle factor is the residual of yields on trend inflation, and we build trend inflation throughout from core CPI (excluding food and energy), following Cieslak and Povala (2015), who argue that core is the better proxy for the persistent component of inflation that bond investors price. Cieslak and Povala (2015) themselves check the alternative in their Table 10, Panel C4, where they rebuild the trend from all-items CPI and report that it delivers a predictive R^2 about nine percent lower on average, with the divergence dating back to the energy-price volatility of the

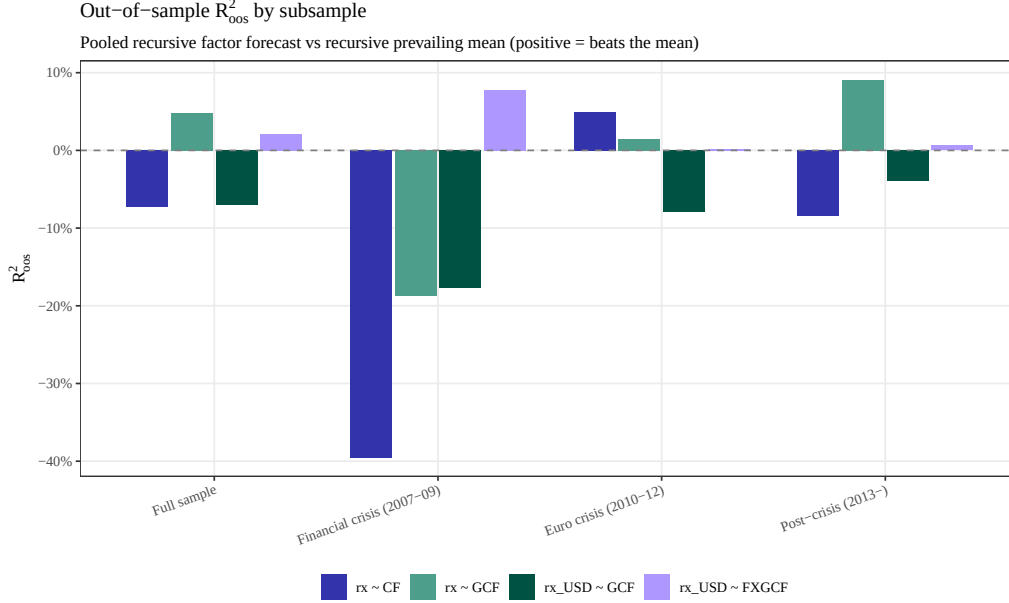


Figure 6.2: Out-of-sample R^2_{oos} by subsample and specification (the short pre-crisis window is omitted; Table 6.2 reports its values). The global cycle factor is above zero in every window except the financial crisis itself.

2000s. In this section, we repeat this test for the G10. We rebuild the entire chain, that is trend inflation, the cycle, and the three factors, on *regular* (all-items, headline) CPI and compare it with the core-CPI baseline on the per-country common sample.

In sample, the two inflation measures lead to very similar factors. The smoothing in the discounted moving average removes most of the food- and energy-price noise, so core and headline trend inflation correlate 0.90 on average across the G10, and the cycle factors they generate correlate 0.95 (local) and 0.92 (global). However, predictability is modestly weaker on headline CPI, exactly as Cieslak and Povala (2015) find. The cross-country mean in-sample R^2 of the local factor falls from 0.249 (core) to 0.229 (headline), and that of the global factor from 0.249 to 0.215, a relative drop of about eight to fourteen percent and thus in the neighbourhood of their nine, while both remain overwhelmingly significant (the global factor's pooled HAC t is 13.9 on core and 14.5 on headline). In Table A.7 and Figure A.4, we can see that the gap is concentrated where the energy pass-through in the data is largest. The United States, whose local R^2 falls from 0.33 to 0.23, accounts for much of the average, whereas a handful of markets (Belgium, France, Japan, Sweden) fit marginally better on headline. The FX-adjusted factor tells the same story (R^2 of 0.139 on core versus 0.126 on headline).

The out-of-sample comparison is where the choice of inflation measure becomes important, and it supports our baseline choice. The global cycle factor's real-time advantage, which is the central finding of Section 4.4, survives on core CPI ($R^2_{oos} = +0.05$) but disappears on headline CPI (-0.02). The local factor remains negative under both, while the FX-adjusted factor is the exception. It is slightly *better* on headline ($+0.035$ versus $+0.021$). We could argue that the energy-price component that contaminates the headline trend also moves exchange rates (Ferraro et al., 2015) and is therefore not pure noise for a dollar-return factor (Table 6.3). For the local-currency factors, the asymmetry makes economic sense. The recursive estimator re-fits

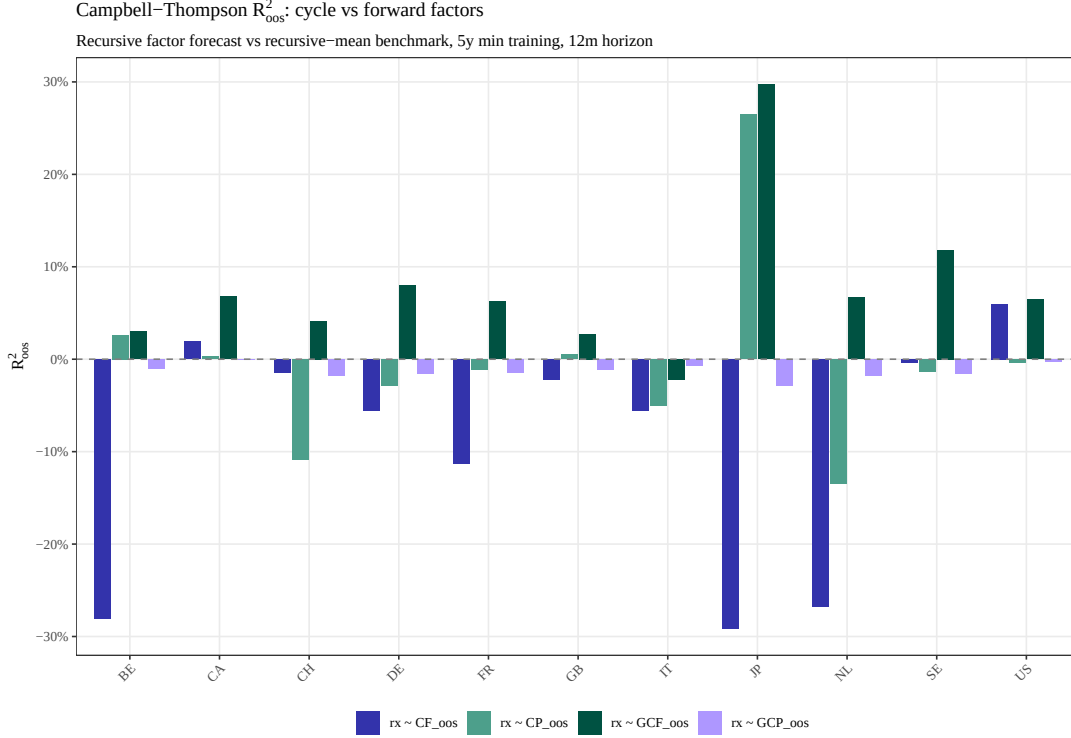


Figure 6.3: Out-of-sample R^2_{oos} by country for the cycle factors (CF, GCF) against the forward-rate factors (CP, GCP), fully recursive against the recursive prevailing mean. Only the global cycle factor is reliably positive.

both the trend and the predictive coefficients on data available only up to each date, so it is far more exposed to the energy-driven noise in headline inflation than the full-sample regression, which averages this noise away. Cieslak and Povala (2015) note that core and all-items inflation diverged in the 2000s because energy-price swings, which do not pass through to the yield curve, inflate the headline trend.

6.5 Sensitivity to Construction Choices

Three construction choices underlie the cycle factor, and it is worth stating how much our results depend on each of them. The first is the exponentially-weighted moving average that defines trend inflation, with decay parameter $v = 0.987$ and window $M = 120$ months. These values follow Cieslak and Povala (2015) directly, and the cycle factor inherits its robustness to them from the nature of the object. Trend inflation is a deliberately slow-moving anchor, so moderate changes in the decay weight or the window length shift the estimated trend by small amounts and leave the cycle, the residual of yields on this trend, and hence the factor largely unchanged. We verify this directly. Table 6.4 rebuilds the whole factor chain over a grid of the decay ($v \in \{0.859, 0.975, 0.987\}$) and the window ($M \in \{60, 120, 180\}$ months). The cross-country mean in-sample R^2 stays within a narrow band (local 0.24–0.29, global 0.25–0.31), and every alternative global factor correlates at least 0.83 with the baseline, rising above 0.98 for the slow-decay specifications closest to it. Only the fastest decay ($v = 0.859$) moves the factor materially, and even then it stays highly correlated with the baseline. At that decay the

Table 6.3
Core versus headline CPI: out-of-sample

Specification	R^2_{OOS}		
	Core	Headline	Difference
$\overline{rx}_{i,t+12} \sim CF_{i,t}$	-0.073	-0.064	-0.009
$\overline{rx}_{i,t+12} \sim GCF_t$	0.048	-0.015	0.062
$\overline{rx}_{i,t+12}^{\text{USD}} \sim FXGCF_t$	0.021	0.035	-0.014

Pooled R^2_{OOS} (3.20) of the fully-recursive factor forecast against the recursive prevailing mean, with the entire chain—trend inflation, the cycle, and the factors—rebuilt on regular (all-items, headline) CPI. Each column is scored over its own available sample; the headline series is shorter in places (Japan ends in 2021). “Difference” is the core value minus the headline value.

Table 6.4
Sensitivity of the cycle factor to the trend parameters v and M

v	M (months)	Half-life (months)	Mean in-sample R^2		$\text{corr}(GCF, GCF^{\text{base}})$
			Local	Global	
0.859	60	5	0.291	0.308	0.832
0.859	120	5	0.291	0.308	0.832
0.859	180	5	0.291	0.308	0.832
0.975	60	27	0.272	0.273	0.944
0.975	120	27	0.262	0.261	0.981
0.975	180	27	0.257	0.257	0.987
0.987	60	53	0.268	0.264	0.960
0.987	120	53	0.254	0.254	1.000
0.987	180	53	0.242	0.245	0.986

Sensitivity of the cycle factor to the trend-inflation decay v and window M (3.3). For each grid point the trend, the yield cycles, the local factor CF , and the global factor GCF are rebuilt from scratch. ‘Mean in-sample R^2 ’ is the cross-country average of the single-factor fits, local ($\overline{rx}_{i,t+12} \sim CF_{i,t}$) and global ($\overline{rx}_{i,t+12} \sim GCF_t$). The last column reports the correlation of each alternative global factor with the baseline factor ($v = 0.987$, $M = 120$; row in bold). The half-life is $\ln(0.5)/\ln(v)$. At the fast decay $v = 0.859$ the window length is immaterial because weights beyond five years are already negligible.

window length is immaterial because the weights beyond five years are already negligible. The second is the maturity menu used to form the cycle and the excess return. The one-, two-, four-, five-, nine-, and ten-year points available in the data, summarised by the duration-standardised, maturity-averaged return (3.8). In Section 4.1, we showed that the predictability holds at each individual maturity, not only for the average, with loadings that are flat-to-rising across the curve. We can claim that the maturity-averaged factor is thus a summary of a curve-wide phenomenon and not an artefact of any single segment. The third is the GDP weighting of the global factor, which we compute each month over the countries in the panel so that the weights sum to one. Because the global factor is dominated by the large core economies whose cycles co-move strongly, the aggregate is not sensitive to the precise weighting of the smaller markets.

The Construction of the FX-Adjusted Factor

A fourth construction choice concerns the FX-adjusted factor itself, and we examine it separately because its construction is not uniquely defined in previous research. The $FXGCF_t$ of Section 3.10

Table 6.5
The FX-adjusted factor under alternative constructions

Construction	Weights	Corr($\cdot$, GCF)	Mean IS R^2	$ t > 1.96$	R^2_{OOS} (pooled)	OOS+
Bottom-up (baseline)	GDP	0.78	0.143	10/11	+0.019	6/11
Bottom-up	$1/n$	0.73	0.141	9/11	+0.015	6/11
Top-down	GDP	0.99	0.095	4/11	+0.002	9/11
Top-down	$1/n$	0.94	0.106	7/11	+0.027	8/11

All four constructions use the same cycle predictors, the one-year cycle $c_{i,t}^{(1)}$ and the average cycle $\bar{c}_{i,t}$, and predict the US-dollar excess return $\overline{rx}_{i,t+12}^{\text{USD}}$. The bottom-up variants fit a USD cycle factor per country (3.10)–(3.11) and aggregate the fitted factors; the top-down variants first aggregate the dollar return and the predictors and fit one regression on the aggregates. Weights are nominal-GDP shares or equal ($1/n$). “Mean IS R^2 ” is the cross-country mean in-sample fit, “ $|t| > 1.96$ ” the number of markets with a significant HAC t -statistic, and the two out-of-sample columns report the pooled fully-recursive R^2_{OOS} statistic and the number of markets with a positive value.

admits a two-by-two design. The factor can be built bottom-up (a USD cycle factor per country, aggregated) or top-down (one regression of the aggregated dollar return on the aggregated predictors), and the aggregation can use GDP or equal ($1/n$) weights. In Table 6.5, we compare all four variants on the identical pipeline.

Two features of the table stand out. The correlation with the unadjusted global factor is set by the construction axis, not by the weighting. The two top-down variants correlate 0.94–0.99 with GCF_t , whereas the two bottom-up variants correlate only 0.73–0.78. A top-down factor projects one aggregate dollar return on the same aggregate cycle information that underlies GCF_t , so it inherits the global cycle almost one for one and has little room to load on currency dynamics. The bottom-up factor, in contrast, lets each country’s regression reweight its own cycles toward the currency-relevant combination before aggregating, and this is where the distinct information comes from. The in-sample recovery for the dollar investor is likewise a bottom-up phenomenon. The mean R^2 is 0.14 for both bottom-up variants, against 0.10 for the top-down ones, and the count of significant markets rises from five to ten as we move from the top-down GDP variant to our baseline. Out of sample, all four variants deliver only small pooled gains (+0.010 to +0.033), and no construction dominates clearly. We adopt the bottom-up GDP-weighted variant as our baseline because it mirrors the construction of GCF_t exactly, carries genuinely distinct information, and is the strongest in-sample specification. The choice of construction does not, however, change the real-time conclusion that the FX-adjusted factor’s out-of-sample edge is small.

We want to also state what is fragile. The local cycle factor’s in-sample predictability does not survive real-time evaluation (Section 4.4). The dollar-investor predictability is regime-dependent and concentrated before 2008 (Section 6.2). The FX-adjusted factor’s strong in-sample recovery also translates into only a small and scheme-sensitive real-time edge, earned mainly in the 2008 window. What survives every test in this chapter is the same single object. The GDP-weighted global cycle factor predicts international bond returns in-sample and out-of-sample, across every subsample except the financial crisis itself and, out of sample, when the macro trend is measured cleanly with core inflation (Section 6.4). It does so more robustly than its local counterpart, its dollar-investor variant, and the forward-rate factor. This global factor, and the economic value it carries (Chapter 5), is therefore the central finding of our thesis. We interpret it further in

Chapter 7.

Chapter 7

Discussion

7.1 Economic Interpretation

The central empirical result of Chapter 4 is that the trend–cycle mechanism of Cieslak and Povala (2015) is not a property of one market but of the asset class as a whole. In every G10 market, the deviation of yields from their trend-inflation anchor predicts next year’s excess return with the same sign structure, namely a negative loading on the one-year cycle and a positive loading on the average cycle. The loadings are also flat to rising across the maturity curve, the signature of a single premium that prices the whole curve. The same reading that Cieslak and Povala (2015) give for the United States now holds with cross-sectional support. Trend inflation pins down where investors expect yields to settle, so the distance of today’s curve from this anchor measures the compensation demanded for holding duration, and this compensation mean-reverts at business-cycle frequency. The uniformity across eleven countries with different monetary histories would be puzzling if the mechanism were market-specific. It is what we should see if the mechanism is fundamental, as the falling-stars evidence of Bauer and Rudebusch (2020) implies. Once the slow-moving macro anchors are accounted for, the risk premium is contained in the cyclical deviation.

The comparison with the forward-rate factor clarifies this interpretation. In sample, the Cochrane and Piazzesi (2005) factor fits better than the cycle factor across the G10 (R^2 of 0.38 against 0.25 on average), which we would expect from a six-predictor combination of prices compared to a two-predictor one. Out of sample, however, the ranking reverses. Of the four candidate factors only the global cycle factor beats the recursive historical mean (Section 6.3). We would suggest that the macro anchoring is therefore essential to the result. Our evidence thus suggests that the cycle performs better in real time.

Our second finding is that a single GDP-weighted global cycle factor matches the explanatory power of eleven national factors and drives the local factor out in eight of eleven markets. This re-establishes the integration result of Dahlquist and Hasseltoft (2013) on the macro-anchored factor family. The economic content of this finding is that the predictable component of government-bond returns is compensation for a common risk. Bond risk premia in the G10 rise and fall together, and a domestic investor in, for example, Sweden is rewarded for exposure to the global cycle, not to a Swedish one. Economically, the inputs to the factor are themselves synchronised.

Inflation trends fell together across the G10 over the sample, business cycles co-move, and monetary policy responded to common shocks, so the national cycles share their informative variation.

We consider the exceptions as informative as the rule. The local factor retains incremental content in Italy, the Netherlands, and Belgium, concentrated in periods of sovereign stress, which suggests that integration is conditional on the prevailing regime rather than a universal property. When sovereign-credit risk reprices a market, the compensation becomes local again, and a GDP-weighted factor dominated by the core economies cannot price it alone. We take this conditionality as a refinement of Dahlquist and Hasseltoft (2013) rather than a contradiction.

Our third finding refines the comparison with the forward-rate literature. For Dahlquist and Hasseltoft (2013), adjusting the global factor for currency risk matters. Their FX-adjusted forward factor correlates only about 0.50 with the unadjusted one and recovers dollar-investor predictability. Once we build the cycle-factor analogue bottom-up, it behaves in a comparable way: our baseline $FXGCF_t$ correlates 0.81 with the unadjusted global factor and substantially raises the dollar investor's in-sample fit (Section 4.3). Two qualifications, however, separate the cycle family from the forward family. First, the recovery depends on the construction. The top-down variant, which projects the aggregate dollar return on the aggregate cycle predictors, is nearly collinear with the unadjusted factor (correlation 0.99) and adds little (Section 6.5). The currency-relevant information in the cycles therefore sits in the cross-section of the country-level USD regressions and is lost once the aggregation happens before the fit. Second, the in-sample recovery translates into only a small real-time edge, and it does not help in timing the aggregate dollar portfolio (Chapter 5). A possible economic explanation could lie in what each factor family retains. The forward-rate factor is built from the level and shape of the forward curve, and cross-country differences in rate levels are the interest differentials that forecast currency returns through the failure of uncovered interest parity (Fama, 1984), so an FX adjustment of a forward factor has carry-relevant information to work with even at the aggregate level. The cycle factor, in contrast, removes the level by construction, because detrending by trend inflation strips out the persistent component and keeps the transitory deviation. The currency-relevant cyclical information that remains is country-specific and regime-dependent, which is why it shows up strongly in the in-sample cross-section, weakly in real time, and hardly at all in the aggregate. The regime evidence supports this reading, as the dollar-investor predictability is concentrated before 2008, when rate differentials were wide, and disappears in the post-crisis regime of compressed, synchronised policy rates (Section 6.2). The FX adjustment seems to work well only in the high-rate regime, in which cross-country differentials make currency returns predictable. In the low-rate regime that fills the second half of our sample, the currency leg collapses onto the bond cycle and the adjustment becomes close to redundant. When the two premia play different roles therefore explains why the FX adjustment matters so much less for the cycle family, in our sample, than for the forward family in Dahlquist and Hasseltoft (2013).

When we assemble the four findings, our evidence reduces to one robust result and a set of conditional ones. The GDP-weighted global cycle factor predicts international bond returns in sample, out of sample, and in every subsample except the 2008 panic. It also beats the

forward-rate alternative in real time and carries economic value for a hedged investor (Chapter 5). The local factors are real in sample but, for nine of eleven markets, cannot be estimated precisely enough to use in real time. Their out-of-sample failure reflects the estimation noise in eleven separately fitted national regressors rather than the absence of an in-sample signal (Cochrane, 2008). The dollar-investor factors, in turn, depend strongly on the prevailing currency regime. We therefore state the result more precisely than “the cycle factor works everywhere”. International bond risk premia contain a common, macro-anchored, slowly-moving component that is large enough and stable enough to estimate in real time, provided that we aggregate across markets, measure the inflation anchor cleanly, and do not use it to forecast exchange rates.

7.2 Limitations

In the process of writing our thesis we faced four limitations that qualify these conclusions. We assume that none of them overturns our results, but each of them limits how far the conclusions should be pushed.

All our headline t -statistics use 18-lag Newey–West HAC standard errors (Newey and West, 1987), whereas Cieslak and Povala (2015) use the reverse-regression delta method of Wei and Wright (2013), which requires the underlying monthly returns and is not identified from the twelve-month overlapping panel we construct (Section 3.12). For highly persistent regressors the two can diverge, and HAC inference at long horizons is known to over-reject in finite samples (Hodrick, 1992). A residual upward bias in the in-sample significance cannot be excluded. Because of this, we base our conclusions primarily on the out-of-sample evidence.

Our panel covers 1989/1994–2024, which corresponds to roughly thirty years of monthly data. The sample is also unbalanced (the nine non-US, non-Japanese markets begin only in December 1994), and the five euro-area members share a currency and a monetary policy from 1999 (and, before then, a single vendor-backfilled euro exchange rate in place of their legacy currencies), so the eleven markets contribute far fewer than eleven independent observations of the “global” cycle. The GDP-weighted factor is to a significant degree a US–euro-area–Japan factor, and the integration finding must be interpreted in that light. Alternative weightings (equal, market-capitalisation, principal-component) were not explored.

Finally, the robustness chapter itself draws the sharpest boundaries. The global factor’s out-of-sample edge is a property of expanding-window estimation and disappears under ten- and fifteen-year rolling windows (Section 6.1). It also depends materially on measuring the trend with core rather than headline inflation (Section 6.4), and it fails inside the 2008 crisis window. The FX-adjusted factor’s small pooled gain is earned mainly in the 2008 window and is sensitive to the training minimum.

7.3 Implications

For the international bond investor. The portfolio prescription that follows from our evidence has three parts. First, we advise hedging the currency. The predictable component of

international bond returns is found in local-currency space. Dollar conversion removes most of it in sample and almost all of it out of sample, and the timing strategy's Sharpe ratio falls from 0.40 hedged to about 0.24 unhedged (Chapter 5). From the standpoint of cycle-based predictability, currency exposure in a G10 bond portfolio therefore adds risk without adding compensation. Second, we advise timing the aggregate rather than the cross-section. The factor that survives in real time is a single global series. It supports varying overall duration exposure, which raised the hedged Sharpe ratio from 0.34 to 0.40 with a certainty-equivalent gain of about half a percent a year, achieved mainly by avoiding drawdowns such as 2022. But the out-of-sample evidence gives no support for country selection on local cycle factors. Third, the limits of the factor should be respected. It is slow, it requires a long estimation history, and it offers no protection in a flight-to-quality panic, when its real-time forecasts were at their worst.

For the term-premium literature. Our findings carry two messages back to the research literature. First, they add G10 evidence that term premia are anchored by slow macro trends. This is the international counterpart of the falling-stars mechanism of Bauer and Rudebusch (2020) and a factor-level complement to the trend-augmented forecasting of Zhang et al. (2021): it is the deviation from the trend, aggregated globally, that is priced, and it is priced in common across markets. Second, our findings suggest that the in-sample horse race between forward-rate and macro-anchored predictors is the wrong contest. In sample the forward factor wins, while in real time only the macro-anchored global factor survives. Studies that evaluate candidate state variables purely in sample will therefore systematically favour price-based over macro-anchored predictors. Out-of-sample evaluation reverses this verdict. A next step, which lies beyond the scope of this thesis, would be to embed the global cycle in a no-arbitrage term structure model with a shared international trend. This would close the gap between the reduced-form evidence we assemble here and the structural falling-stars framework.

Chapter 8

Conclusion

In our thesis, we asked whether the cycle factor of Cieslak and Povala (2015) applies to international government bonds. To answer this question, we constructed the local cycle factor for eleven G10 markets, aggregated the national factors into a GDP-weighted global cycle factor, and additionally built a novel FX-adjusted global cycle factor for an unhedged US-dollar investor. We evaluated this three-factor hierarchy in three empirical phases, both in-sample and under a fully recursive out-of-sample pipeline, and we validated our empirical engine by replicating the published US and international results (Chapter B). In this chapter, we summarise our answers to the research question and its three sub-questions, restate our contribution, and outline directions for future work.

Our answer to the main question is affirmative. In all eleven markets, the local cycle factor predicts one-year excess returns with the same sign pattern that Cieslak and Povala (2015) document for the United States, significantly so in ten, with an average in-sample R^2 of 25% and loadings that are uniform across the maturity curve. We therefore conclude that the trend-cycle mechanism is not specific to the United States but a property of the asset class. This is in line with the falling-stars evidence of Bauer and Rudebusch (2020), according to which the predictable component of bond returns appears once the slow-moving macro anchors are accounted for.

Regarding our first sub-question, on the role of the local relative to the global cycle factor, we find that the predictability is overwhelmingly global. The GDP-weighted global cycle factor is significant in every market and matches the explanatory power of the eleven separately estimated local factors. Regarding our second sub-question, we find that the global factor does subsume the local factors in most markets. Once the global factor is included, genuinely local content survives only in Italy, the Netherlands, and Belgium, concentrated in periods of sovereign stress. Similar to the findings of Dahlquist and Hasseltoft (2013) for the forward-rate family, we can therefore conclude that government-bond risk premia are internationally integrated, although sovereign influences may dominate predictability. Because five of the eleven markets share the euro and a common monetary policy, they furnish fewer than eleven independent tests of the global cycle, so this integration result should be read with that in mind.

For the unhedged US-dollar investor, our results are mixed. Dollar conversion erodes most of the global factor's predictive power. The FX-adjusted global cycle factor, built bottom-up from the country-level USD regressions, recovers much of it in sample. It substantially raises the average dollar-investor fit and correlates only 0.81 with the unadjusted factor, comparable in

spirit to the 0.50 that Dahlquist and Hasseltoft (2013) report for the forward-rate family. We also find, however, that this recovery depends on the construction of the FXGCF and that it translates into only a small edge that does not help in timing the aggregate dollar portfolio. The practical implication we draw from this is that the dollar investor should hedge the currency exposure, since the exploitable component of international bond returns is a local-currency phenomenon.

The finding we regard as most important concerns real time. Out of sample, the local factors lose to the recursive historical mean almost everywhere, while the global cycle factor beats it in ten of eleven markets. This edge holds across crisis subsamples, beats the global forward-rate factor, and depends on measuring the inflation trend with core rather than headline inflation. It also carries economic value. A timing strategy based on the global factor raises the Sharpe ratio of a hedged global bond portfolio from 0.34 to 0.40 with a positive certainty-equivalent gain (Chapter 5).

With our thesis, we thus contribute a unified, macro-anchored global cycle factor together with its FX-adjusted counterpart, and we provide the first G10-wide evidence on the international validity of the Cieslak and Povala (2015) mechanism. Summarising our findings, we conclude that international bond risk premia contain a common, macro-anchored, and slowly moving component that is large enough and stable enough to estimate in real time, provided it is aggregated across markets, its inflation anchor is measured with core CPI, and it is not asked to forecast exchange rates.

Our conclusions are subject to the limitations we discussed in Chapter 7, in particular the inference under overlapping returns, the largely homogeneous macro regime covered by our sample, and the dependence of the real-time result on expanding-window estimation and core inflation. Because of this, we see several directions for future work. First, a richer maturity menu and longer samples would allow a more precise estimation of the factor loadings and would add independent observations of the global cycle. Second, alternative weightings of the global factor, such as equal or market-capitalisation weights, could show how much the integration result depends on the GDP weighting. Third, the global cycle factor could be embedded in a no-arbitrage term structure model with a shared international trend, which would close the gap between our reduced-form evidence and the structural framework of Bauer and Rudebusch (2020). We leave these extensions to future research.

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Appendix A

Appendix

A.1 Additional Tables

This section collects detailed tables that support the main text: the per-maturity Phase I estimates, the cross-country yield correlations, the correlations among the cycle factors, and the in-sample subsample and inflation-measure estimates.

Table A.1

Phase I by maturity: predicting individual excess returns with the cycle factor

Country	$n = 2$			$n = 5$			$n = 10$		
	$\hat{\beta}$	t	R^2	$\hat{\beta}$	t	R^2	$\hat{\beta}$	t	R^2
Belgium	0.83	(4.88)	0.272	1.16	(4.64)	0.302	1.01	(3.96)	0.251
Germany	0.75	(3.70)	0.203	1.19	(4.16)	0.265	1.05	(3.81)	0.228
France	0.59	(2.58)	0.059	1.19	(4.11)	0.134	1.22	(4.84)	0.149
Italy	0.81	(3.83)	0.316	1.15	(4.05)	0.278	1.04	(3.93)	0.232
Netherlands	0.79	(4.77)	0.302	1.19	(5.51)	0.358	1.03	(4.57)	0.282
Switzerland	0.61	(1.02)	0.038	1.34	(1.96)	0.092	1.05	(1.61)	0.059
United Kingdom	0.74	(3.71)	0.199	1.19	(4.28)	0.269	1.07	(3.62)	0.229
Sweden	0.86	(3.87)	0.245	1.10	(3.34)	0.218	1.04	(3.24)	0.205
Canada	0.87	(3.85)	0.268	1.14	(4.48)	0.316	0.99	(3.76)	0.271
Japan	0.57	(1.97)	0.126	1.23	(2.60)	0.167	1.20	(3.20)	0.203
United States	0.75	(3.35)	0.233	1.14	(4.33)	0.314	1.11	(5.20)	0.347
G10 panel	0.78	(14.32)	0.253	1.16	(17.25)	0.271	1.06	(15.44)	0.243

The dependent variable is the duration-standardized individual excess return $rx_{i,t+12}^{(n)}/n$ for $n \in \{2, 5, 10\}$ years. Each maturity block reports the regression of $rx_{i,t+12}^{(n)}/n$ on the local cycle factor $CF_{i,t}$ of (3.6), as in (3.13). Newey–West HAC t -statistics (12-month overlap) in parentheses; R^2 is in-sample. “G10 panel” pools all countries with country fixed effects. The table mirrors Cieslak and Povala (2015), Table 4, Panel A: a single return-forecasting factor prices the whole curve, with loadings flat-to-rising in maturity.

A.2 Additional Figures

This section collects supporting figures referenced in the main text.

Table A.2**Cross-country correlation of the ten-year zero-coupon yield**

Country	BE	DE	FR	IT	NL	CH	GB	SE	CA	JP	US
Belgium	1.00										
Germany	0.98	1.00									
France	0.99	0.99	1.00								
Italy	0.89	0.84	0.88	1.00							
Netherlands	0.99	1.00	1.00	0.85	1.00						
Switzerland	0.95	0.97	0.95	0.81	0.97	1.00					
United Kingdom	0.94	0.96	0.96	0.84	0.96	0.90	1.00				
Sweden	0.95	0.96	0.96	0.90	0.96	0.95	0.95	1.00			
Canada	0.94	0.97	0.95	0.84	0.96	0.95	0.95	0.98	1.00		
Japan	0.91	0.91	0.92	0.87	0.90	0.85	0.93	0.92	0.89	1.00	
United States	0.90	0.94	0.92	0.75	0.93	0.89	0.95	0.91	0.96	0.90	1.00

This table reports pairwise-complete correlations of the monthly ten-year zero-coupon yield $y_t^{(10)}$ across countries, computed over the full per-country sample (see Table 3.1). Column labels are the ISO codes of the countries in the first column. Correlations are uniformly high and tightest within the euro-area bloc. After Dahlquist and Hasseltoft (2013), Table 1.

Table A.3**Summary statistics of the non-yield inputs**

Country	Core inflation (%)		FX return vs USD (%)		GDP weight (%)
	Mean	SD	Mean	SD	
Belgium	1.96	1.15	−0.36	9.72	1.3
Germany	1.50	1.04	−0.36	9.72	9.8
France	1.31	0.75	−0.36	9.72	6.9
Italy	1.94	1.20	−0.36	9.72	5.6
Netherlands	2.10	1.16	−0.36	9.72	2.2
Switzerland	0.50	0.69	1.47	8.92	1.6
United Kingdom	2.01	1.21	−0.67	8.53	7.3
Sweden	1.48	2.23	−1.10	11.38	1.3
Canada	1.91	0.97	0.03	7.50	3.9
Japan	0.45	1.28	−0.05	10.70	18.7
United States	2.63	1.18	—	—	47.1

Summary statistics of the three non-yield inputs, each computed over the same per-country yield sample as Table 3.2. Core inflation is the year-on-year change in the core CPI, in percent. The FX return is the twelve-month log change in the US-dollar price of the local currency $s_{i,t} - s_{i,t-12}$, in percent; the five euro-area members share the euro, and the United States is the numeraire (no currency leg). GDP weight is the mean of the time-varying GDP weight $w_{i,t}$ (3.7), in percent.

Table A.4**Correlations among the local cycle factors and with the global factor**

	BE	DE	FR	IT	NL	CH	GB	SE	CA	JP	US
Belgium (BE)	1.00										
Germany (DE)	0.92	1.00									
France (FR)	0.81	0.79	1.00								
Italy (IT)	0.36	0.26	0.23	1.00							
Netherlands (NL)	0.89	0.86	0.60	0.51	1.00						
Switzerland (CH)	0.40	0.64	0.55	0.10	0.34	1.00					
United Kingdom (GB)	0.82	0.88	0.65	0.40	0.82	0.48	1.00				
Sweden (SE)	0.79	0.93	0.69	0.12	0.75	0.72	0.79	1.00			
Canada (CA)	0.80	0.87	0.66	0.03	0.71	0.57	0.74	0.91	1.00		
Japan (JP)	0.74	0.84	0.50	0.41	0.78	0.60	0.77	0.76	0.62	1.00	
United States (US)	0.70	0.84	0.67	−0.02	0.63	0.66	0.69	0.89	0.88	0.63	1.00
<i>GCF</i>	0.84	0.94	0.75	0.21	0.80	0.66	0.84	0.94	0.89	0.78	0.96
GDP weight (%)	1.3	9.8	6.9	5.6	2.2	1.6	7.3	1.3	3.9	18.7	47.1

Pairwise-complete correlations between the eleven local cycle factors $CF_{i,t}$ over the common sample. The *GCF* row reports each local factor's correlation with the GDP-weighted global cycle factor, and the final row the country's mean GDP weight $w_{i,t}$ in percent. The global factor correlates most strongly with the large core economies (the United States, 0.96; Germany, 0.94), but the pattern is not purely a weight effect: Sweden and Canada correlate 0.94 and 0.89 despite small weights, because their cycles co-move with the global business cycle, whereas Italy is the outlier (0.21). Across countries the correlation between a factor's *GCF* loading and its GDP weight is only 0.26.

Table A.5**Currency return versus the local cycle factor**

Country	corr(FX ret, <i>CF</i>)	SD of FX ret (%)
Belgium	0.03	2.62
Germany	0.14	2.62
France	0.21	2.61
Italy	0.16	2.56
Netherlands	−0.04	2.62
Switzerland	0.34	2.41
UK	0.17	2.28
Sweden	0.20	3.11
Canada	0.17	2.03
Japan	0.09	2.96
US	—	0.00

For each country, the twelve-month currency-return component of the US-dollar excess return— $rx_{i,t+12}^{\text{USD}} - rx_{i,t+12}$ in the duration-standardized units of (3.9)—is correlated with the local cycle factor $CF_{i,t}$ over the country's estimation sample. The last column reports the standard deviation of that currency component. The US carries no currency leg, so its currency return is identically zero. The currency return is at most weakly related to the bond cycle in every market (absolute correlation never exceeds 0.34), so currency conversion adds largely independent noise rather than being absorbed by the bond factor.

Table A.6
In-sample predictability across subsamples

Subsample	Months	$\bar{r}\bar{x} \sim CF$		$\bar{r}\bar{x} \sim GCF$		$\bar{r}\bar{x}^{\text{USD}} \sim GCF$		$\bar{r}\bar{x}^{\text{USD}} \sim FXGCF$	
		R^2	t	R^2	t	R^2	t	R^2	t
Full sample	412	0.254	(17.09)	0.254	(13.61)	0.086	(6.63)	0.143	(8.05)
Pre-crisis (−2007:06)	210	0.238	(10.24)	0.164	(9.93)	0.121	(4.79)	0.291	(7.67)
Financial crisis (2007–09)	30	0.175	(1.70)	0.054	(1.91)	0.051	(1.53)	0.118	(−2.46)
Euro crisis (2010–12)	36	0.506	(6.75)	0.387	(2.41)	0.258	(1.45)	0.265	(0.68)
Post-crisis (2013–)	136	0.133	(6.68)	0.242	(11.51)	0.040	(−1.64)	0.041	(−2.08)

Each cell pair reports the cross-country mean single-factor in-sample R^2 and, in parentheses, the pooled country-fixed-effects Newey–West HAC t -statistic on the factor, estimated on the indicated subsample, dated by the forecast-origin month. $\bar{r}\bar{x}$ is the local-currency and $\bar{r}\bar{x}^{\text{USD}}$ the US-dollar duration-standardized, maturity-averaged one-year excess return; $CF_{i,t}$ and GCF_t are the local and global cycle factors and $FXGCF_t$ the FX-adjusted global factor, i.e. the specifications (3.13), (3.15), (3.18), and (3.19). “Months” is the number of forecast origins in the window. The two crisis windows are short, so their HAC t -statistics are necessarily noisier than the full-sample figures.

Table A.7
Core versus headline CPI: in-sample

	N	Correlation		In-sample R^2			Pooled t	
		Trend	Factor	Core	Headline	ΔR^2	Core	Headline
Belgium	353	0.60	0.96	0.301	0.309	−0.007		
Germany	353	0.91	0.95	0.256	0.232	0.024		
France	353	0.90	0.90	0.126	0.176	−0.050		
Italy	353	0.98	0.95	0.287	0.243	0.043		
Netherlands	353	0.78	0.91	0.344	0.304	0.039		
Switzerland	353	0.98	0.96	0.073	0.055	0.018		
United Kingdom	353	0.96	0.99	0.262	0.231	0.031		
Sweden	353	0.99	0.99	0.238	0.242	−0.004		
Canada	353	0.85	0.97	0.320	0.290	0.031		
Japan	378	0.98	0.97	0.209	0.211	−0.001		
United States	412	0.94	0.87	0.326	0.229	0.097		
Mean (CF)	361	0.90	0.95	0.249	0.229	0.020		
GCF	412		0.92	0.249	0.215	0.033	(13.85)	(14.53)
$FXGCF$	412		0.93	0.139	0.126	0.013	(7.91)	(6.89)

Trend inflation, the cycle, and the factors are rebuilt from regular (all-items, headline) CPI in place of core CPI and compared with the core-CPI baseline on the per-country common sample. The “Correlation” columns report the core-versus-headline correlation of trend inflation $\pi_{i,t}^e$ and of the resulting cycle factor. “Core” and “Headline” are the in-sample predictive R^2 of the excess return on the factor under each inflation measure, and ΔR^2 is their difference (positive means core fits better, as in Cieslak and Povala (2015), Table 10, Panel C4). Country rows refer to the local cycle factor $CF_{i,t}$ and “Mean (CF)” averages them; the GCF and $FXGCF$ rows report the cross-country mean R^2 for the global factors together with the pooled country-fixed-effects Newey–West HAC t -statistics in parentheses (“Pooled t ”). The excess return is the local-currency $\bar{r}\bar{x}_{i,t+12}$ for CF and GCF and the US-dollar $\bar{r}\bar{x}_{i,t+12}^{\text{USD}}$ for $FXGCF$.

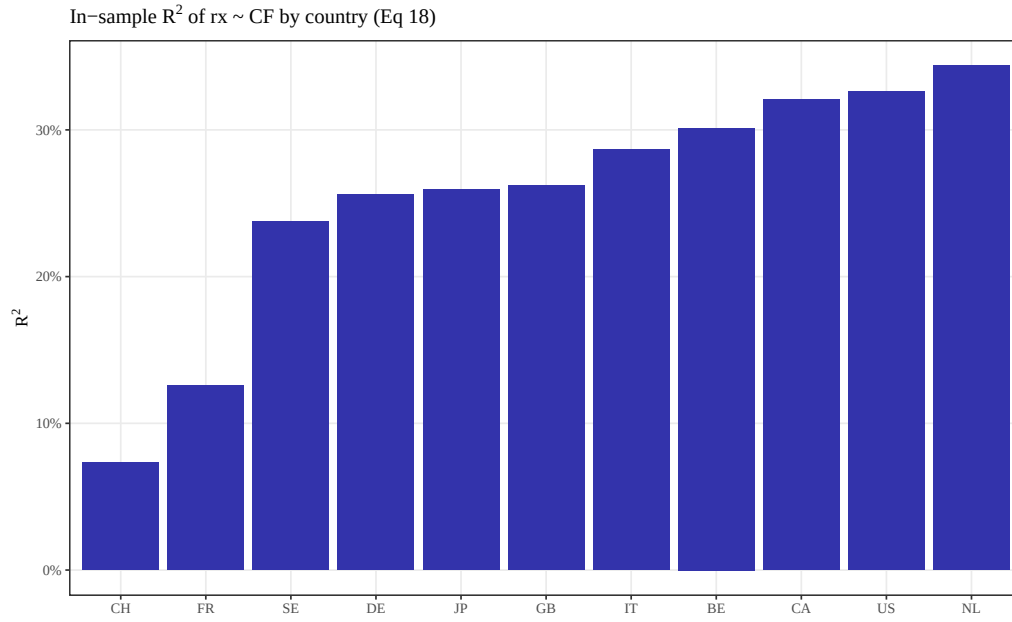


Figure A.1: In-sample R^2 of the local predictive regression (3.13), $\bar{rx}_{i,t+12} \sim CF_{i,t}$, by country.

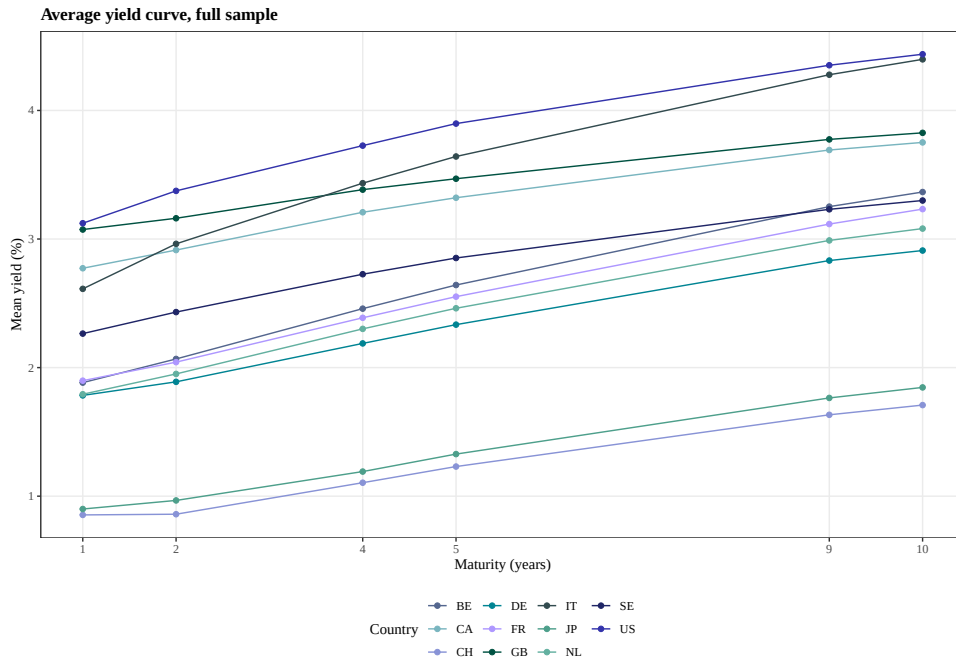


Figure A.2: Average zero-coupon yield curve by country over the sample.

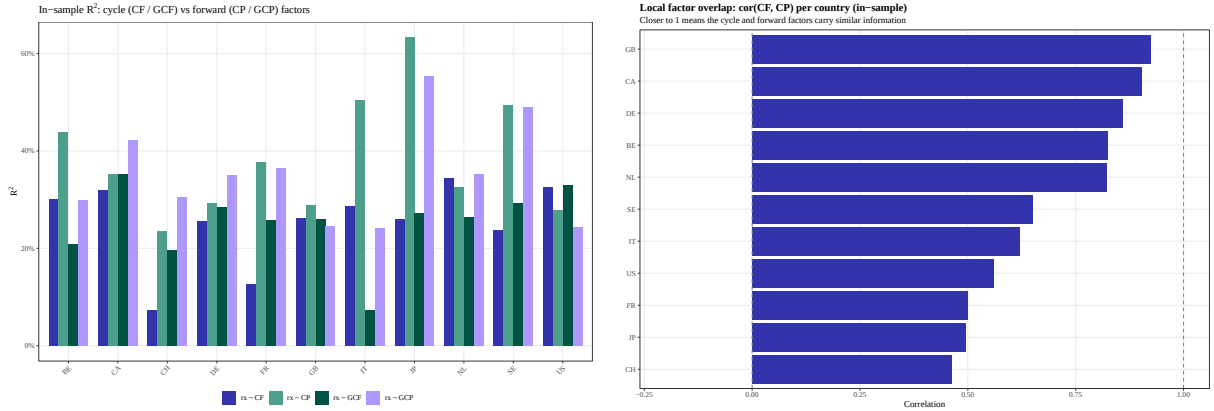


Figure A.3: Left: in-sample R^2 by country for the cycle factors (CF, GCF) against the forward-rate factors (CP, GCP). Right: per-country correlation between the local cycle factor and the local forward factor.

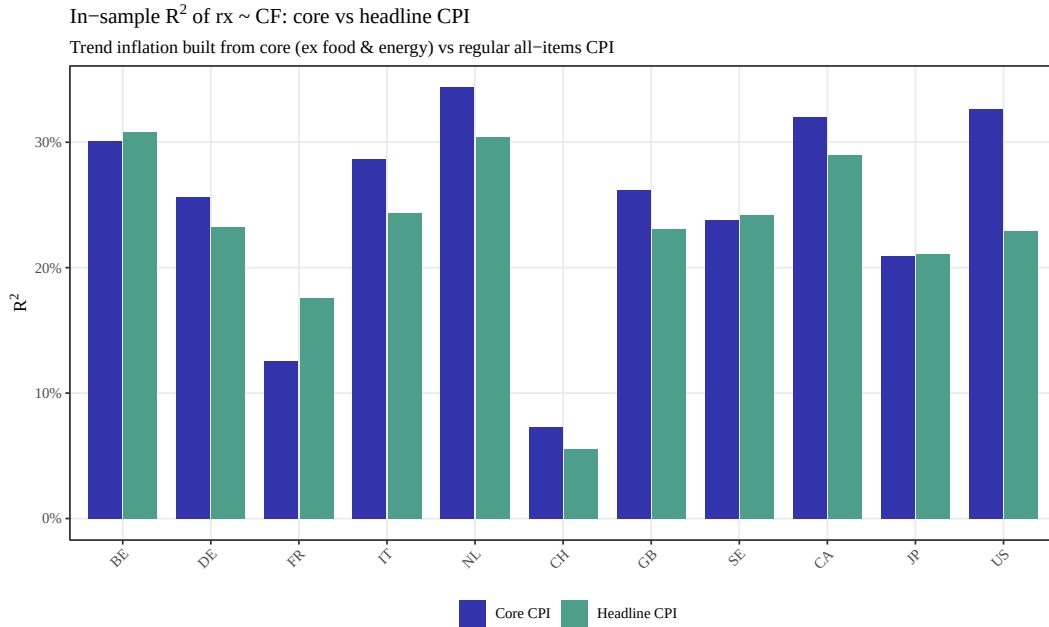


Figure A.4: In-sample R^2 of the local predictive regression $\overline{rx}_{i,t+12} \sim CF_{i,t}$ by country, with the cycle factor built on core (ex food and energy) versus regular all-items CPI.

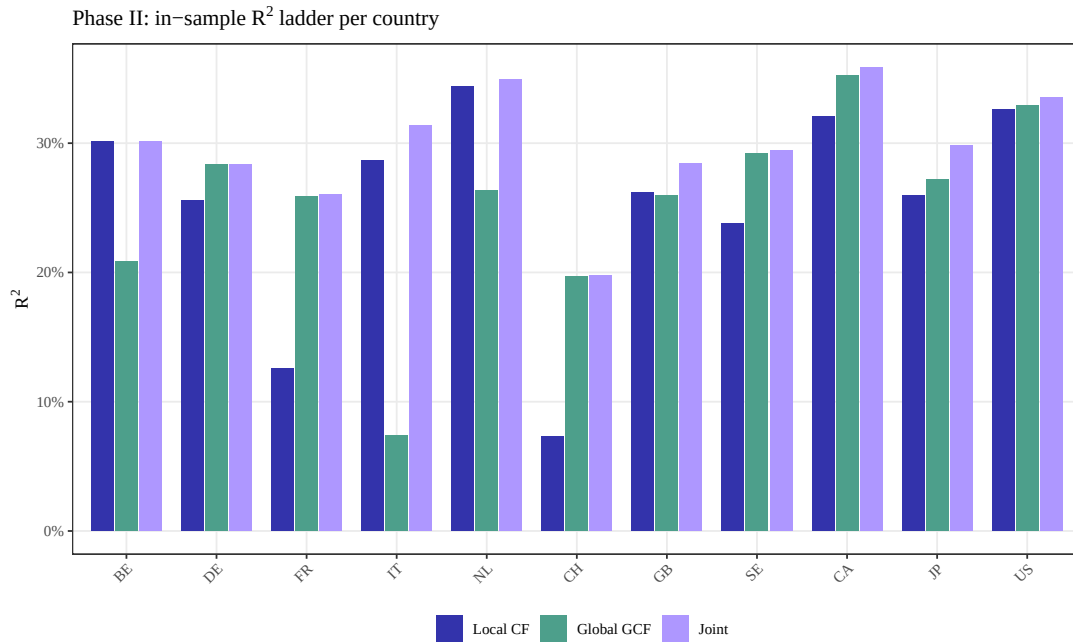


Figure A.5: Phase II in-sample R^2 ladder by country: the local factor alone (3.13), the global factor alone (3.15), and the joint horse race (3.14).

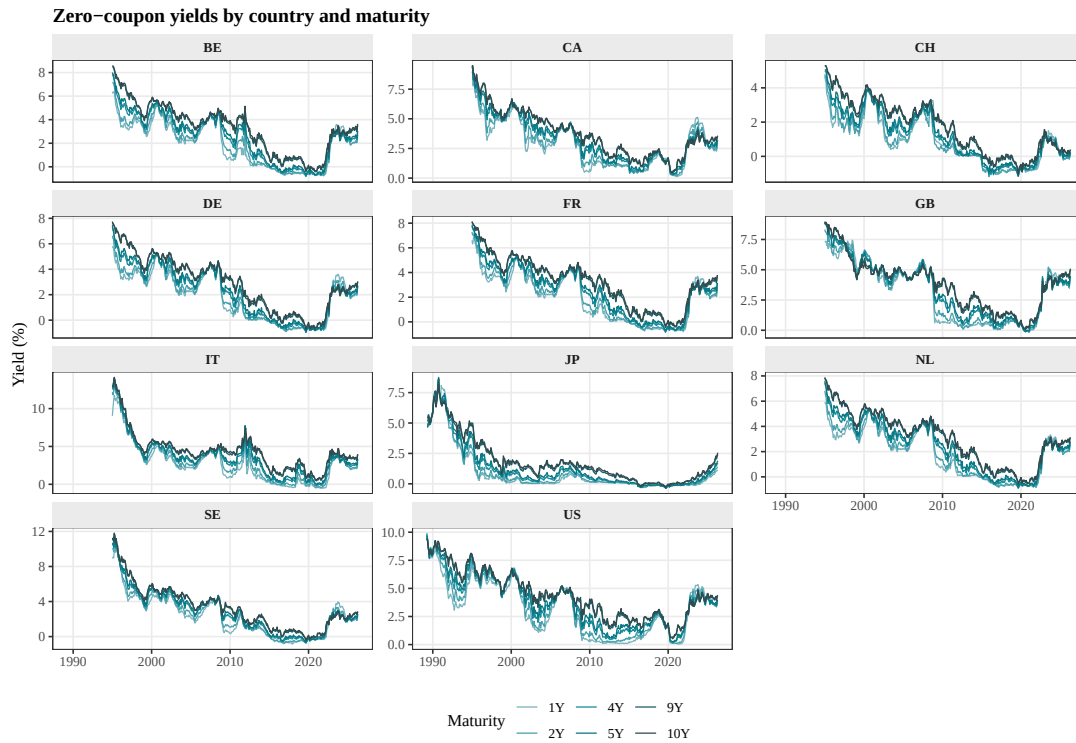


Figure A.6: Ten-year zero-coupon yields across the G10, 1989–2026. The secular decline in nominal yields and the post-2021 reversal are common across countries.

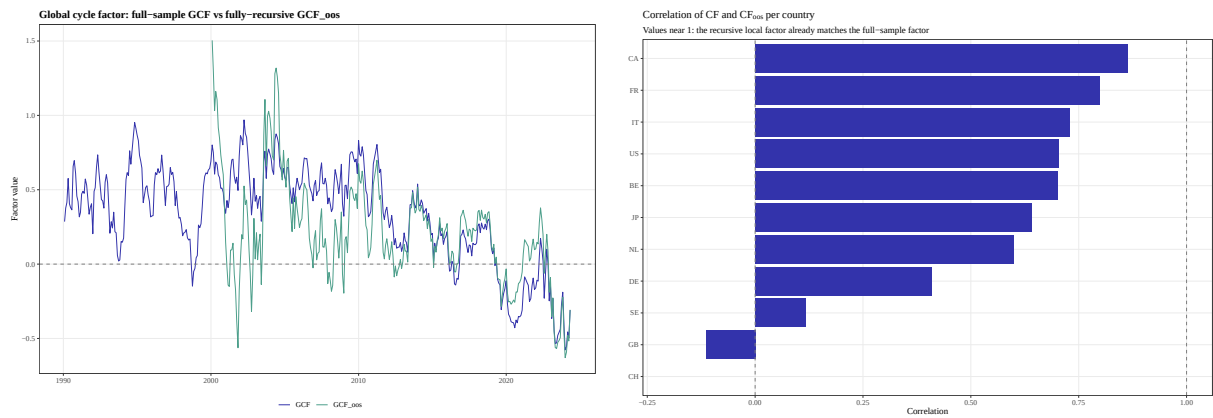


Figure A.7: Left: the full-sample global cycle factor GCF_t against the fully-recursive GCF_t^{oss} ; convergence after the trend-inflation burn-in signals a stable real-time estimator. Right: per-country correlation between the full-sample and recursive *local* cycle factors, where a value closer to one means that the full-sample look-ahead added little.

Appendix B

Replication and Validation

Before we take the macro-anchored cycle factor to the G10, we first have to show that our pipeline produces comparable results. In this appendix, we therefore validate the entire pipeline, namely the trend-inflation construction (3.3), the yield-cycle decomposition (3.2), the cycle factor (3.6), the forward-rate factor of Cochrane and Piazzesi (2005), and the global aggregation (3.7), against the two studies it builds on. In Section B.1, we reproduce the United States evidence of Cieslak and Povala (2015), who introduced the cycle mechanism. In Section B.2, we reproduce the international evidence of Dahlquist and Hasseltoft (2013), who established the global integration of bond risk premia through the forward-rate factor. This exercise is, however, not a mechanical re-run, since our data, our sample window, and our inference differ from the originals. The relevant test is therefore agreement on signs, magnitudes. We restrict the United States return regressions to January 1990–December 2014 (300 months) to align with the window of Cieslak and Povala (2015). The yield-cycle decomposition of Table B.1, which needs no returns, uses the full US yield history from March 1989. Inference uses the block bootstrap of Section 3.12 and Newey–West HAC standard errors, with 18 lags for the Cieslak–Povala tables and 12 lags for the Dahlquist–Hasseltoft tables, matching each original specification.

B.1 Cieslak–Povala (2015)

The cycle mechanism starts by splitting each yield into a slow trend component, driven by trend inflation, and a transitory cycle. In Table B.1 we reproduce Table 1 of Cieslak and Povala (2015). Panel A regresses each US yield on trend inflation (3.2): the slope is large and significant at every maturity (1.63–1.88, HAC t from 5.8 to 12.8), and the share of yield variation the trend explains rises from 58% (one year) to 84% (ten). Figure B.1 shows the same split for four representative markets (Germany, the UK, Japan, and the US).

Panel B confirms a single cycle factor: adjacent-maturity cycle correlations are near unity (0.99) and fall to 0.74 between the one- and ten-year cycles, and the cycles are far less persistent than the yields (half-life $55 \rightarrow 10$ months versus ~ 46 –56).

The fitted factor CF_t (3.6) correlates 0.69 with the average cycle in our US sample, close to the 0.61 of Cieslak and Povala (2015) (Figure B.2).

In Table B.2 the forward curve alone explains 25% of return variation and adding trend inflation lifts this to 46% (trend loading $t = -3.6$). The two-predictor cycle representation

Table B.1

Replication of Cieslak–Povala (2015) Table 1: yields and trend inflation

<i>Panel A. Yields on trend inflation</i>					
Maturity	$a_n \times 100$	$t(a_n)$	b_n	$t(b_n)$	$\bar{R}^2$
1y	−174.4	−1.70	1.77	5.78	0.58
2y	−173.1	−1.85	1.88	6.95	0.65
4y	−114.5	−1.47	1.84	8.38	0.72
5y	−82.9	−1.18	1.80	9.31	0.75
9y	11.9	0.24	1.67	12.26	0.83
10y	32.5	0.69	1.63	12.79	0.84

<i>Panel B. Cycle and yield properties</i>				
Maturity	Cycle SD	Cycle half-life	Yield SD	Yield half-life
1y	1.61	55.5	2.50	56.4
2y	1.49	36.2	2.52	52.3
4y	1.23	23.3	2.33	48.0
5y	1.11	18.3	2.24	45.9
9y	0.82	11.2	1.98	45.5
10y	0.78	10.3	1.92	45.8

The table replicates Cieslak and Povala (2015), Table 1, on United States data, 1989m3–2014m12. Panel A reports the maturity-by-maturity projection of yields on trend inflation, $y_t^{(n)} = a_n + b_n \tau_t^{\text{CPI}} + \varepsilon_t^{(n)}$ (3.2), where τ_t^{CPI} is the discounted moving average of realised core CPI inflation (3.3); intercepts are scaled by 100 and expressed in basis points, and t -statistics are Newey–West adjusted (18 lags). Panel B reports the standard deviation (in percent) and the AR(1) half-life (in months), computed as $\ln(0.5)/\ln(|\hat{\psi}|)$ for AR(1) coefficient $\hat{\psi}$, of the maturity-specific cycle $c_t^{(n)}$ defined as the residual of (3.2) and of the corresponding yield $y_t^{(n)}$. Because the sample is shorter than the 1975–2014 window of Cieslak and Povala (2015), intercept levels differ and the cycle half-lives are shorter; the slopes and $\bar{R}^2$ align with the original.

recovers 27% with the opposite-signed loadings that define the mechanism, and the BIC places essentially all posterior weight on the yields-plus-trend specification.

Read against a simulated expectations-hypothesis null (Table B.3, ??), whose 95th percentile reaches 15%–19% at our 300-month window, the realised 27%–46% sits far in the right tail. Our Monte Carlo reproduces the shape of the null but not the exact published percentiles.

Finally, Table B.4 shows the single cycle factor loading positively and significantly at every maturity (0.69/1.37/1.42 at two/five/ten years), with the opposite-signed cycle structure and the long end most predictable ($R^2 = 32\%$ at ten years). We thus reproduce the US evidence in its essentials.

B.2 Dahlquist–Hasseltoft (2013)

The international step rests on a different factor. Dahlquist and Hasseltoft (2013) work with the forward-rate return-forecasting factor of Cochrane and Piazzesi (2005) rather than with the macro-anchored cycle. They build a GDP-weighted global version of it and show that the global factor subsumes the local ones, which is the integration result that we later re-establish for the cycle factor. By validating their evidence, we therefore also validate our implementation of the Cochrane–Piazzesi factor and of the global aggregation. The G10 yields underlying this exercise match the summary statistics of their Table 1 in level and dispersion: average yields rise

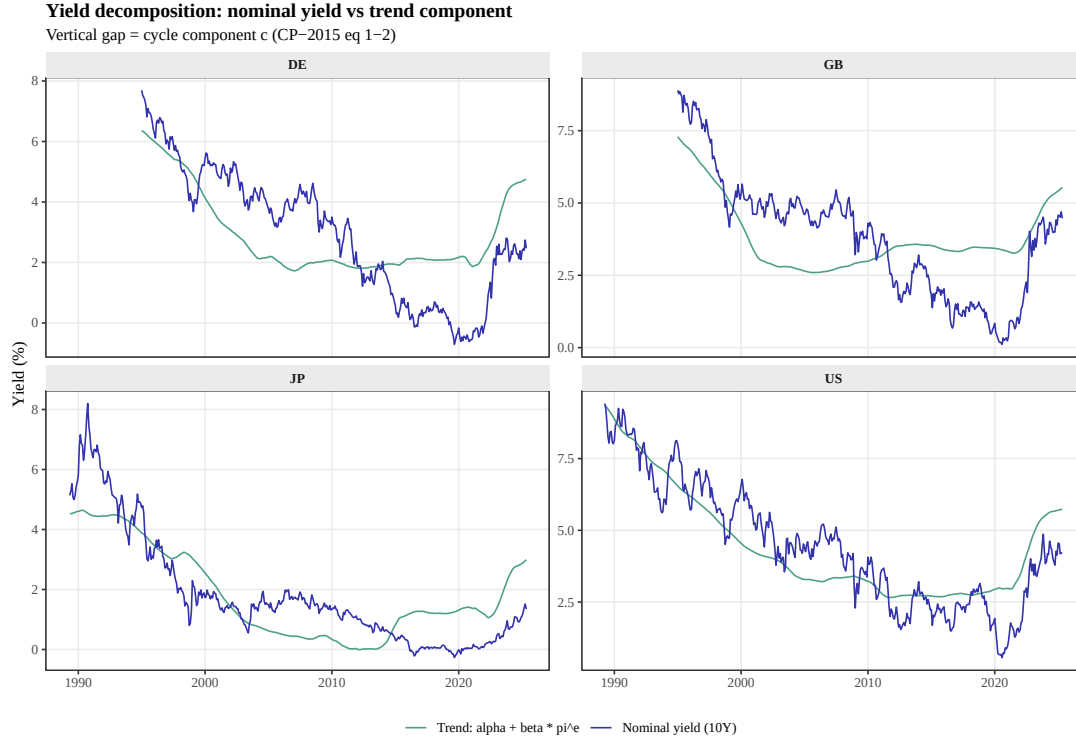


Figure B.1: Yield-cycle decomposition for four representative markets (Germany, the UK, Japan, and the US): the ten-year yield, its trend-inflation component, and the residual cycle (3.2). Four markets are shown for legibility. The same decomposition is applied to all eleven G10 markets in the analysis that follows.

with maturity in every market, they are highest in Italy and the United States and lowest in Switzerland and Japan, and the ten-year yields are strongly correlated across countries. The latter is the data feature that makes a common factor plausible.

Dahlquist and Hasseltoft (2013), Table 3, documents that the local forward factors correlate strongly with the GDP-weighted global factor. In Table B.5 we reproduce this, with local-global correlations of 0.76–0.97 (highest for the large core markets, lowest for Belgium and Italy).

In Table B.6 we reproduce Table 4, contrasting the single forward-spot-spread regression of Fama and Bliss (1987) with the full Cochrane–Piazzesi factor. The CP factor dominates in almost every market and maturity (e.g. the United States at two years, 30% against 8%), with the strikingly high Japanese fit (R^2 up to 69%) that Dahlquist and Hasseltoft (2013) also note.

In Table B.7 we reproduce Table 6: the global forward factor is significant in every market and maturity, the orthogonalised local factor adds little outside Italy and Japan, and the joint test rejects everywhere.

Table B.8 reproduces Table 7, the US-dollar investor’s regressions. The unadjusted global forward factor predicts dollar returns only weakly, whereas the FX-adjusted global factor, built top-down on the average dollar return, restores predictability (HAC t between 2.0 and 4.4. The euro at ten years rises from 7% to 22%). This motivates the FX-adjusted global cycle factor of our thesis. For the cycle family, however, the same top-down adjustment is nearly collinear with the unadjusted global cycle factor in the G10, which is why we build our FX-adjusted cycle factor bottom-up instead (Section 3.10, Section 6.5). We take up this difference in Chapter 7.

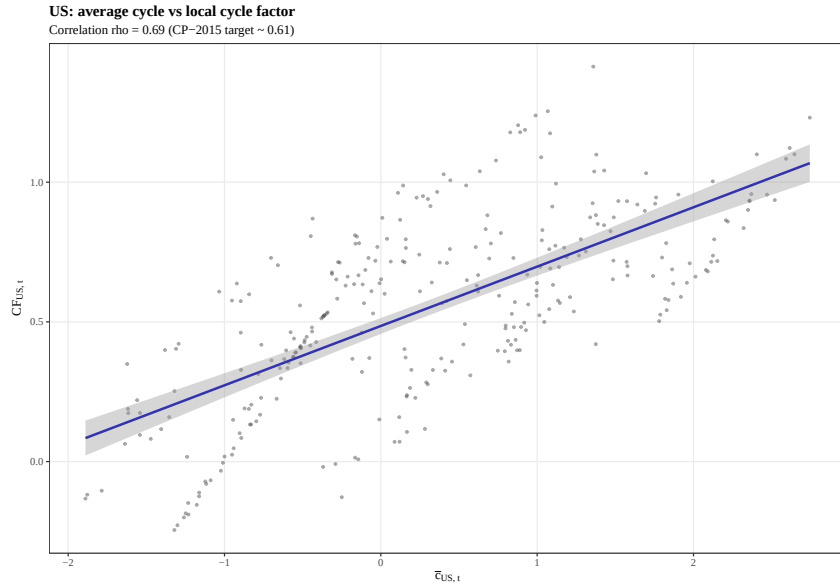


Figure B.2: The return-forecasting cycle factor CF_t against the average cycle $\bar{c}_t$ for the United States. The correlation is 0.69, compared to the 0.61 reported by Cieslak and Povala (2015), Figure 2.

Taken together, the two replications establish that our engine reproduces both pillars: the macro-anchored cycle mechanism of Cieslak and Povala (2015) on US data and the global-integration evidence of Dahlquist and Hasseltoft (2013) on the G10, including the FX-adjusted construction. The differences from the originals, namely a sample beginning in 1990 and HAC and bootstrap inference, do not disturb the qualitative agreement. With the engine validated, we turn in Chapter 4 to applying the macro-anchored cycle factor, rather than the forward-rate factor, to the international cross-section.

Table B.2

Replication of Cieslak–Povala (2015) Table 2, Panel A: predictive regressions

Regressor	(1) Yields		(2) Yields, τ^{CPI}		(3) $y^{(1)}, \bar{y}$		(4) $y^{(1)}, \bar{y}, \tau^{\text{CPI}}$		(5) $c^{(1)}, \bar{c}$	
	Coef.	t	Coef.	t	Coef.	t	Coef.	t	Coef.	t
$y_t^{(1)}$ or $c_t^{(1)}$	0.50	(0.93)	0.23	(0.50)	−0.39	(−1.94)	−0.79	(−3.92)	−0.59	(−3.59)
$y_t^{(2)}$	−2.37	(−1.92)	−0.82	(−0.91)						
$y_t^{(4)}$	4.70	(1.31)	1.70	(0.82)						
$y_t^{(5)}$	−4.35	(−1.43)	−3.34	(−1.78)						
$y_t^{(9)}$	8.60	(3.36)	9.13	(3.00)						
$y_t^{(10)}$	−6.92	(−3.34)	−6.22	(−2.35)						
τ_t^{CPI}			−0.94	(−3.58)			−0.70	(−2.33)		
$\bar{y}_t$ or $\bar{c}_t$					0.52	(2.39)	1.23	(4.20)	1.01	(4.13)
$\bar{R}^2$	0.25		0.45		0.17		0.31		0.27	
Wald	47.67		56.10		12.69		24.41		17.26	
Wald p	0.000		0.000		0.002		0.000		0.000	
Rel. prob. (BIC)	0.86		1.00		0.84		0.92		0.90	

The dependent variable is the duration-standardised, maturity-averaged one-year excess return $\bar{r}_{t+12}$ (3.8) for the United States, 1990m1–2014m12 (300 observations). Column (1) uses the yields at the six available maturities; column (2) adds trend inflation τ_t^{CPI} ; columns (3) and (4) use the one-year yield and the average yield $\bar{y}_t$, without and with trend inflation; column (5) uses the one-year cycle $c_t^{(1)}$ and the average cycle $\bar{c}_t$. t -statistics in parentheses are Newey–West adjusted (18 lags). Wald is the statistic of the joint test that all slope coefficients are zero, with its p -value below. Rel. prob. (BIC) is the relative posterior model probability $\exp\{(\text{BIC}_{\text{best}} - \text{BIC}_i)/2\}$, equal to one for the best-fitting model (Section 3.12). The table replicates Cieslak and Povala (2015), Table 2, Panel A.

Table B.3

Predictive R^2 under the expectations-hypothesis null (CP Table 2, Panel B)

		Percentiles of $\bar{R}^2$		
		5th	50th	95th
ϕ_τ	ϕ_r			
0.800	0.75	−0.003	0.007	0.045
0.975	0.75	−0.002	0.019	0.096
0.999	0.75	−0.002	0.022	0.102
0.975	0.60	−0.003	0.015	0.083
0.975	0.75	−0.002	0.019	0.102
0.975	0.90	−0.002	0.029	0.125

Each row reports the 5th, 50th, and 95th percentiles of the adjusted R^2 from the predictive regression of $\bar{r}_{t+12}$ on the latent factors (τ_t, r_t) in data simulated from the calibrated expectations-hypothesis economy of ??, in which risk premia are constant ($\lambda = 0$); 5,000 simulations of length $T = 470$ months, the sample length of Cieslak and Povala (2015). The upper block varies the persistence of trend inflation ϕ_τ holding $\phi_r = 0.75$; the lower block varies the persistence of the real-rate factor ϕ_r holding $\phi_\tau = 0.975$ (the blocks are simulated independently, so the shared grid point differs slightly between them). The 95th percentile rises with both persistences, reproducing the shape of the null distribution in Cieslak and Povala (2015), Table 2, Panel B; its level lies below their 0.19–0.23 because the original calibration is underspecified, a gap documented in ?? rather than closed.

Table B.4

Replication of Cieslak–Povala (2015) Table 4: individual returns on the cycle factor

	$rx^{(2)}$	$rx^{(5)}$	$rx^{(10)}$
<i>Panel A. Single cycle factor</i>			
CF_t	0.69 (2.53)	1.37 (3.91)	1.42 (5.06)
$\bar{R}^2$	0.11	0.25	0.32
$\Delta\bar{R}^2$	0.25	0.18	0.21
<i>Panel B.1. Short and own-maturity cycle</i>			
$c_t^{(1)}$	-0.70 (-1.82)	-0.61 (-3.48)	-0.40 (-4.39)
$c_t^{(n)}$	0.84 (2.09)	1.02 (4.10)	1.08 (5.65)
$\bar{R}^2$	0.09	0.25	0.43
<i>Panel B.2. Own-maturity cycle only</i>			
$c_t^{(n)}$	0.08 (1.08)	0.20 (1.56)	0.44 (2.85)
$\bar{R}^2$	0.03	0.06	0.17

The dependent variable is the duration-standardised individual one-year excess return $rx_{t+12}^{(n)}/n$ for $n \in \{2, 5, 10\}$ years; United States, 300 observations. Panel A regresses the return on the single cycle factor CF_t (3.6); t -statistics in parentheses are Newey–West adjusted (18 lags), and the bracketed interval is the 5th–95th percentile range of the HAC t -statistic under the stationary block bootstrap of Politis and Romano (1994) (mean block length 12, 5,000 replications). $\Delta\bar{R}^2$ is the additional adjusted R^2 obtained when the single factor is replaced by the unrestricted regression on the six yields and trend inflation. Panel B.1 regresses the return on the one-year cycle $c_t^{(1)}$ and the own-maturity cycle $c_t^{(n)}$; Panel B.2 uses $c_t^{(n)}$ alone. The table replicates Cieslak and Povala (2015), Table 4.

Table B.5

Correlations between local and global Cochrane–Piazzesi factors (DH Table 3)

	BE	DE	FR	IT	NL	CH	GB	SE	CA	JP	US	Global
BE	1.00											
DE	0.76	1.00										
FR	0.80	0.87	1.00									
IT	0.82	0.75	0.84	1.00								
NL	0.81	0.92	0.90	0.81	1.00							
CH	0.65	0.84	0.76	0.69	0.80	1.00						
GB	0.66	0.79	0.81	0.67	0.84	0.71	1.00					
SE	0.66	0.83	0.80	0.77	0.82	0.81	0.73	1.00				
CA	0.65	0.89	0.80	0.69	0.83	0.87	0.78	0.88	1.00			
JP	0.45	0.59	0.54	0.57	0.60	0.75	0.46	0.70	0.62	1.00		
US	0.62	0.81	0.77	0.59	0.80	0.78	0.83	0.80	0.85	0.77	1.00	
Global	0.76	0.92	0.90	0.79	0.92	0.87	0.88	0.89	0.91	0.84	0.97	1.00

The table reports pairwise correlations of the monthly local Cochrane–Piazzesi forward-rate factors $CP_{i,t}$ (Cochrane and Piazzesi, 2005) and the GDP-weighted global factor GCP_t over the full sample, 1990–2024; only the lower triangle is shown. The table replicates Dahlquist and Hasseltoft (2013), Table 3.

Table B.6**Fama–Bliss and Cochrane–Piazzesi regressions by country (DH Table 4)**

Country	n	Fama–Bliss			Cochrane–Piazzesi		
		b	t	$\bar{R}^2$	b	t	$\bar{R}^2$
BE	2	1.12	(4.74)	0.25	1.69	(7.06)	0.41
	5	1.72	(4.31)	0.22	5.78	(7.26)	0.44
	10	3.14	(5.09)	0.25	9.99	(5.92)	0.35
DE	2	0.67	(2.16)	0.08	1.49	(3.89)	0.22
	5	1.46	(4.27)	0.17	5.87	(4.73)	0.29
	10	2.75	(4.36)	0.18	10.80	(4.35)	0.27
FR	2	0.64	(2.75)	0.07	1.51	(4.06)	0.29
	5	1.64	(4.53)	0.18	5.65	(4.85)	0.36
	10	2.79	(4.92)	0.21	11.14	(5.16)	0.37
IT	2	0.86	(3.92)	0.17	1.47	(8.53)	0.46
	5	1.32	(2.35)	0.06	5.73	(8.34)	0.48
	10	0.69	(0.70)	0.01	11.17	(7.33)	0.47
NL	2	0.87	(2.91)	0.12	1.43	(3.82)	0.23
	5	1.67	(4.56)	0.21	5.76	(4.69)	0.32
	10	3.21	(5.43)	0.25	11.32	(4.41)	0.32
CH	2	0.35	(1.64)	0.05	1.18	(2.61)	0.11
	5	1.07	(2.92)	0.08	6.44	(5.49)	0.27
	10	1.67	(2.08)	0.07	11.19	(4.53)	0.21
GB	2	0.70	(2.97)	0.10	1.31	(3.31)	0.17
	5	1.20	(3.23)	0.13	5.89	(4.35)	0.29
	10	1.08	(2.14)	0.06	11.69	(4.38)	0.30
SE	2	0.53	(2.54)	0.05	1.66	(6.38)	0.47
	5	1.34	(3.37)	0.10	5.47	(8.08)	0.45
	10	3.33	(3.86)	0.19	10.77	(9.02)	0.45
CA	2	0.46	(2.55)	0.05	1.66	(4.14)	0.27
	5	1.12	(4.04)	0.14	5.62	(5.66)	0.34
	10	1.87	(3.50)	0.14	10.43	(5.32)	0.33
JP	2	0.14	(0.28)	0.01	1.35	(6.62)	0.62
	5	1.99	(5.09)	0.34	6.71	(10.15)	0.69
	10	0.63	(1.28)	0.04	9.84	(7.85)	0.45
US	2	0.66	(2.23)	0.08	1.84	(4.51)	0.30
	5	0.87	(2.29)	0.07	5.49	(4.82)	0.25
	10	1.34	(2.48)	0.08	9.80	(4.64)	0.23

The dependent variable is the raw one-year excess return $rx_{i,t+12}^{(n)}$ (3.1) for $n \in \{2, 5, 10\}$ years in each of the eleven markets. The Fama–Bliss regression projects the return on the forward–spot spread $f_{i,t}^{(n)} - y_{i,t}^{(1)}$ (Fama and Bliss, 1987); the Cochrane–Piazzesi regression projects it on the local forward-rate factor $CP_{i,t}$ (Cochrane and Piazzesi, 2005). t -statistics in parentheses are Newey–West adjusted (12 lags); adjusted R^2 reported. The table replicates Dahlquist and Hasseltoft (2013), Table 4.

Table B.7
Local and global Cochrane–Piazzesi regressions (DH Table 6)

Country	n	Local		Global		Joint			Wald p
		b_{CP}	$\bar{R}^2$	b_{GCP}	$\bar{R}^2$	$b_{CP^\perp}$	b_{GCP}	$\bar{R}^2$	
BE	2	1.69	0.41	2.24	0.28	1.47	2.24	0.41	[0.00]
	5	5.78	0.44	7.44	0.28	5.30	7.44	0.44	[0.00]
	10	9.99	0.35	13.75	0.26	8.14	13.75	0.36	[0.00]
DE	2	1.49	0.22	2.03	0.32	−0.86	2.03	0.33	[0.00]
	5	5.87	0.29	7.39	0.36	−0.41	7.39	0.35	[0.00]
	10	10.80	0.27	12.82	0.29	3.13	12.82	0.30	[0.00]
FR	2	1.51	0.29	2.23	0.35	0.06	2.23	0.35	[0.00]
	5	5.65	0.36	7.44	0.35	3.28	7.44	0.37	[0.00]
	10	11.14	0.37	13.64	0.31	10.02	13.64	0.37	[0.00]
IT	2	1.47	0.46	2.43	0.25	1.66	2.43	0.47	[0.00]
	5	5.73	0.48	8.64	0.21	7.23	8.64	0.50	[0.00]
	10	11.17	0.47	17.71	0.23	13.25	17.71	0.48	[0.00]
NL	2	1.43	0.23	2.02	0.31	−0.63	2.02	0.32	[0.00]
	5	5.76	0.32	7.37	0.35	1.39	7.37	0.35	[0.00]
	10	11.32	0.32	13.58	0.31	7.17	13.58	0.33	[0.00]
CH	2	1.18	0.11	1.12	0.18	−0.36	1.12	0.18	[0.01]
	5	6.44	0.27	5.34	0.32	1.52	5.34	0.33	[0.00]
	10	11.19	0.21	9.83	0.29	0.18	9.83	0.28	[0.00]
GB	2	1.31	0.17	1.72	0.20	0.30	1.72	0.20	[0.00]
	5	5.89	0.29	6.47	0.23	5.36	6.47	0.29	[0.00]
	10	11.69	0.30	12.68	0.23	11.12	12.68	0.29	[0.00]
SE	2	1.66	0.47	2.80	0.45	1.05	2.80	0.49	[0.00]
	5	5.47	0.45	9.58	0.46	2.54	9.58	0.48	[0.00]
	10	10.77	0.45	18.59	0.45	5.75	18.59	0.47	[0.00]
CA	2	1.66	0.27	2.30	0.34	−0.19	2.30	0.34	[0.00]
	5	5.62	0.34	7.51	0.40	0.45	7.51	0.40	[0.00]
	10	10.43	0.33	14.13	0.40	0.05	14.13	0.40	[0.00]
JP	2	1.35	0.62	1.00	0.44	1.33	1.00	0.62	[0.00]
	5	6.71	0.69	5.45	0.59	5.05	5.45	0.71	[0.00]
	10	9.84	0.45	8.69	0.46	5.14	8.69	0.49	[0.00]
US	2	1.84	0.30	1.86	0.26	2.38	1.86	0.30	[0.00]
	5	5.49	0.25	5.49	0.22	7.69	5.49	0.25	[0.00]
	10	9.80	0.23	9.93	0.21	12.18	9.93	0.23	[0.00]

For each country and maturity $n \in \{2, 5, 10\}$ years, the one-year excess return $rx_{i,t+12}^{(n)}$ is regressed on the local Cochrane–Piazzesi factor $CP_{i,t}$ alone (Local); on the GDP-weighted global factor GCP_t alone (Global); and on both jointly (Joint), where the local factor is first orthogonalised with respect to the global one (denoted $CP_{i,t}^\perp$) so that its coefficient measures the incremental local content—the forward-factor analogue of the horse race (3.14)–(3.15). Adjusted R^2 is reported for each specification; the Wald p -value in brackets tests the joint significance of the two slopes in the joint regression. All inference is Newey–West adjusted (12 lags). The table replicates Dahlquist and Hasseltoft (2013), Table 6.

Table B.8

US-dollar excess returns on the global and FX-adjusted factors (DH Table 7)

Pair	n	GCP_t			$FXGCP_t$		
		b	t	$\bar{R}^2$	b	t	$\bar{R}^2$
EUR/USD	2	-0.57	(-0.13)	-0.00	6.23	(2.29)	0.10
	5	4.79	(0.97)	0.02	8.82	(3.00)	0.17
	10	10.18	(1.76)	0.07	11.35	(3.37)	0.22
CHF/USD	2	-2.16	(-0.46)	0.00	6.43	(2.78)	0.12
	5	2.07	(0.43)	0.00	8.62	(3.46)	0.19
	10	6.56	(1.26)	0.03	10.63	(3.96)	0.25
GBP/USD	2	6.28	(1.93)	0.05	2.52	(0.96)	0.02
	5	11.00	(2.85)	0.15	5.14	(1.84)	0.08
	10	17.17	(3.24)	0.24	6.95	(2.12)	0.10
JPY/USD	2	7.33	(1.88)	0.07	5.98	(2.82)	0.09
	5	11.78	(2.95)	0.16	7.99	(3.58)	0.14
	10	15.02	(3.89)	0.23	9.68	(4.43)	0.19
CAD/USD	2	4.37	(1.42)	0.03	4.32	(2.02)	0.08
	5	9.59	(2.76)	0.13	7.17	(3.18)	0.19
	10	16.21	(4.25)	0.26	9.98	(3.94)	0.26
SEK/USD	2	6.99	(1.23)	0.03	5.98	(1.76)	0.06
	5	13.81	(2.22)	0.11	8.32	(2.32)	0.11
	10	22.87	(3.44)	0.25	10.74	(2.71)	0.14

The dependent variable is the one-year US-dollar excess return (3.9) earned by a United States investor who holds the foreign n -year bond unhedged, $n \in \{2, 5, 10\}$ years. The return is regressed on the unadjusted global Cochrane–Piazzesi factor GCP_t and, separately, on the FX-adjusted global factor $FXGCP_t$, constructed top-down on the GDP-weighted average dollar return in the manner of Section 3.10. The euro pair uses German bonds; all other pairs use the respective own market. t -statistics in parentheses are Newey–West adjusted (12 lags); adjusted R^2 reported. The table replicates Dahlquist and Hasseltoft (2013), Table 7.